

VCS

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Introduction

VCS

Architectures, Platforms, Operations, and GitOps

A practical path from your first commit to running a forge and shipping with GitOps. You start as a Git user, then you install and compare hosting platforms, then you operate them, then you let Git drive the cluster.

VCS

Git foundations	mental model → install → first repository
Everyday workflows	commits, branches, rebase, remotes, hooks
Power skills	internals, reflog, bisect, repair
Hosting platforms	GitHub lock-in, GitLab, Forgejo, Gerrit, ...
Operating a forge	identity, CI, storage, HA, air-gap
GitHub Actions	workflows as a product
GitOps	Flux, Argo CD, Argo Rollouts

The sidebar is the full TOC (chapters 1–66 plus appendices). **Every chapter starts with Theory** (invariants, a diagram, what can break). **Parts I–II** have prose for the original pages and for chapter 7 (reset, restore, and checkout). The other newer core sub-pages are still stubs. Later parts are stubs until a writing pass fills them, except chapter 25 (Gitea and Forgejo), which is prose. The map lives in `content/_planning/SYLLABUS.md`.

Who this book is for

Software engineers, DevOps, platform, and SRE readers on **Linux**. You can start with no Git. Terms are defined on first use. Depth ramps: user → installer → admin → operator → GitOps practitioner.

Commands, paths, and package names assume a POSIX shell on Linux (**bash** is fine). Examples use `/home/...`, `~/.gitconfig`, and distro packages (**pacman**, **apt**, **dnf**). The Git commands themselves are the same on other Unix systems; this book does not walk macOS or Windows.

What you will master

- Git’s object model, everyday commands, and recovery
- How a Git hosting platform is built, and how GitLab, Forgejo, Bitbucket, Gerrit, SourceHut, and GitHub differ
- Identity, CI, packages, supply-chain controls, backups, and air-gap
- GitHub Actions as one CI product
- GitOps with Flux and Argo CD, then progressive delivery with Argo Rollouts

How to read this so it sticks

Work front to back in Part I. Type every command in a throwaway directory. Later parts assume you can commit, branch, and push. Platform labs reuse the same three desk repositories: `desk-web`, `desk-api`, `desk-infra`.

The companion kit

Required for Parts I–II: Git 2.45 or newer on Linux (examples checked against 2.55), a terminal, and a text editor (`nvim` is the default in config examples).

Required from the GitOps parts onward: Docker, kind or K3s, `kubectl`, and Helm. Install those when you reach Part VII. You do not need a cluster to finish Git foundations.

Conventions

Commands are complete. Run them from the directory the page names. Output is representative; hashes and timestamps on your machine will differ.

Heredocs create files without opening an editor:

```
cat > README.md <<'EOF'
# desk-api
EOF
```

The quoted `'EOF'` keeps `$variables` and backticks literal. Drop the quotes only when you want the shell to expand them.

A **repository** (repo) is a directory Git is tracking. A **forge** is a Git hosting platform: the website and services around repositories (issues, pull requests, CI, packages). A **remote** is another copy of a repository you fetch from and push to.

Acknowledgements

K19G. Built as part of the books library.

Git Foundations

Mental Model

Theory

Theory: Snapshots, Names, and a Graph

This chapter is the model the rest of the book assumes. Commands later only add objects and move names.

Invariants

1. A **commit** names a complete tree, not a patch. Unchanged files reuse blob hashes.
2. History is a **DAG**: each commit points at parent(s). Nothing in the object store is rewritten in place.
3. A **branch** is a movable name for one commit. **HEAD** is a name for “where I am,” usually a pointer to a branch name.
4. Everyday work uses **three trees**: working directory, index, **HEAD**. Status is the report of how they differ.
5. **Distribution** means every clone is a repository. A forge is extra software around that fact.

names (refs)	objects (content-addressed)
HEAD → main → C	C commit → tree T → blobs
↑	B commit
B ← A	A commit

If the name moves and the objects stay, you have not lost history; you have lost a pointer. Reflog (chapter 17) is the local journal of pointer motion.

Why this model

Centralized systems made the server the only full history. Git inverts that: the laptop is complete. Cheap branches follow from “a branch is a file containing a hash.” Integrity follows from hashing content. Hosting platforms, CI, and GitOps cannot escape these invariants; they only add names, ACLs, and jobs.

What can break

- Treating Git like a file-sync tool (last write wins, no merge).
- Treating GitHub as Git (issues, Actions, and PRs are forge objects).
- Committing on a **detached HEAD** and switching away without a branch name.
- Assuming `git pull` is a primitive; it is `fetch` plus `integrate`.

How to read this chapter

Landscape is *why a VCS*. DAG, three trees, objects, refs, and distribution are *how Git thinks*. Detached HEAD and revision ranges are the first notations you will type. Install and the first repository apply this model on Linux.

The Version Control Landscape

The Version Control Landscape

After this page you can say what version control is for, why Git is the usual tool, and what a forge is.

Why you need a history

Without version control, a desk project grows copies:

```
desk-api/  
  app.py  
  app_backup.py  
  app_final.py  
  app_final_v2.py  
  app_really_final_fixed.py
```

You lose which file is current, who changed the login line, and any way to undo Tuesday without also undoing Wednesday. Two people editing `app.py` overwrite each other.

A **version control system** (VCS) keeps one tree of files, a history of snapshots, and a way to merge parallel work.

Three generations

Generation	Idea	Examples
Local	History on one disk	SCCS, RCS
Centralized	One server is the repository	CVS, Subversion, Perforce
Distributed	Every clone is a full repository	Git, Mercurial

Local tools recorded deltas next to the files. Fine for one person on one machine.

Centralized tools put the real history on a server. You check out a working copy. Committing needs the network. If the server dies, history you do not have locally is gone.

Distributed tools give every clone the whole history. `git commit` is a local operation. A **remote** is another copy you push to when you choose. That is why Git works on a train, and why rewriting commits you already shared is a people problem.

Why Git

Git won the ecosystem. The properties that matter for this book:

- A commit is a **snapshot** of the whole tree, not a pile of patches as the primary model.
- A **branch** is a cheap name for one commit. Merging and rebasing are graph operations.
- Object names are **hashes** of content. Bit flips show up as missing or mismatched objects.
- Log, diff, commit, branch, and merge all work without a network.
- Hosting platforms speak Git remotes. Learning Git is not loyalty to one website.

Git vs Subversion, the comparison that still comes up:

	Git	Subversion
Architecture	Distributed	Centralized
Offline commit	Yes	No
Branch	A name on a commit	A copy in the tree
Typical remote	Any number	One canonical server

Mercurial is also distributed. Git is what forges, CI systems, and GitOps engines speak.

What a forge is

A **repository** is a directory Git is tracking (the `.git` folder plus your files).

A **forge** is a Git hosting platform: the website and services around repositories. Issues, pull requests, CI, packages, and user accounts live on the forge. GitHub, GitLab, Forgejo, Bitbucket, Gerrit, and SourceHut are forges. This book treats Git first, then the platforms.

Git is the tool. The forge is the product you log into. GitOps (later) is the idea that the same Git history should drive what runs in production.

What Git is not

- Not a backup by itself. A single laptop clone can die; you still need remotes and backups.
- Not a file-sync tool. Dropbox overwrites. Git records snapshots and asks you to merge.
- Not GitHub. You can use Git for years with a forge you run yourself, or with no forge at all.

Next

The next pages are how Git thinks: a graph of commits, three trees on disk, four object types, names for those objects, and what “distributed” means for a server.

The Directed Acyclic Graph

The Directed Acyclic Graph

After this page you can draw a tiny Git history as a graph and say what a branch is.

Snapshots with parents

A **commit** is a snapshot of the whole project at one instant, plus metadata (author, message, time) and a pointer to the commit that came before it — its **parent**.

Two commits in a row:

```
A ← B
```

B's parent is A. Time flows left to right in this book's diagrams; the parent pointer points backward, toward history.

A chain of commits is a **directed acyclic graph** (DAG): directed because each commit points at parents, acyclic because a commit cannot be its own ancestor.

A branch is a name

A **branch** is a movable name for one commit. **main** is a file that contains a hash. When you commit, Git writes a new snapshot whose parent is the current tip, then slides the name forward.

```
A ← B ← C      main
```

main now names C. A and B are still there. Nothing was overwritten.

A second line of work

Start a branch named **feature** while **main** is at C, then add two commits:

```
A ← B ← C      main
          ↑
          D ← E  feature
```

C has two children. Git is fine with that. A **merge commit** has two parents:

```

A ← B ← C ← M      main
           ↑       ↑
           D ← E      feature

```

M ties the two lines together. **Rebase** (chapter 9) copies D and E onto a new parent instead. New hashes, similar diffs. You will do that after first-repository muscle memory is in place.

HEAD

HEAD is “where I am.” Most of the time it points at a branch name, and the branch name points at a commit:

```
HEAD → refs/heads/main → C
```

Commands that “move HEAD” are moving that name, or (when HEAD is **detached**) pointing HEAD at a raw commit hash. Detached HEAD shows up when you check out a tag or a hash. New commits on a detached HEAD are easy to lose when you switch away. Create a branch before you leave, or find them later with the reflog (chapter 17).

```

git log --oneline --graph --all
git rev-parse HEAD main
git status

```

`git status` says HEAD detached at ... when it is.

Look at the graph

Once you have a repository (next chapter’s neighbor, chapter 3):

```
git log --oneline --graph --decorate --all
```

`--decorate` prints the names (`main`, `HEAD`, tags) next to the commits they point at. The graph is the truth. Commands add objects and move names. If you hold the graph, `reset`, `rebase`, and `reflog` stop feeling like magic.

The Three-Tree Model

The Three-Tree Model

After this page you can name Git's three snapshots of the same project and say which command copies between them.

Three snapshots, one project

Everyday Git is three versions of the same tree:

Name	On disk	Role
Working directory	your files	What you edit
Index (staging area)	<code>.git/index</code>	What the <i>next</i> commit will contain
HEAD commit	an object in <code>.git/objects</code>	Last committed snapshot on this branch

```
working directory --git add-->  index --git commit-->  HEAD
                        <--git restore--      <--git restore --staged / git reset--

~/desk/desk-api/           .git/index           .git/objects
README.md, src/...         next commit        last commit (HEAD)
```

- `git add` copies working-tree bytes into the index (and writes **blob** objects).
- `git commit` writes a **tree** from the index, then a **commit** pointing at that tree and at the current HEAD, then slides the branch name forward.
- `git restore <file>` copies from the index (or a commit) onto the working tree.
- `git restore --staged <file>` copies from HEAD into the index, leaving your working-tree edit in place.

There is no fourth “GitHub tree.” A remote is another repository’s names plus objects you have fetched.

Why the index exists

The index lets the next commit be a chosen subset of what is on disk. You can edit five files and commit two.

```
echo "ok" >> README.md
echo "wip" >> notes.txt
git add README.md
git status
# README.md is staged; notes.txt is not
```

`git commit -a` copies **already tracked** files from the working tree into the index and commits. New files still need `git add`. Prefer explicit `git add` until that distinction is boring.

status is the three-tree report

```
git status
```

Git compares:

- working tree vs index → “Changes not staged for commit”
- index vs HEAD → “Changes to be committed”
- untracked files → paths the index does not know

When all three match, `status` is clean. That is the happy state after a commit.

reset in one sentence

`git reset` moves HEAD and optionally the index and working tree. Soft / mixed / hard are which of the three trees follow. Chapter 7 (reset) and chapter 17 (reflog) use this on purpose. For now, prefer `git restore` to throw away an unstaged edit, and `git restore --staged` to unstage.

Check it

```
git status
git diff          # working tree vs index
git diff --staged # index vs HEAD
```

If `git diff` is empty and `git diff --staged` is not, the next commit already contains the change. If both are empty and `status` still lists a file, it is untracked — `git add` it.

Object Types

Object Types

After this page you can name Git's four object types and say what a hash is identifying.

Content-addressed storage

Git stores objects in `.git/objects`. The name of an object is a **hash** of its content. Change one byte of a file → new blob, new tree, new commit. Unchanged files keep the same blob hash across commits, which is why a one-line edit is cheap.

Historically the hash is SHA-1. Repositories can use SHA-256. Commands look the same. This book prints short hashes (`a1b2c3d`) the way `git log --oneline` does.

Four types

Type	Payload
blob	file bytes (no filename)
tree	name → object (a blob or a nested tree)
commit	a tree + parent commit(s) + author + message
tag	annotated pointer + message + optional signature

Filenames live in **trees**, not in blobs. Rename a file: new tree, same blob.

```
commit C
tree T
  README.md → blob B1
  src/      → tree T2
  main.py   → blob B2
parent → commit B
author Ada Desk <ada@example.com>
message "Add ping"
```

A **lightweight tag** is only a name (a ref). An **annotated tag** is a real tag object. Chapter 12 covers tagging in daily use.

Packfiles are compression

Git also stores **packfiles**: many objects packed together with deltas, for disk. That is compression, not the data model. You still think in snapshots. Chapter 16 opens the object store with `git cat-file`. You can peek now:

```
git cat-file -t HEAD      # usually "commit"
git cat-file -p HEAD      # the commit payload
git rev-parse HEAD^{tree} # the tree that commit names
```

`-p` pretty-prints. `-t` prints the type.

Integrity

Because names are hashes, a silent bit flip in a blob does not match the name the tree recorded. `git fsck` (chapter 21) walks the graph and reports missing or malformed objects. Forges and GitOps engines rely on this: the commit you pulled is the tree the author hashed.

What this buys a server

A Git hosting platform is, at minimum, a store of these objects plus names (branches, tags) plus auth. Chapter 23 is that architecture. The object model you just learned is the constraint the server cannot wish away: clones fetch objects and update names.

References: Branches, Tags, and HEAD

References: Branches, Tags, and HEAD

After this page you can say what a ref is, where it lives, and how HEAD relates to a branch.

A ref is a name for a hash

A **ref** (reference) is a name that points at a commit (or, for annotated tags, at a tag object). Git stores them as files under `.git/refs/` and, when there are many, in `.git/packed-refs`.

Ref	Typical path	Meaning
Branch	<code>.git/refs/heads/main</code>	Movable name for a line of work
Tag	<code>.git/refs/tags/v1.0</code>	Name meant to stay put
Remote-tracking	<code>.git/refs/remotes/origin/main</code>	Last-seen tip of a branch on a remote
HEAD	<code>.git/HEAD</code>	Where you are

```
git rev-parse main
git rev-parse HEAD
cat .git/HEAD
# ref: refs/heads/main
```

`git rev-parse` turns a name into a hash. Use it when a diagram and the disk disagree in your head.

Branches move, tags mostly do not

```
git branch                # local branches
git branch feature        # new name at current HEAD
git switch feature        # HEAD now points at feature
git switch -c hotfix      # create and switch
```

`git switch` is the modern command for changing branches (`git checkout` still works; this book uses `switch` for branches and `restore` for files).

A tag is a bookmark:

```
git tag v0.1                # lightweight: a name at HEAD
git tag -a v0.1 -m "first"  # annotated: a tag object
```

Moving a published tag surprises people who already fetched it. Moving a branch is normal.

HEAD is usually a pointer to a pointer

```
.git/HEAD          → ref: refs/heads/main
.git/refs/heads/main → a1b2c3d4...
```

Commit while on `main`: Git writes a new commit whose parent is `a1b2c3d4...`, then overwrites `refs/heads/main` with the new hash. `HEAD` did not change its text; the branch it names did.

Detached HEAD: `.git/HEAD` contains a raw hash instead of `ref: refs/heads/...`. You get there with `git switch --detach`, or by checking out a tag. To keep work from a detached `HEAD`:

```
git switch -c keep-this
```

That creates a branch at the current commit and attaches `HEAD` to it.

Remote-tracking names

After you add a remote named `origin` and fetch (chapter 11):

```
refs/remotes/origin/main
```

That is *your* last photograph of `main` on that remote. It updates on fetch. You do not commit onto it. Your local `main` is a separate name you push.

Plumbing vs porcelain

`git branch`, `git switch`, `git tag` are **porcelain** (human commands). `git update-ref` and `git symbolic-ref` are **plumbing** (the same operations, no sugar). This book stays on porcelain until internals (chapter 16). Knowing refs are files is enough to understand why creating a branch is cheap: Git writes a 41-byte file.

The Distributed Model

The Distributed Model

After this page you can explain why every clone is a repository, and what that implies for a hosting server.

Every clone is a full copy

When you `git clone`, you get:

- every commit reachable from the branches and tags you fetched
- a working tree checked out at one branch (usually `main`)
- a remote named `origin` pointing at the URL you cloned

There is no “thin client.” `git log`, `git diff`, `git commit`, `git branch`, and `git merge` talk to `.git/` on your disk. The network is for **fetch** (download objects and update remote-tracking names) and **push** (upload objects and ask the other side to move a name).

your laptop		a forge
<code>.git/objects</code>		<code>.git/objects</code>
<code>refs/heads/main</code>	<code>←</code>	<code>refs/heads/main</code>
<code>refs/remotes/origin/main</code>	<code>→</code>	(plus issues, CI, packages)

Both sides are Git repositories. The forge adds a website, accounts, and extra services. Git itself does not need them.

Implications

Offline work is real. You can commit on a plane. Push when you have a network.

Several remotes are normal. `origin` is a convention, not a law. You can push the same `desk-api` history to a laptop USB clone, a Forgejo server, and GitHub. Chapter 22 uses that as an independence move.

The server does not own history uniquely. If the forge vanishes, every up-to-date clone still has the Git objects. What you lose are issues, pull-request comments, CI logs, and packages — the forge’s extras. That split is the rest of this book.

Rewriting shared history is a social operation. `git commit --amend` and rebase change hashes. On your laptop before anyone fetched, that is fine. After others based work on those hashes, you are asking them to reset. Chapter 9 is the skill; this page is the reason it matters.

Fetch is not pull. `git fetch` updates `origin/main` on disk. `git pull` is fetch plus a merge or rebase into your current branch. Chapter 11 separates them. Prefer fetch until pull's extra step is a choice you can name.

What a hosting platform must speak

A Git server has to:

1. Store objects and refs.
2. Speak the Git protocol over SSH and/or HTTP (smart protocol: the client and server negotiate which objects to send).
3. Decide who may move which refs (authorization).
4. Run **hooks** on receive (the server-side programs that can reject a push).

Everything else — pull requests, CI, registries, SAML — is forge software on top of that. Chapter 23 draws the boxes. Chapter 3 stays on your laptop: `git init`, `add`, `commit`.

Detached HEAD

Detached HEAD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What detached HEAD is, how you get there, how to keep the work.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Revisions and Ranges

Revisions and Ranges

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- HEAD~ vs HEAD~, A..B vs A...B, @{u}, reflog syntax preview.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Install and Config

Theory

Theory: Identity and Config are Data, Not Login

Git does not have accounts. It has **key-value config** and **author strings** baked into commits. A forge later *matches* those strings to a user.

Invariants

1. Config is three layers: system (`/etc/gitconfig`), global (`~/.gitconfig`), local (`.git/config`). **Later overrides earlier.**
2. `user.name` and `user.email` are copied into every commit object. Changing config does not rewrite old commits.
3. Authentication to a **remote** (SSH key or HTTPS token) is separate from commit identity.
4. `init.defaultBranch` only affects **new** repositories.

```
/etc/gitconfig      machine
~/.gitconfig        you
.git/config          this repo    ← wins

user.email    commit.author    (history)
SSH key       git@host          (transport)
```

Why layers

One person, two emails (home and work) is a local or `includeIf` problem, not a second Git install. One laptop, several forges, is several keys in `~/.ssh/config`, not several `user.names`.

What can break

- Commits with `$USER@$HOSTNAME` because identity was never set (`user.useConfigOnly` refuses that).
- Pushing with HTTPS using an account password; most forges want a **token**.
- Setting `core.autocrlf` on Linux and fighting LF files.
- Assuming `core.editor` is optional and then hanging in vim on `git commit` with no `-m`.

How to read this chapter

Install is PATH and version. Layers are which file won (`--show-origin`). Identity is what history will say. SSH vs HTTPS is how a remote will know you. Editor, aliases, and updates are the Linux daily driver.

Installing Git

Installing Git

After this page you have a current `git` on your PATH on Linux and can print its version.

Check what you already have

```
git --version
which -a git
```

This book wants **2.45 or newer** on **Linux**. Examples were checked against **2.55**. You need `git switch`, `git restore`, and (later) SSH commit signing.

Most desktop distros already ship a new enough Git. If `which git` prints `/usr/bin/git` and the version is 2.45+, you are done.

Install from the distro

Arch Linux and Omarchy

```
pacman -Q git
# missing or too old:
sudo pacman -S git
```

Debian and Ubuntu

```
sudo apt update
sudo apt install git
```

Fedora

```
sudo dnf install git
```

Enterprise images sometimes freeze Git at 2.25. If `git --version` is older than 2.45, use your distro's backports or a current tarball from git-scm.com so `switch` and `restore` exist.

Line endings and the filesystem

Linux stores **LF** line endings and executable bits. Leave `core.autocrlf` unset. Leave `core.fileMode` at the default (`true`) so `chmod +x` is recorded. Those defaults are why hooks and scripts in later chapters work with `chmod +x hook.sh`.

What else you need for Parts I–II

Tool	Role
A Linux terminal	Where you run <code>git</code> (<code>bash</code> or another POSIX shell)
A text editor	<code>nvim</code> in this book's config examples; any editor is fine

Docker, Kubernetes, and a forge account wait until later parts.

Set `core.editor` once you know which editor you type in (next page). Until then, `git commit -m "message"` avoids opening one.

Confirm

```
git --version
which git
echo "$HOME"
```

`$HOME` should be `/home/<you>`. Then set identity and the three config layers on the next page.

Config Layers

Config Layers

After this page you can say which file a Git setting comes from, and set the handful of keys this book assumes.

Three layers

Layer	Flag	Typical file	Who
System	<code>--system</code>	<code>/etc/gitconfig</code>	every user on the machine
Global	<code>--global</code>	<code>~/.gitconfig</code>	you, every repo
Local	<code>--local</code> (default)	<code>.git/config</code>	this repo only

Later keys **override** earlier ones. Work email in a company repo is a local `user.email`, not a second global identity you forget to switch.

wins

<code>.git/config</code>	local	this repo
<code>~/.gitconfig</code>	global	this user
<code>/etc/gitconfig</code>	system	this machine

reads bottom → top; last value for a key wins

```
git config --list --show-origin --show-scope
git config --show-origin user.email
```

`--show-origin` prints the file that won. `--unset` removes a key at the layer you name:

```
git config --global --unset core.editor
```

Set the global layer once

```
git config --global init.defaultBranch main
git config --global core.editor "nvim"
git config --global pull.ff only
git config --global rerere.enabled true
```

```
git config --global diff.algorithm histogram
git config --global merge.conflictstyle zdiff3
```

Key	Why
init.defaultBranch main	git init creates main, not master
core.editor	git commit without -m opens this editor (code --wait, vim, nvim)
pull.ff only	git pull fast-forwards or fails; you merge or rebase on purpose
rerere.enabled	reuse recorded conflict resolutions
diff.algorithm histogram	better diffs on refactored files
merge.conflictstyle zdiff3	conflict markers include the common ancestor

pull.ff only means a pull that would create a merge commit fails instead. That is safer than the old default. Chapter 11 is remotes; this setting is safe to apply now.

Leave core.fileMode at the default so Git records executable bits. Set it to **false** only on a filesystem that cannot store those bits.

Repo-only overrides

```
cd desk-api
git config user.email "ada@work.example"
# writes .git/config
```

Inspect:

```
git config --local --list --show-origin
```

includeIf for home vs work

```
# ~/.gitconfig
[user]
    name = Ada Desk
    email = ada@example.com

[includeIf "gitdir:~/work/"]
    path = ~/.gitconfig-work

# ~/.gitconfig-work
[user]
```

```
email = ada@work.example
```

Any repo under `~/work/` picks up the work email without a per-repo command. Paths in `gitdir:` should be absolute or start with `~/`.

See the result

```
git config --list --show-origin --show-scope | less
```

Identity (`user.name`, `user.email`) is the next page. Those two keys go into every commit.

Identity, Email, and Essential Settings

Identity, Email, and Essential Settings

After this page every commit you make records a name and email you chose.

Identity is metadata, not a login

`user.name` and `user.email` are written **into every commit**. They are not a password and not a forge account. Forges *match* the email to an account when they display history. A typo produces “unknown author” on the website. `git commit` still succeeds.

```
~/.gitconfig
user.name = Ada Desk
user.email = ada@example.com

commit object
  tree    ...
  parent  ...
  author  Ada Desk <ada@example.com> 1774... +0530
  committer Ada Desk <ada@example.com> ...
  message ...
```

```
git config --global user.name "Ada Desk"
git config --global user.email "ada@example.com"
```

Use the email the forge should associate with you. For a first laptop, any address you control is fine. For work, the company address, often via `includeIf` from the previous page.

Refuse to commit if identity is missing, instead of guessing `$USER@$HOSTNAME`:

```
git config --global user.useConfigOnly true
```

Check what will be recorded

```
git config user.name
git config user.email
git config --show-origin user.email
```

After you have a commit (chapter 3):

```
git log -1 --format=full
```

Author: is `user.name` <`user.email`> from the time of the commit. Changing `~/.gitconfig` later does not rewrite old commits. Chapter 6 is amending; chapter 20 is signing.

Signing is later

Signed commits (SSH or GPG) prove the commit came from a key you hold. Set identity here first. Signing is chapter 20.

Author vs committer

A commit has two people fields: **author** (who wrote the patch) and **committer** (who made the commit object). They match until you cherry-pick, rebase, or apply a patch from someone else. `git log` shows the author by default. You do not need to set `GIT_COMMITTER_NAME` for Part I.

Essential settings, one list

If you jumped here from install, this is the whole global block this book expects:

```
git config --global user.name "Ada Desk"
git config --global user.email "ada@example.com"
git config --global user.useConfigOnly true
git config --global init.defaultBranch main
git config --global core.editor "nvim"
git config --global pull.ff only
```

Replace the name, email, and editor with yours. Then pick SSH or HTTPS on the next page.

SSH vs HTTPS

SSH vs HTTPS

After this page you can authenticate to a forge over SSH or HTTPS and know which URL form you are looking at.

Two ways to talk to a remote

	SSH	HTTPS
URL	<code>git@host:owner/repo.git</code>	<code>https://host/owner/repo.git</code>
Secret	Private key on disk	Password, personal access token, or credential helper
Typical use	Daily push/fetch on a machine you own	CI, quick clone, hosts that block SSH

Both speak the Git smart protocol. The difference is how the server decides who you are. Part I does not require a forge. Set this up before chapter 11 so the first push is not a scavenger hunt.

```
Linux laptop                                forge
~/.ssh/id_ed25519  SSH                      authorized keys
(private)                                git@host
id_ed25519.pub      or HTTPS+token  + Git objects

git push / git fetch
```

SSH keys

Generate an **ed25519** key if you do not have one:

```
ssh-keygen -t ed25519 -C "ada@example.com" -f ~/.ssh/id_ed25519
```

Press Enter for no passphrase, or set one (then **ssh-agent** remembers it for the session):

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Public key (this is what you paste into the forge):

```
cat ~/.ssh/id_ed25519.pub
```

The private file `~/.ssh/id_ed25519` stays on your machine. Never commit it. Never paste it into a ticket.

On the forge: account settings → SSH keys → paste the `.pub` line. GitHub, GitLab, and Forgejo all have this screen. The label can be `laptop`.

Test (GitHub as an example host; use your forge's hostname):

```
ssh -T git@github.com
```

A greeting that includes your username means the key was accepted. A password prompt means the server did not see the key (`ssh -v` shows which files it tried).

`~/.ssh/config` for several hosts:

```
Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519

Host forgejo.home
  HostName 192.168.68.200
  User git
  Port 2222
  IdentityFile ~/.ssh/id_ed25519
```

A later Forgejo lab uses a port like 2222. The `Host` alias lets you write `git@forgejo.home:ada/desk-api.git`.

HTTPS and tokens

```
git clone https://github.com/example/desk-api.git
```

Pushing over HTTPS asks for a username and a **personal access token** (not your account password on most forges). Store it in a credential helper:

```
git config --global credential.helper libsecret # GNOME Keyring / libsecret, if installed
# memory-only fallback (forgets at reboot or after a timeout):
git config --global credential.helper cache
```

CI systems often use HTTPS with a job-scoped token. Humans on a laptop usually have an easier time with SSH.

Pick one default

For this book's desk repos on a machine you own: **SSH**. Clone and remote URLs will look like:

```
git@host:owner/desk-api.git
```

You can change a remote later (chapter 11):

```
git remote -v  
git remote set-url origin git@host:owner/desk-api.git
```

Chapter 3 stays local: `git init` does not need SSH or HTTPS.

Editor, Pager, and Difftool

Editor, Pager, and Difftool

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `core.editor`, `core.pager`, `less`, `nvimdiff` / `vimdiff` on Linux.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Aliases

Aliases

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git config alias.*`, shell aliases vs Git aliases, useful starter set.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Updating Git on Linux

Updating Git on Linux

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Distro packages vs building from source; checking `git --version`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

First Repository

Theory

Theory: The Add/Commit Cycle is the Product

A repository is useful the moment you can take bytes on disk, choose a subset, and name a snapshot. Everything else (branch, rebase, push) is names and copies of that operation.

Invariants

1. `git init` creates `.git/` and a symbolic `HEAD`. The first **commit** creates `refs/heads/main`.
2. `git add` copies working-tree bytes into the **index** (and writes blobs). It does not create a commit.
3. `git commit` writes a tree from the index, then a commit, then slides the current branch name.
4. `git status` is three comparisons: working vs index, index vs `HEAD`, untracked vs index.
5. Undo of **uncommitted** work is `restore`. Undo of the **last unpublished** commit is `--amend`. Undo of **published** history is `revert` (later).

```
edit  →  add  →  commit
WD      index  HEAD / main

status reports which of the three disagree
```

Why the index exists

Without an index, every commit would be “whatever is on disk.” The index is how you commit two of five edited files, or one hunk of a file (`add -p`). That is also why `.gitignore` exists: untracked paths never enter the index unless you force them.

What can break

- `git add .` of `.env` and secrets (ignore first).
- `git commit -a` on a **new** file (it only auto-stages tracked paths).
- `git init` in `$HOME` and then every subdirectory looking like a repo.
- Amending after someone else fetched that hash.

How to read this chapter

Init, add/commit, status/log/diff, and undo are the loop. The hands-on page is that loop on `desk-api`. Ignore, interactive add, `rm/mv`, and “what not to commit” are the same loop with sharper edges.

Initializing a Repository

Initializing a Repository

After this page you have an empty Git repository on `main` and you know what `git init` created.

A directory becomes a repository

```
mkdir -p ~/desk/desk-api
cd ~/desk/desk-api
git init
```

Output looks like:

```
Initialized empty Git repository in /home/ada/desk/desk-api/.git/
```

If you set `init.defaultBranch main` (chapter 2), the empty branch is named `main`. Confirm:

```
git status
```

On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)

`~/desk/desk-api` is the **working tree**. `~/desk/desk-api/.git/` is the **repository** (objects, refs, config). You edit files next to `.git`, not inside it.

<code>~/desk/desk-api/</code>	working tree (you edit these)
<code>README.md</code>	
<code>src/</code>	
<code>.git/</code>	repository
<code>HEAD</code>	ref: refs/heads/main
<code>config</code>	
<code>objects/</code>	empty until first commit
<code>refs/heads/</code>	main appears after first commit

This book's recurring desk is three repos: `desk-web`, `desk-api`, `desk-infra`. Part I uses `desk-api` as the tiny project. Create the other two the same way when a later chapter asks.

What git init wrote

```
ls -a .git
```

You will see at least `config`, `HEAD`, `objects/`, `refs/`. `HEAD` currently says `ref: refs/heads/main` even though `refs/heads/main` does not exist yet. The first commit creates that file.

```
cat .git/HEAD
git rev-parse --git-dir
# .git
```

`--git-dir` is useful when you are in a subdirectory later.

init vs clone

Command	When
<code>git init</code>	You are starting a project that has no Git history yet
<code>git clone <url></code>	You are copying an existing repository, including its history

`clone` is chapter 11. For a brand-new desk project, `init` is the right verb.

Already have files?

`git init` in a directory that already has source files is fine. Git does not add them until you `git add`. The next page is that loop.

init in the wrong place

If you run `git init` in your home directory, every `git status` from a subdirectory without its own `.git` will try to use `~/.git`. Remove a mistaken repo with:

```
# only if you meant to throw it away and there are no commits you need
rm -rf .git
```

That deletes Git's data for that directory, not your files. The first-commit hands-on page builds `desk-api` from scratch so you can see a clean `init`.

The Add/Commit Cycle

The Add/Commit Cycle

After this page you can take a file from the working tree to a commit on `main`.

The loop

```
edit files → git add → git commit → git status (clean)
```

That is the whole product: copy chosen files into the index, snapshot the index, name the snapshot with a message, slide `main` forward.

```
working tree          index          HEAD / main

README.md -add→ README.md -commit→ commit C
ping.py                                tree T

                                main
```

One file, one commit

From an initialized `desk-api` (previous page):

```
cd ~/desk/desk-api

cat > README.md <<'EOF'
# desk-api

HTTP API for the desk.
EOF

git add README.md
git status
git commit -m "Add README"
git log --oneline
```

`git add` stages `README.md`. `git commit -m` writes the commit without opening an editor. `git log --oneline` prints one line: a short hash and the message.

status is now clean. Working tree, index, and HEAD match.

A second commit

```
mkdir -p src
cat > src/ping.py <<'EOF'
def ping():
    return "ok"
EOF

git add src/ping.py
git commit -m "Add ping"
git log --oneline
```

Two commits. `src/` exists in the tree object because Git stored a tree named `src` pointing at the `ping.py` blob. You did not `git add` the directory separately; adding the file is enough.

What `-m` is doing

Every commit needs a message. `-m "Add README"` is the short form. Without `-m`, Git opens `core.editor`. Multi-paragraph messages:

```
git commit -m "Add ping" -m "Returns a fixed ok string so the health check has a function to
```

The first `-m` is the subject. The second is the body. Chapter 6 is Conventional Commits (`feat:`, `fix:`). For Part I, a short imperative sentence is enough: **Add ping**, **Fix empty name**, **Document env vars**.

git add shapes

```
git add README.md           # one file
git add src/ping.py src/app.py # several
git add src                  # a directory, recursively
git add .                    # everything from here down, still respects .gitignore
git add -p README.md         # hunk by hunk (interactive)
```

`.gitignore` is a file of patterns Git will leave untracked. Create it when you have build output or secrets to skip:

```
cat > .gitignore <<'EOF'
__pycache__/
.env
EOF
```

```
git add .gitignore
git commit -m "Ignore bytecode and env files"
```

Patterns are matched from the file's directory downward. `/build` at the repo root is only `build/` at the root; `build/` is every directory named `build`.

`commit -a`

```
git commit -a -m "Touch tracked files"
```

Stages **already tracked** files and commits. New files still need `git add`. Prefer explicit `git add` until that is automatic in your head.

After the cycle

```
git status
git log --oneline
```

Clean status, two or three commits. Next page: how to *read* what you just did (`status`, `log`, `diff`).

status, log, and diff

git status, git log, git diff

After this page you can read the three-tree report, walk history, and see a patch before you commit it.

git status — where you are

```
cd ~/desk/desk-api
git status
git status -sb          # short: branch + one letter per file
```

-sb prints ## main and lines like M README.md (modified) or ?? notes.txt (untracked). Use it when the long form is noise.

git status is the three-tree report from chapter 1: working tree vs index vs HEAD.

git diff	git diff --staged	git log
working vs index	index vs HEAD	HEAD chain
working tree	index	HEAD
		A ← B ← C main
extra	staged	

git log — the DAG as a list

```
git log
git log --oneline
git log --oneline --graph --decorate --all
git log -3          # last three
git log --oneline -- README.md  # commits that touched this path
git log -p -1       # last commit plus its patch
```

--oneline is the default view in this book. --graph --decorate --all is the view that matches the mental-model diagrams.

Useful format:

```
git log --format='%h %an %s'
```

%h short hash, %an author name, %s subject. `git help log` lists more placeholders. You will use `--format` in hooks and CI later; for now, `--oneline` is enough.

git diff — the patch

```
git diff                # working tree vs index (unstaged)
git diff --staged        # index vs HEAD (what the next commit will contain)
git diff HEAD           # working tree vs HEAD (staged and unstaged together)
git diff HEAD~1 HEAD    # last commit vs its parent
git diff main feature   # two branches (once you have them, chapter 8)
```

Empty `git diff` and non-empty `git diff --staged` means everything is already in the index. Empty both, plus an untracked file in `status`, means you still need `git add`.

DiffTool

For a side-by-side view in the terminal:

```
git config --global diff.tool nvimdiff    # or vimdiff, meld
git difftool README.md
git difftool --staged
```

`difftool` opens each changed file in the tool you named. `mergetool` is the sibling for conflicts (chapter 8). Stay on `git diff` until you want the split view. The patch is the same data.

A one-minute drill

```
echo "note" >> README.md
git status
git diff
git add README.md
git diff                # empty
git diff --staged        # the note
git restore --staged README.md
git restore README.md    # discard the note
git status              # clean
```

`restore` is the next page, on purpose. Here you only need to *see* the three diffs change as the file moves.

Undoing Mistakes

Undoing Mistakes

After this page you can discard an unstaged edit, unstage a file, and rewrite the last commit **before** anyone else has it.

Three common “undo”s

You did	You want	Command
Edited a tracked file, not staged	Throw the edit away	<code>git restore <file></code>
<code>git add</code> too soon	Keep the edit, leave it out of the next commit	<code>git restore --staged <file></code>
Last commit message (or last commit contents) is wrong, not pushed	Replace that commit	<code>git commit --amend</code>

These are safe on a laptop branch nobody has fetched. After `git push`, amending rewrites a hash other people may have. Chapter 9 and chapter 11 cover that case. Chapter 17 (reflog) is the safety net when you went too far.

<code>restore <file></code>	<code>restore --staged</code>	<code>commit --amend</code>
<code>working ← index</code>	<code>index ← HEAD</code>	<code>A ← B ← C</code>
		<code>A ← B ← C' (C gone from main)</code>

Discard unstaged work

```
cd ~/desk/desk-api
echo "oops" >> README.md
git restore README.md
```

The working-tree file matches the index again. The “oops” line is gone. Git cannot get it back from Git; your editor’s undo might.

Restore one file from HEAD even if it was staged:

```
git restore --source=HEAD --worktree --staged README.md
```

That is “make this file look like the last commit, in both index and working tree.”

Unstage, keep the edit

```
echo "keep me" >> README.md
git add README.md
git restore --staged README.md
git status
```

The line is still in the file. It is not in the index. The next `git commit` will not include it until you `git add` again.

Older docs say `git reset HEAD README.md` for the same unstage. This book uses `git restore --staged`.

Amend the last commit

You just committed and noticed a typo in the message:

```
git commit --amend -m "Add ping helper"
```

You forgot a file:

```
git add src/ping.py
git commit --amend --no-edit
```

`--no-edit` keeps the old message. `--amend` replaces the last commit with a new one (new hash) whose parent is the same as before. `git log` no longer shows the old hash as `main`. The old commit is still in the reflog for about 30 days (chapter 17).

Amend only when:

- it is the latest commit on this branch, and
- you have not pushed it, or you are sure no one else built on it.

Revert is a new commit

`git revert <hash>` adds a commit that undoes a previous one. Use it on history other people already have. Chapter 10 is revert and cherry-pick. For a local last-commit mistake, amend. For an old published mistake, revert.

rm and mv

```
git rm src/old.py           # stop tracking and delete the file
git rm --cached secrets.env  # stop tracking, keep the file on disk
git mv src/ping.py src/health.py
git status
git commit -m "Rename ping to health"
```

`git mv` is `mv` plus `git add` of the rename. Git detects renames in diffs even if you `mv` and `add` by hand; `git mv` is the tidy form.

If you are lost

```
git status
git log --oneline -5
```

Status tells you which of the three trees disagree. Log tells you where `main` points. The hands-on page next builds a tiny `desk-api` from scratch so you can practice the whole loop, including one amend.

Hands-On: Build a Tiny Project

Hands-On: Build a Tiny Project

Build `desk-api` from nothing: init, two commits, a bad edit, a restore, an amend, a clean log.

```
init → add/commit README → add/commit ping
      → restore an "oops" → amend the last message
      → leave .env untracked
```

```
A ← B ← C      main      status clean
```

Setup

Use a throwaway directory. If `~/desk/desk-api` already exists from earlier pages, either work there or pick `~/desk/desk-api-lab`.

```
mkdir -p ~/desk/desk-api
cd ~/desk/desk-api
git init
git status
```

Expect On branch `main` and No commits yet.

Commit 1 — README

```
cat > README.md <<'EOF'
# desk-api

HTTP API for the desk.
EOF

git add README.md
git commit -m "Add README"
git log --oneline
```

One line of history.

Commit 2 — a tiny module

```
mkdir -p src
cat > src/ping.py <<'EOF'
def ping():
    return "ok"
EOF

cat > .gitignore <<'EOF'
__pycache__/
.env
EOF

git add src/ping.py .gitignore
git status
git commit -m "Add ping and gitignore"
```

Read it

```
git status
git log --oneline --graph --decorate
git show --stat HEAD
```

`git show` prints the last commit's metadata and patch. `--stat` is the short file list.

Break it on purpose

```
echo "oops" >> README.md
git status
git diff
git restore README.md
git status
```

Clean again. The “oops” line is gone.

Stage, unstage, restage

```
echo "Health: GET /ping" >> README.md
git add README.md
git diff --staged
git restore --staged README.md
```

```
git diff
git add README.md
git commit -m "Document /ping"
```

Amend the message

That last message is vague. Replace it (still local, unpushed):

```
git commit --amend -m "Document the ping endpoint in README"
git log --oneline
```

Three commits. The amended hash is new. `git status` is clean.

Ignore a secret-shaped file

```
echo 'TOKEN=dev' > .env
git status
```

`.env` should be untracked (your `.gitignore` lists it). If it shows as untracked, the ignore pattern is working. If Git wants to add it, the `.gitignore` commit is missing — add the file and commit.

Leave `.env` untracked. Do not `git add .env`.

Done when

```
git status
git log --oneline
```

- Status: clean, on `main`
- Log: three commits (README, ping, document endpoint)
- `.env` exists on disk and is untracked

You now have the loop this book's later Git chapters assume: `init`, `add`, `commit`, `status`, `log`, `diff`, `restore`, `amend`.

Part II starts with commit anatomy and Conventional Commits. Remotes and forges wait until you can do this drill without looking at the page.

.gitignore in Depth

.gitignore in Depth

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Patterns, negation, / vs glob, `.git/info/exclude`, global excludes.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Interactive Add

Interactive Add

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git add -p`, hunks, splitting hunks, what staging a hunk means.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git rm and git mv

git rm and git mv

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stop tracking, keep on disk, rename detection.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

What Not to Commit

What Not to Commit

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets, build artifacts, `.env`, binaries; first look at LFS.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Inspecting History

Theory

Theory: History is a Query over the DAG

`git log` is not a file. It is a **walk** of commit objects from one or more tips, filtered by paths, dates, pickaxe, and ranges. Blame is a walk of *lines* through that DAG.

Invariants

1. Reachability: `git log B` shows commits reachable from B by following parents.
2. **Two-dot** `A..B` = reachable from B, not from A (what a feature branch adds).
3. **Three-dot** `A...B` = reachable from A or B, not from the merge base (symmetric difference).
A pull-request diff is often `git diff A...B` (merge-base to B).
4. `git blame` attributes each line of a file to the commit that last touched it; `-C` also hunts copies.
5. Pickaxe `-S/-G` filters commits by whether a string's count or a regex's hunk changed.

```
A ← B ← C ← G      main
    ↑
    D ← E          feat

main..feat      = D, E
main...feat     = G, D, E      (both sides since split)
```

Why this matters on a forge

A PR “files changed” list is this query, computed on the server. Reviewers who can run `git log main..HEAD` and `git diff main...HEAD` on Linux are not trapped in the UI.

What can break

- Reading `A..B` as “between A and B inclusive on a line” (it is set difference of reachability).
- `git blame` on a file that was reformatted in one giant commit (whitespace; `-w`).
- `--follow` only works when you name **one** path.

How to read this chapter

Log flags, ranges, show, blame, pickaxe, shortlog, then a lab that finds a line's origin in `desk-api`.

Advanced git log

Advanced git log

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--oneline --graph --decorate --all, --format, --since`, path limiting.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Two-Dot and Three-Dot Ranges

Two-Dot and Three-Dot Ranges

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A..B vs A...B; what a pull request actually diffs.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git show and –stat

git show and -stat

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git show, --stat, --name-only, --name-status.`

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git blame

git blame

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git blame`, `-L`, `-C` copy detection, ignoring whitespace.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Pickaxe and –follow

Pickaxe and `--follow`

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `-S`, `-G`, `--follow` across renames.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git shortlog

git shortlog

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git shortlog`, contributor summaries.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hands-On: History Archaeology

Hands-On: History Archaeology

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Find who changed a line and when a string entered `desk-api`.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Ignore, Attributes, and LFS

Theory

Theory: The Index is a Filter; LFS is a Pointer

Git will store any blob you add. Ignore rules and attributes are **policy on what enters the index and how it is treated**. LFS replaces a large blob with a tiny pointer blob and puts the real bytes elsewhere.

Invariants

1. `.gitignore` applies to **untracked** files. A tracked file stays tracked until `git rm --cached`.
2. Precedence: command-line exclude → `.git/info/exclude` → `.gitignore` in-tree → global excludes file.
3. `gitattributes` attach behavior to paths: text vs binary, eol, diff, merge, filter (clean/smudge).
4. Git LFS stores a **pointer file** (text, few hundred bytes) in Git, and the payload in an LFS object store the forge must provide.

<code>git add huge.bin</code>	without LFS	blob of huge.bin in <code>.git/objects</code>
<code>git add huge.bin</code>	with LFS	blob of pointer version <code>https://git-lfs...</code> payload LFS store (not in packfiles)

Why Git is a bad backup of videos

Every clone downloads every version of every blob (unless partial clone / LFS / sparse). A 200 MB binary edited ten times is ~2 GB of packfile. LFS and “do not commit build output” are the same theory: Git’s object store is for source, not for arbitrary bytes.

What can break

- Ignoring `.env` after it was committed (must `rm --cached`).
- CRLF filters on a Linux-only repo.
- Pushing LFS pointers to a forge with no LFS storage; clones get pointers, not files.

How to read this chapter

Patterns, attributes, archive, LFS, then a lab. Platform chapters later must store LFS if you use it.

gitignore Patterns

gitignore Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Per-repo, global, and `.git/info/exclude`; precedence.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

gitattributes

gitattributes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Text/binary, eol, ident, filter, diff, merge drivers.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

export-ignore and git archive

export-ignore and git archive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `export-ignore`, `git archive` for tarballs.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Git LFS

Git LFS

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When blobs are too big; pointers; `git lfs install` on Linux; what a forge must store.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hands-On: Ignore and LFS

Hands-On: Ignore and LFS

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Ignore `.env` and `__pycache__`; optional LFS track of a sample binary.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Everyday Workflows

Commits and History

Theory

Theory: A Commit is Immutable; Messages are for Humans and Machines

Once written, a commit object is byte-for-byte fixed. `--amend` and `rebase` do not edit it; they write a **new** object and slide a name. That is why hashes change and why shared history is a social problem.

Invariants

1. Commit hash = hash of (tree, parents, author, committer, message). Change any field → new hash.
2. Author and committer can differ (`rebase`, `cherry-pick`).
3. Conventional Commits are a **convention**, not a Git feature. Hooks and forges can enforce them.
4. Unstaging moves a path from index toward `HEAD`; it does not delete working-tree bytes.

```
amend
```

```
A ← B ← C      main      C has message "stuff"
A ← B ← C'     main      C' new object; C remains in reflog
```

Why messages matter

`git log --grep '^fix'` only works if subjects are structured. Changelog generators and `feat!:` breaking-change scanners parse the same text. The object store does not care; your future self and CI do.

What can break

- Amending a commit already on `origin/main`.
- Empty subject or WIP on a branch you intended to merge.
- Trailers with the wrong spelling (`Co-authored-by` is magic for some forges).

How to read this chapter

Anatomy, Conventional Commits, amend, unstage, lab; then templates, split/empty, trailers/mailmap, `git show`.

Commit Anatomy

Commit Anatomy

After this page you can read a commit object and say what each field is for.

A commit is a snapshot plus metadata

```
cd ~/desk/desk-api
git cat-file -p HEAD
```

You will see something like:

```
tree 4b825dc642cb6eb9a060e54bf8d69288fbee4904
parent 9fceb02d0ae598e95dc970b74767f19372d61af8
author Ada Desk <ada@example.com> 1774000000 +0530
committer Ada Desk <ada@example.com> 1774000000 +0530
```

```
Document the ping endpoint in README
```

Field	Meaning
tree	Hash of the directory snapshot this commit names
parent	Previous commit (omitted on the first commit; two lines on a merge)
author	Who wrote the patch, plus timestamp
committer	Who created this commit object, plus timestamp
blank line, then body	The message

The **name** of the commit is the hash of this object. Change the message, the tree, or a parent → new hash. That is why amend and rebase rewrite history: they write new objects.

Author and committer

They match until you rebase, cherry-pick, or apply someone else's patch. `git log` shows the **author** by default. `git log --format=fuller` shows both.

```
git log -1 --format=fuller
```


Short names

```
git rev-parse HEAD
git rev-parse HEAD~1      # first parent, one step back
git rev-parse HEAD^       # same as HEAD~1 on a non-merge
git rev-parse --short HEAD # abbreviated, unique in this repo
```

On a merge commit, `HEAD^1` is the first parent (the branch you were on) and `HEAD^2` is the branch you merged in. `HEAD~n` walks first-parent only. Chapter 10 uses that when reverting a merge.

The tree under it

```
git ls-tree -r HEAD
```

Each line is a mode, type, hash, and path. Blobs have no filename inside them; the tree supplies the name. Two commits that share an unchanged `src/ping.py` share the same blob hash.

What `git show` adds

```
git show HEAD
git show --stat HEAD
git show --name-only HEAD
```

`git show` pretty-prints the commit and, by default, the diff against its first parent. That diff is computed; Git stored the tree, not the patch.

Next page: how to write the message so later `git log` and automation can parse it.

Conventional Commits

Commit Messages That Matter: Conventional Commits

After this page you can write a subject line other people and tools can parse.

Shape of a message

```
type(scope): short summary

Optional body: why, not what. Wrap at 72 characters.

Optional footer:
Fixes: #12
BREAKING CHANGE: /ping now returns JSON
```

The **subject** is one line, imperative, about 50 characters, no trailing period. Git itself does not enforce this. Forges, changelog generators, and `git log --grep` benefit when you do.

Types this book uses

Type	When
feat	A user-visible capability
fix	A bug
docs	README, comments, this book's desk docs
chore	Tooling, ignore files, no user-facing change
refactor	Same behavior, different code
test	Tests only
ci	Pipeline files (later parts)

scope is optional: `feat(api):`, `fix(web):`. For `desk-api` a scope of `ping` or `auth` is enough.

Examples that match the desk:

```
feat(ping): return JSON from /ping
fix(ping): treat empty name as 400
docs: document GET /ping in README
```

```
chore: ignore .env and __pycache__
```

Why bother

```
git log --oneline --grep '^fix'  
git log --oneline --grep '^feat(ping)'
```

Later, platform governance (chapter 31) can require this shape in a server-side hook. A client-side `commit-msg` hook (chapter 13) can reject a bad subject before the object exists.

What to put in the body

The subject is *what*. The body is *why* and any context `git log` should keep:

```
git commit -m "fix(ping): treat empty name as 400" -m "The handler returned 200 and an empty"
```

Skip the body when the subject is the whole story.

Trailers

```
Signed-off-by: Ada Desk <ada@example.com>  
Co-authored-by: Sam Desk <sam@example.com>  
Fixes: #18
```

`git interpret-trailers` reads these. Signed-off-by is a project convention (Linux kernel, some forges). Use it when the project asks. Co-authored-by is how forges attribute a second author on a squash.

A local alias

```
git config --global alias.cm '!f() { git commit -m "$1"; }; f'
```

Aliases are optional. The skill is the subject line, not the shortcut.

Amending a Commit Message

Amending and Fixing an Incorrect Commit Message

After this page you can replace the last commit's message, and you know when that is safe.

The last commit only

```
cd ~/desk/desk-api
git log -1 --format='%s'
git commit --amend -m "docs: document GET /ping in README"
git log -1 --format='%h %s'
```

`--amend` writes a **new** commit object with the same tree and parent (unless you also staged files) and slides the branch name to it. The old hash disappears from `git log`. It is still in `git reflog` for about 30 days (chapter 17).

before	after
A ← B ← C main	A ← B ← C' main
message "stuff"	C still in reflog

Keep the old message and only restage files:

```
git add src/ping.py
git commit --amend --no-edit
```

Open the editor with the old message loaded:

```
git commit --amend
```

`core.editor` (chapter 2) is what opens. On Linux with `nvim`, you edit, save, quit, and Git takes the buffer.

When this is safe

Safe when **all** of these hold:

- it is HEAD (the latest commit on this branch)

- you have not pushed, or you are the only clone
- nobody else based work on the old hash

After a push, amending means a force-push (chapter 11). On `main` shared with a team, prefer a new commit: `git commit -m "docs: fix typo in README"`.

Older than HEAD

`--amend` cannot retitle commit `HEAD~3` by itself. Interactive rebase can (chapter 9, `reword`). For now, live with old subjects or wait for rebase.

Recover if you amended by mistake

```
git reflog
git reset --soft HEAD@{1}
```

`--soft` moves `HEAD` back and leaves the index and working tree as they are. Chapter 7 is reset in full. The hash you just left is in the reflog (chapter 17). This one-liner is enough if you just amended and panicked.

Identity in an amended commit

Amend rewrites the **committer** date by default. The author stays. To keep both timestamps:

```
git commit --amend --no-edit --date="$(git log -1 --format=%aD)"
```

Rarely needed. Know that `git log` after an amend shows “now” as the committer time.

Unstaging a File

Unstaging an Added File

After this page you can take a path out of the index and leave the working-tree edit in place.

The index is the next commit

You `git add` too much. The file should stay on disk. It should not be in the next snapshot.

```
cd ~/desk/desk-api
echo "scratch" >> README.md
echo "debug" >> src/ping.py
git add README.md src/ping.py
git status
git restore --staged src/ping.py
git status
```

README.md is still staged. `src/ping.py` is modified, not staged. `git commit` will only record the README line.

after git add -A			after restore --staged ping.py		
working	index	HEAD	working	index	HEAD
ping.py	ping.py	(old)	ping.py	(old)	(old)
README	README	(old)	README	README	(old)
			still on disk, not in next commit		

Restore vs reset

Command	Index	Working tree
<code>git restore --staged <file></code>	Matches HEAD	Unchanged (your edit stays)
<code>git restore <file></code>	Unchanged	Matches the index (edit discarded)
<code>git restore --source=HEAD --staged --worktree <file></code>	Matches HEAD	Matches HEAD

This book uses `git restore`. Older guides use `git reset HEAD <file>` for unstage; same index effect.

Unstage everything

```
git restore --staged .  
git status
```

All paths return to “modified” or “untracked.” Nothing is deleted.

Untracked files

`git add` of a new file copies it into the index. `git restore --staged` on that path leaves it **untracked** (the index no longer knows it). The file remains on disk. `git clean` would delete untracked files; that is chapter 12. Leave `git clean` alone until you can name what it removes.

Partial unstage

```
git restore --staged -p src/ping.py
```

Git offers hunks. `n` keeps the hunk staged, `y` unstages it. Same interactive idea as `git add -p`.

Check before you commit

```
git diff --staged  
git diff
```

`--staged` is the next commit. The other is leftover work. Both should match what you intend. The hands-on page next amends a real **desk-api** commit so this is muscle memory.

Hands-On: Amend and Rewrite

Hands-On: Amend and Rewrite a Commit

Practice on ~/desk/desk-api. If that tree is missing, run chapter 3's tiny-project page first.

```
commit "stuff"      A ← B ← C
amend subject       A ← B ← C'      C in reflog only
amend add file      A ← B ← C''     tree now includes version.py
```

1. A sloppy commit

```
cd ~/desk/desk-api

echo "Returns a fixed string." >> README.md
git add README.md
git commit -m "stuff"
git log -1 --oneline
```

2. Fix the subject

```
git commit --amend -m "docs: note that ping returns a fixed string"
git log -1 --format='%h %s'
```

The hash changed. The tree is the same if you staged nothing else.

3. Forgot a file

```
cat > src/version.py <<'EOF'
VERSION = "0.1.0"
EOF
git add src/version.py
git commit --amend --no-edit
git show --name-only --oneline HEAD
```

version.py is now in the same commit as the docs line.

4. Staged too much, then unstaged

```
echo "WIP" >> src/ping.py
echo "also" >> README.md
git add -A
git restore --staged src/ping.py
git diff --staged
git commit --amend --no-edit
git restore src/ping.py
git status
```

The WIP line is gone. README’s “also” is in HEAD if it was staged. Status is clean.

5. Confirm the graph

```
git log --oneline -5
git status
```

You should be on `main`, clean, with a conventional subject on `HEAD`. Next chapter: branches, so this `main` history can grow a second line of work.

Commit Templates

Commit Templates

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `commit.template`, `prepare-commit-msg` preview.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Empty and Split Commits

Empty and Split Commits

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--allow-empty`; splitting a commit with `reset + add -p`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Trailers and mailmap

Trailers and mailmap

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Signed-off-by, Co-authored-by, `git interpret-trailers`, `.mailmap`.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Reading a Commit with git show

Reading a Commit with git show

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git show, --pretty=fuller, --format=raw.`

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Reset, Restore, and Checkout

Theory

Theory: Reset Moves Names; Restore Copies Bytes; Revert Adds a Commit

Three different operations get called “undo.” They act on different trees and different audiences.

Invariants

Tool	What it moves	History
<code>git restore</code>	Working tree and/or index, from a source	Names unchanged
<code>git reset</code> (no path)	HEAD (and maybe index, working tree)	Branch name slides; old commit may become unreachable
<code>git reset</code> (with path)	Index (mixed) for those paths	HEAD stays
<code>git revert</code>	Nothing; writes a new commit	History goes forward

Soft / mixed / hard is **which of the three trees follow HEAD** when you reset a commit:

	HEAD	index	worktree	
<code>--soft</code>	moves	stays	stays	
<code>--mixed</code>	moves	matches HEAD	stays	(default)
<code>--hard</code>	moves	matches HEAD	matches HEAD	

Why checkout split

Old `git checkout` meant both “switch branch” and “restore file.” Git 2.23+ splits them: **switch** for branches, **restore** for files. This book uses the split so a command’s name is its theory.

What can break

- `reset --hard` on a shared `main`.
- Confusing mixed reset (unstage all) with hard (throw away tracked work).
- Expecting `--hard` or `restore` to delete untracked files. They stay. Removing them is `git clean`.

How to read this chapter

Three modes, the comparison table as practice, path reset, switch/restore/checkout, then a lab that walks one commit through soft, mixed, and hard on Linux.

Soft, Mixed, and Hard Reset

Soft, Mixed, and Hard Reset

After this page you can predict `git status` after `--soft`, `--mixed`, and `--hard`, and you can put the branch name back with `ORIG_HEAD`.

One commit, three trees

Start from a clean `main` on `~/desk/desk-api` (chapter 3). This page adds one scratch commit, tries each mode, and removes that commit at the end.

```
cd ~/desk/desk-api
git switch main
git status
```

Ignored files such as `.env` are fine. Commit or restore any other uncommitted work before you continue.

```
echo "scratch" >> README.md
git add README.md
git commit -m "lab: reset modes"
```

```
A ← B ← S      main
HEAD, index, and worktree all name S
```

`S` is the scratch commit. Each mode below starts from `S`. Git records the old tip in `ORIG_HEAD` whenever a reset moves the branch, so `git reset --hard ORIG_HEAD` returns to `S`.

`--soft` moves the name

```
git reset --soft HEAD~1
git status
```

```
HEAD → B
index still S      "Changes to be committed"
worktree still S
```

`main` now names `B`. The scratch line is staged. `git log` no longer shows `lab: reset modes`. The commit object stays in the reflog (chapter 17).

```
git reset --hard ORIG_HEAD
git log --oneline -1
```

You are back on the scratch commit. The line is in README.md, and status is clean.

--mixed moves the name and the index

--mixed is the default. `git reset HEAD~1` and `git reset --mixed HEAD~1` are the same command.

```
git reset HEAD~1
git status
git diff
```

Git prints Unstaged changes after reset:.

```
HEAD → B
index matches B      the scratch line is not staged
worktree still S     "Changes not staged for commit"
```

The patch is in the working tree. The next commit includes it only after `git add`.

```
git reset --hard ORIG_HEAD
```

--hard moves all three

```
git reset --hard HEAD~1
git status
grep scratch README.md || echo "scratch line is gone"
```

```
HEAD, index, and worktree all match B
tracked files match B
```

The line is gone from disk. Untracked files are still on disk. --hard rewrites tracked files to the commit you named and leaves untracked and ignored files alone.

```
git reset --hard ORIG_HEAD
```

The scratch commit is the branch tip again, and the line is back. ORIG_HEAD is the immediate undo. After other commands have moved HEAD, the reflog is the record (chapter 17). The lab at the end of this chapter recovers with HEAD@{1} on purpose.

Remove the scratch commit

```
git reset --hard HEAD~1
git status
git log --oneline -5
```

lab: `reset` modes is off the log. `README.md` has no scratch line. The object remains in the reflog until it expires.

Where this is safe

Use these modes on a local branch nobody else has fetched. On a `main` other people already have, moving the name and pushing is a force-push (chapter 11). A published undo is `git revert` (chapter 10). The next page is how to choose restore, reset, or revert.

restore vs reset vs revert

Restore, Reset, and Revert

After this page you can pick the undo that matches what is wrong and who else has the commit.

Three audiences

uncommitted
working tree or index
restore

unpublished
only this clone has the hash
reset or amend

published
someone else may have it
revert

Situation	Command	Branch tip
Edited a tracked file, not staged	<code>git restore <file></code>	stays
Staged too soon	<code>git restore --staged <file></code>	stays
Last local commit should be redone	<code>git reset --soft HEAD~1</code> , then commit	parent; changes stay staged
Last local commit should be an unstaged edit	<code>git reset HEAD~1</code>	parent; changes stay in the worktree
Last local commit and its files should disappear	<code>git reset --hard HEAD~1</code>	parent; tracked files match the parent
Last local commit needs a new message or one more file	<code>git commit --amend</code>	new hash, same parent (chapter 6)
A shared commit should be undone	<code>git revert <hash></code>	moves forward (chapter 10)

`restore` copies bytes. The branch name stays. `reset` with no path slides the branch name. `revert` adds a commit.

Amend and soft reset

Both replace an unpublished tip.

Amend keeps one commit at the tip: same parent, new hash. You are still one commit ahead of the parent. Chapter 6 is amend.

Soft reset takes the tip off the branch and leaves that tree staged. You can drop a file, split the change, or commit it again with an ordinary message. The previous page is that mode.

Revert goes forward

`git revert` writes a new commit whose patch is the inverse of an older one. The old hash stays reachable. Use it when the commit is on a shared `main`, or when you are unsure whether anyone fetched it. Chapter 10 includes un-merging with `git revert -m 1`.

```
published:  A ← B ← C ← R      R undoes C; C is still reachable
local only: A ← B              reset moved main off C; C is in the reflog
```

A test you can apply

Ask where the hash lives.

- In the working tree or the index only: **restore**.
- On this branch, and you have not pushed it (or there is no remote yet): **reset** or **amend**.
- On a forge, or a teammate has pulled: **revert**.

Chapter 11 is the force-push that follows a reset of something you already pushed. On shared `main`, `revert`.

Resetting Paths

Resetting Paths

After this page you can change one path in the index while the branch name stays put.

A path means the branch stays

With no path, `git reset` moves HEAD. With a path, HEAD stays and Git copies that path from a commit into the index only.

```
cd ~/desk/desk-api
git switch main
git status
BEFORE=$(git rev-parse --short HEAD)
echo "path lab" >> README.md
git add README.md
git commit -m "lab: path reset"
git reset HEAD~1 -- README.md
git rev-parse --short HEAD
echo "parent was $BEFORE"
git status
git diff -- README.md
git diff --staged -- README.md
```

The short hash is the commit you just made, `lab: path reset`. It is not `$BEFORE`. Status lists `README.md` in two places:

- **Changes to be committed.** The index holds the parent's blob.
- **Changes not staged for commit.** The working tree still has the `path lab` line.

That is one edit, seen from both sides. `git diff` shows the line added. `git diff --staged` shows the line removed. Git also prints `Unstaged changes after reset:`.

```
HEAD stays on "lab: path reset"
index README      ← parent blob
worktree README   ← still has "path lab"
```

Older notes unstage with `git reset HEAD -- README.md`. This book unstages with `git restore --staged` (chapters 3 and 6). The path form is how you aim the index at an older blob, and how you read a script that still uses it.

Put the file back, then drop the commit

HEAD is still the scratch commit. Copy that commit's README.md into the index and the working tree:

```
git restore --source=HEAD --staged --worktree -- README.md
git status
```

Status is clean. The log still shows lab: path reset.

```
git reset --hard HEAD~1
git log --oneline -3
```

The scratch commit is off main, and README.md has no path lab line. The path reset did not do that. The last command is a branch reset from the first page of this chapter.

Whole tree versus one path

```
git reset HEAD~1
    main slides to the parent

git reset HEAD~1 -- README.md
    main stays; the index path changes

git restore --source=HEAD~1 --staged --worktree -- README.md
    main stays; index and worktree take the old blob
```

The third command is how you make a file look like an older commit and then record that with a new commit. The older commit stays in history. **switch** and **restore** on the next page are the same split: names versus bytes.

switch, restore, and checkout

switch, restore, and checkout

After this page you can change branches with `git switch` and files with `git restore`, and you can translate an old `git checkout` line.

Two commands, two jobs

Until Git 2.23, `git checkout` did both jobs. It still does, including in Git 2.55. Checking out a branch moves `HEAD`. Checking out a path copies bytes. One word for both is how a file gets overwritten when the goal was to change branches.

`git switch` moves `HEAD` and refuses a path. `git restore` copies bytes and leaves `HEAD` where it is.

Older command	Command in this book
<code>git checkout main</code>	<code>git switch main</code>
<code>git checkout -b feat/x</code>	<code>git switch -c feat/x</code>
<code>git checkout -- README.md</code>	<code>git restore README.md</code>
<code>git checkout HEAD -- README.md</code>	<code>git restore --source=HEAD --staged --worktree -- README.md</code>
<code>git checkout <hash></code>	<code>git switch --detach <hash></code>

Scripts and old notes still say `checkout`. Translate them with the table. Type `switch` and `restore` yourself.

Switch branches

```
cd ~/desk/desk-api
git switch main
git switch -c feat/switch-lab
git switch main
git switch -
git branch --show-current
```

`git switch -` uses `@{-1}`, the checkout you were on before this one. The last command prints `feat/switch-lab`.

```
git switch main
git branch -d feat/switch-lab
```

`-d` deletes a branch whose tip is already reachable from `HEAD`. You are on `main`, and `feat/switch-lab` names that same commit, so the delete succeeds.

A path is not a branch

```
git switch README.md
```

Git stops with fatal: invalid reference: README.md. The file is unchanged.

Local edits that would be overwritten

```
git switch -c feat/switch-lab
echo "other side" >> README.md
git add README.md
git commit -m "lab: other side"
git switch main
echo "main side" >> README.md
git switch feat/switch-lab
```

Git aborts. The message still says the files would be overwritten by checkout. `README.md` on `main` still ends with `main side`. `feat/switch-lab` still has `other side`.

```
git restore README.md
git branch -D feat/switch-lab
git status
```

`-D` deletes a branch even when `main` does not contain its commits. Use it here to discard the throwaway `lab: other side` commit. When you need to keep the uncommitted line and still change branches, stash it (chapter 12).

Detached HEAD

```
git switch --detach HEAD
git switch main
```

`--detach` points `HEAD` at a commit and at no branch. Chapter 1 is the trap: a commit made there needs a new branch before you leave it. From a detached `HEAD` with no new commit, `git switch main` attaches you again.

The lab next walks one scratch commit through soft, mixed, and hard, then brings it back from the reflog.

Hands-On: Reset Lab

Hands-On: Reset Lab

Walk one commit through soft, mixed, and hard on `desk-api`. Lose the line on purpose, and take it back from the reflog.

Setup

```
cd ~/desk/desk-api
git switch main
git status
START=$(git rev-parse HEAD)
echo "$START"
```

Status is clean, aside from ignored files such as `.env`. If `git switch main` refuses, restore or commit the work left from the previous page.

```
echo "lab line" >> README.md
git add README.md
git commit -m "lab: scratch line"
```

Soft

```
git reset --soft HEAD~1
git status
git diff --staged -- README.md
```

The line is under “Changes to be committed”. `git log --oneline` does not show `lab: scratch line`.

```
git commit -m "lab: scratch line"
```

Same tree, new hash. The next mode starts from a commit again.

Mixed

```
git reset HEAD~1
git status
git diff -- README.md
```

Git prints `Unstaged changes after reset:`. The line is in the file and not in the index.

```
git add README.md
git commit -m "lab: scratch line"
```

Hard, and an untracked file

```
echo "not tracked" > notes.tmp
git reset --hard HEAD~1
git status
grep "lab line" README.md || echo "line gone"
ls notes.tmp
```

Tracked files match the commit in `$START`. `notes.tmp` is still in the directory. `--hard` left it there.

Recover with the reflog

```
git reflog -5
```

The first line is this reset (`HEAD@{0}`). The next line (`HEAD@{1}`) is `commit: lab: scratch line`. `HEAD@{1}` means “where `HEAD` was one move ago.” That is a different question from `HEAD~1`, the parent of the current commit. Chapter 17 is that difference in full.

```
git reset --hard HEAD@{1}
grep "lab line" README.md
git log --oneline -1
```

The line is back. `main` names the scratch commit again.

Leave the repo as you found it

```
git reset --hard "$START"
rm -f notes.tmp
git status
git log --oneline -5
```

`$START` is the tip from the start of the lab, so this last reset also drops `lab: scratch line`.

Done when

- `git status` is clean (an ignored `.env` is fine)
- `notes.tmp` is gone
- the log has no `lab: scratch line`
- `README.md` has no `lab line`
- you saw that line staged, then unstaged, then missing, then restored from `HEAD@{1}`

Chapter 8 creates branches from this same `desk-api`. A reset of shared `main` waits; merging and rebasing come first.

Branching and Merging

Theory

Theory: A Branch is a Name; a Merge is a Node with Two Parents

Creating a branch does not copy files. It writes a ref. Merging either **slides** that ref (fast-forward) or **writes a commit with two parents** (a join).

Invariants

1. `git switch -c feat` = write `refs/heads/feat` at current HEAD, point HEAD at it.
2. Fast-forward is legal iff the current tip is an ancestor of the other tip.
3. A three-way merge diffs **base**→**ours** and **base**→**theirs** and applies both to the base. The merge base is the unique best common ancestor (usually).
4. Conflict markers are unmerged index stages (1 base, 2 ours, 3 theirs) spilled into the working tree.
5. During a merge, `--ours` is HEAD. During a rebase, `--ours` is the **upstream**. That swap is the theory, not a bug.

fast-forward	three-way merge
A ← B ← C main	A ← B ← C ← M main
← D ← E	↑ ↑
	D ← E

Why merge commits exist

They record *that two lines of work met*, and they keep both sides' hashes stable. Rebase prefers a straight first-parent chain and pays with new hashes. Neither changes the fact that Git stores snapshots.

What can break

- Merging with uncommitted work that overlaps the incoming tree.
- Resolving markers but forgetting `git add` (merge stays open).
- `git merge --abort` after the merge commit exists (too late).

How to read this chapter

Create/switch, merge types, three-way, conflicts, fast-forward, lab; then mergetool, ours/theirs, orphan, comparing branches, rerere.

Creating and Switching Branches

Branch to Create a Safe Working Area

After this page you can create, switch, list, and delete branches on Linux without touching `main`.

A branch is a name

```
cd ~/desk/desk-api
git branch
git switch -c feat/ping-json
git branch
```

`git switch -c` creates `feat/ping-json` at current HEAD and points HEAD at it. `git branch` (no args) lists local names; `*` is where HEAD is.

```
git switch main
git switch feat/ping-json
```

`git switch` is the modern command. `git checkout` still works; this book uses `switch` for branches.

Name them so later pages stay readable

```
feat/...    a capability
fix/...     a bug
chore/...   tooling
```

Slashes are fine on Linux. The name is a file under `.git/refs/heads/` (or a line in `packed-refs`). `feat/ping-json` is the path `refs/heads/feat/ping-json`.

Work on the branch

```
git switch feat/ping-json
# edit, add, commit as in chapter 3
git log --oneline --graph --all --decorate
```

`main` stays where you left it. Commits on `feat/ping-json` are extra nodes in the DAG.

```

A ← B ← C                main
                        D ← E    feat/ping-json  HEAD

```

List and see

```

git branch -v                # last commit on each
git branch --show-current
git log --oneline main..HEAD # commits on this branch that main does not have

```

A..B means reachable from B, not from A.

Delete

```

git switch main
git branch -d feat/ping-json    # refuses if not merged
git branch -D feat/ping-json    # delete anyway

```

-d is the default you want. -D is when you meant to throw the line of work away.

Rename

```

git branch -m feat/ping-json feat/ping

```

Renames the current branch if you omit the old name. After you publish a branch, renaming locally does not rename it on the remote; chapter 11.

Stuck with a dirty working tree

Git refuses to switch if the switch would overwrite uncommitted edits. Commit, stash (chapter 12), or restore first.

```

git status
git switch main

```

Merge Types

Merge Types: Fast-Forward, Recursive, and Octopus

After this page you can name the three merge shapes Git actually writes.

Fast-forward

`main` is an ancestor of `feat/ping-json`. Merging can just slide `main` forward. No new commit.

```
A ← B ← C                main
      ← D ← E            feat/ping-json

# after git switch main && git merge feat/ping-json

A ← B ← C ← D ← E        main, feat/ping-json

git switch main
git merge feat/ping-json
```

`git log` shows a straight line. `git merge --no-ff` forces a merge commit even when a fast-forward is possible (Git Flow likes that; trunk-based usually does not).

Recursive (two parents)

The branches diverged. Git writes a **merge commit** with two parents.

```
A ← B ← C ← M                main
      ↑       ↑
      D ← E            feat/ping-json
```

The default strategy for two heads is `ort` (the current “recursive” replacement). You rarely pass `-s`. Git finds a **merge base** (common ancestor), diffs both sides against it, and applies both diffs to the base.

```
git switch main
git merge feat/ping-json -m "merge feat/ping-json"
git log --oneline --graph --decorate
```

Octopus (three or more parents)

```
git merge topic-a topic-b topic-c
```

One merge commit with several parents. Git only does this when there are **no** overlapping conflicts. If two topics touch the same hunk, octopus refuses; merge them pairwise instead. You will almost never need octopus on `desk-api`. Know the name so `git log --graph` with a commit that has three parents is not a mystery.

```

A ← B ← C ← M      main      M has three parents
  ↑   ↑   ↑
  D   E   F      three topic tips
```

See the parents

```
git cat-file -p HEAD
git rev-parse HEAD~1 HEAD~2
```

`HEAD~1` is the first parent (the branch you had checked out). `HEAD~2` is what you merged in.

Abort

```
git merge --abort
```

Returns the three trees to the pre-merge state while a merge is in progress (`MERGE_HEAD` exists). After the merge commit exists, abort is too late; `git reset` (chapter 7) or `git revert -m 1` (chapter 10).

Three-Way Merge

Three-Way Merge Explained

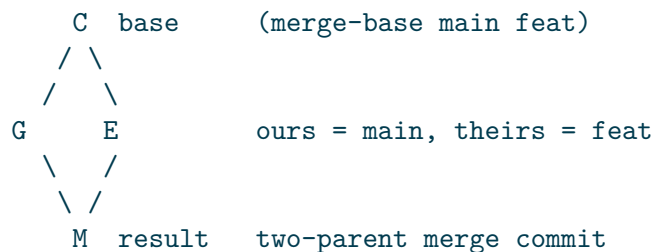
After this page you can draw the three inputs Git uses when two branches diverged.

The three trees

Call them:

Name	What
Base	Merge base: latest common ancestor of both tips
Ours	The branch you have checked out (HEAD)
Theirs	The branch you named in <code>git merge</code>

Git diffs base→ours and base→theirs, then applies both diffs to the base. That is a **three-way merge**. A two-way merge (ours vs theirs only) cannot tell “we both added the same line” from “they added a line we never had.”



See the base

```
cd ~/desk/desk-api
git merge-base main feat/ping-json
git log --oneline --graph main feat/ping-json
```

`merge-base` prints one hash. `git log --graph` should make that node obvious.

Stages in the index during a conflict

When both sides changed the same hunk, Git records three stages for that path in the index:

Stage	Content
1	base
2	ours
3	theirs

```
git ls-files -u
```

`git checkout --ours path / --theirs path` copies stage 2 or 3 into the working tree. You still `git add` after you edit. The next page is the full conflict loop.

Why this matters for a forge

A pull request / merge request is this same three-way merge, run on the server when you click Merge. The conflict you see in the UI is stages 2 and 3 plus markers. Learning it locally means the UI is not a new theory.

zdiff3 markers

With `merge.conflictstyle zdiff3` (chapter 2) a conflicted file shows the common ancestor as well as ours and theirs. That extra block is the base. Read it before you delete markers.

Merge Conflicts

Merge Conflicts: In Git and in Teams

After this page you can finish a conflicted merge on Linux and leave a clean `main`.

What a conflict is

Both branches changed the same region of a file (or one side changed a file the other deleted). Git writes **conflict markers** into the working tree and leaves the merge unfinished.

```
<<<<<<< HEAD
return "ok"
=====
return {"status": "ok"}
>>>>>>> feat/ping-json
```

With `zdiff3` you also get an ancestor block (`||||||`).

The loop

```
git switch main
git merge feat/ping-json
# Git stops and prints "fix conflicts and then commit"

git status
# both modified: src/ping.py
```

Edit the file in `nvim`. Remove every `<<<<<<<`, `=====`, `>>>>>>>` (and `||||||` if present). Leave the code you want.

```
git add src/ping.py
git status          # "all conflicts fixed"
git commit          # default message is "Merge branch 'feat/ping-json'"
```

Git opens `core.editor` for the merge message if you omit `-m`. A conventional subject is optional here; merge commits are often left as Git's default so `git log --first-parent` stays readable.

Abort vs continue

```
git merge --abort      # give up; back to pre-merge
```

After `git add`, there is no `--continue` for merge; `git commit` finishes it. *Rebase does use `--continue`* (chapter 9).

Binary files and deletes

`git status` will say deleted by them or added by us. Pick a side:

```
git restore --ours --worktree --staged path
git restore --theirs --worktree --staged path
git rm path
```

Then `git add` / `git commit` as usual.

Teams

A conflict is a **data** problem (two diffs) plus a **people** problem (whose behavior should win). On a forge, the same markers appear in the pull-request UI. Resolve on your laptop, commit, push. Do not resolve only in the UI if you cannot run the tests; Linux is where **desk-api** actually runs.

rerere (chapter 2) records how you resolved a hunk and reapplies it if the same conflict comes back. Leave it enabled.

After the merge

```
git log --oneline --graph --decorate --all
git branch -d feat/ping-json
```

`-d` succeeds once the work is reachable from `main`.

Fast-Forward for Clean History

Creating a Fast-Forward Merge for a Clean History

After this page you can keep `main` a straight line when the feature branch is ahead of it.

When a fast-forward is possible

`main` has not moved since you branched. Every commit on the feature is a descendant of `main`.

before <code>--ff-only</code>	after <code>--ff-only</code>
<pre>A ← B ← C main ← D ← E feat</pre>	<pre>A ← B ← C ← D ← E main feat</pre>

```
cd ~/desk/desk-api
git switch main
git merge --ff-only feat/ping-json
```

`--ff-only` **fails** if Git would need a merge commit. That is the setting you already put in `pull.ff` only for pulls. Use it on purpose for local merges when you want a linear `main`.

If it fails

Someone committed to `main` while you worked. Options:

1. Merge anyway (`git merge feat/ping-json`) → merge commit, history shows the join.
2. Rebase the feature onto `main` (chapter 9), then `--ff-only`.
3. `git merge --no-ff feat/ping-json` → always a merge commit (Git Flow release style).

See it

```
git log --oneline --graph --decorate
```

A fast-forward looks like a single chain. A merge commit is a join with two parents.

first-parent

```
git log --first-parent --oneline
```

Walks only ~ 1 . After `--no-ff` merges, this is “what landed on `main`.” After a fast-forward, it is the same as a normal log. Chapter 15 picks a branching strategy; this page is the mechanism that strategy will call.

Hands-On: A Real Merge Conflict

Hands-On: Simulate a Real Merge Conflict

Two branches edit `src/ping.py` on Linux. You merge, resolve markers, and delete the feature branch.

Setup

Start from a clean `~/desk/desk-api` (chapter 3 hands-on).

```
cd ~/desk/desk-api
git switch main
git status
```

Branch A — JSON ping

```
git switch -c feat/ping-json
cat > src/ping.py <<'EOF'
def ping():
    return {"status": "ok"}
EOF
git add src/ping.py
git commit -m "feat(ping): return a dict"
git switch main
```

Branch B — string change on main

```
cat > src/ping.py <<'EOF'
def ping():
    return "OK"
EOF
git add src/ping.py
git commit -m "feat(ping): shout the status"
```

`main` and `feat/ping-json` have diverged.


```
A ← B ← C ← shout      main          ping.py returns "OK"
                        dict      feat/ping-json  ping.py returns {"status": "ok"}
```

Merge and conflict

```
git merge feat/ping-json
git status
cat src/ping.py
```

You should see markers. Open the file in `nvim` (or your editor) and make it:

```
def ping():
    return {"status": "OK"}
```

No marker lines left.

```
git add src/ping.py
git commit -m "merge feat/ping-json"
git log --oneline --graph --decorate -8
git branch -d feat/ping-json
```

Done when

- `git status` is clean on `main`
- `src/ping.py` returns a dict with "OK"
- `feat/ping-json` is deleted
- `git log --graph` shows a merge commit (two parents)

If you want to retry: `git reset --hard` to the commit before the merge (only on this lab clone), or copy the repo and run the page again. Chapter 9 rebases this same divergence into a straight line instead.

mergetool

mergetool

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git mergetool`, `nvimdiff`, `meld` on Linux.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

ours, theirs, and Renames

ours, theirs, and Renames

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--ours` / `--theirs`, rename detection, `merge.renamelimit`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Orphan Branches

Orphan Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git switch --orphan` for gh-pages-style roots.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Comparing Branches

Comparing Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git log A..B`, `git diff A...B`, `git merge-base`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

rerere

rerere

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Reuse recorded resolution; when it helps a long-lived feature.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Rebasing

Theory

Theory: Rebase Copies Commits; It Does Not Slide Them

A rebase walks a range of commits, takes each patch against its parent, and applies it onto a new base. Each successful apply writes a **new** commit (new hash, usually same author, you as committer).

Invariants

1. Original commits remain in the object store until GC; the branch **name** moves to the copied tip.
2. Conflicts happen **once per copied commit**, not once at a join.
3. Interactive rebase is the same machine plus a todo list (**pick**, **squash**, **fixup**, **reword**, **edit**, **drop**).
4. `pull --rebase` = `fetch` + rebase local-only commits onto `@{u}`.
5. `--onto` is “take this range and replay it on that commit,” the general form of rebase.

before		after <code>git rebase main</code>	
A ← B ← C ← G	main	A ← B ← C ← G	main
↑			
D ← E	feat	D' ← E'	feat

Why teams fight about it

Merge preserves hashes and adds a node. Rebase rewrites hashes and can keep `main` linear. Unpublished feature branches are the safe audience. Published `main` is not.

What can break

- Rebasing commits other people have based work on (force-push incident).
- `--ours`/`--theirs` confusion vs merge.
- Empty commits after rebase (`--skip` vs `keep`).

How to read this chapter

Why, interactive, conflicts, pull `-rebase`, vs merge, lab; then `--onto`, autosquash, `--exec`, `--update-refs`.

Why Rebase

Why Rebase? Clean History vs Merge Commits

After this page you can say what rebase copies, and why the hashes change.

Replay commits on a new parent

You branched from `main` at `C`. `main` moved to `G`. Rebase copies your commits `D` and `E` as `D'` and `E'` with parent `G`:

```
A ← B ← C ← G          main
                ← D ← E    feat/ping-json

# git switch feat/ping-json && git rebase main

A ← B ← C ← G          main
                ← D' ← E' feat/ping-json
```

`D'` has a new hash. The patch is similar; the parent is `G`. Git then fast-forwards or you `--ff-only` merge into `main`.

```
cd ~/desk/desk-api
git switch feat/ping-json
git rebase main
```

What you gain

- `main` can stay a single chain (fast-forward).
- Reviewers read your commits on top of current `main`, not an old base.
- `git log --first-parent` on `main` after `--ff-only` is your feature commits, not a merge node.

What you pay

- New hashes. Anyone who fetched `D` and `E` now has stale names.
- Conflicts are replayed **per copied commit**, not once in a merge commit.

Rebase **unpublished** work (your feature branch). Merge **published** history you share (`main` that others pulled). Chapter 11 is force-push if you already pushed the old `D` and `E`.

rebase vs merge in one line

Merge adds a join. Rebase copies a chain onto a new base. Both produce a tree that contains both sides of the work. The graph looks different. The next pages are interactive rebase, conflicts, `pull --rebase`, and when to pick which.

Interactive Rebase

Interactive Rebase: Squash, Reorder, Fixup, Polish

After this page you can rewrite a short chain of local commits on Linux without a GUI.

Open the todo list

```
cd ~/desk/desk-api
git switch feat/ping-json
git rebase -i HEAD~3
```

Git writes a file and opens `core.editor` (nvim in this book). Each line is a commit, oldest first:

```
pick a1b2c3d feat(ping): return a dict
pick d4e5f6a docs: mention JSON
pick 98ab765 fix typo
```

You edit the **verbs**, save, quit. Git replays from the top.

Verbs you will use

Verb	Effect
pick	Keep this commit
reword	Keep, edit the message
edit	Stop here so you can amend
squash	Fold into the previous commit; editor for a combined message
fixup	Fold into the previous commit; keep the previous message
drop	Omit this commit

Reorder lines to reorder commits. Delete a line to drop (same as `drop`).

Linux without the editor (labs)

For a scripted squash of everything after the first of the last three:

```
GIT_SEQUENCE_EDITOR="sed -i '2,\$s/^pick /fixup /'" git rebase -i HEAD~3
```

`sed -i` on GNU `sed` (Linux) edits in place. This is for labs and hooks. Day to day, use `nvim`.

Stop and amend (edit)

Git stops at that commit. You `git add` / `git commit --amend`, then:

```
git rebase --continue
```

Abort

```
git rebase --abort
```

Back to the pre-rebase branch tip. Safe while the rebase is in progress.

Range

`git rebase -i main` rebases each commit reachable from `HEAD` but not from `main`. That is the usual “update my feature onto main” todo list.

Do this on a feature branch. Interactive rebase of shared `main` rewrites hashes everyone already has.

Rebase Conflicts

Rebase Conflicts: How They Differ from Merge Conflicts

After this page you can finish a conflicted rebase one commit at a time.

One commit at a time

A merge conflict happens **once**, at the join. A rebase conflict happens **each time** a copied commit does not apply onto the new parent.

```
merge: one fight at M

A ← B ← C ← M
    ↑      ↑
    D ← E

rebase: a fight at D', then maybe again at E'

A ← B ← C ← G ← D' ← E'
                stop   stop
```

Markers look the same (<<<<<<). `git status` during a rebase says `rebase in progress` and names the commit being applied.

```
git rebase main
# CONFLICT (content): Merge conflict in src/ping.py
git status
```

The loop

1. Edit files, remove markers.
2. `git add` the resolved paths.
3. `git rebase --continue` (not `git commit`, unless Git asks you to).
4. Repeat until Git prints `Successfully rebased`.

```
nvim src/ping.py
git add src/ping.py
```

```
git rebase --continue
```

If the commit becomes empty (your patch is already in the new base):

```
git rebase --skip
```

Give up:

```
git rebase --abort
```

Ours / theirs are swapped from merge

During **merge**, **--ours** is the branch you had checked out (**main**). During **rebase**, you checked out the feature and Git replays onto **main**: **--ours** is the **upstream** (the new base), **--theirs** is the commit being replayed.

That swap surprises everyone once. Prefer reading the markers over **--ours** until it is boring.

rerere

With **rerere.enabled** (chapter 2), Git records the resolution. A later rebase that hits the same hunk applies it again. On a long-lived feature branch that rebases onto **main** daily, this is the difference between pain and a two-second continue.

After success

```
git log --oneline --graph --decorate main feat/ping-json
```

A straight line from **main** through your copied commits. Then `git switch main && git merge --ff-only feat/ping-json`.

Pull with Rebase

Pull with Rebase When Your Branch Is Out of Date

After this page you can update a local branch from a remote without a merge commit from `git pull`.

`pull` is `fetch` plus `integrate`

```
git pull
```

means `git fetch` then, by default, a merge of `@{u}` (the upstream) into `HEAD`. If both you and the remote moved, you get a merge commit that only exists because you pulled.

after fetch, both moved

```
origin/main    A ← B ← C ← G
local main     A ← B ← C ← D ← E
```

```
git pull                (merge)    A ← B ← C ← G ← M
                                   D ← E
```

```
git pull --rebase       A ← B ← C ← G ← D' ← E'
```

Chapter 2 set:

```
git config --global pull.ff only
```

Then `git pull` **fails** instead of merging. You choose:

```
git pull --rebase
# or
git fetch
git rebase @{u}
```

`--rebase` copies your local-only commits onto the updated upstream, then fast-forwards your branch name.

When this is the right shape

You committed two fixes on `main` locally. Someone else pushed to `origin/main`. `git pull --rebase` replays your two fixes on top of theirs. History stays linear. Conflicts are per commit (previous page).

When to merge instead

You want a merge commit that records “I integrated upstream at this point.” Feature branches that you share, or `main` policies that require merge commits, use `git pull --no-rebase` or an explicit `git merge`.

Set it for this branch only

```
git config pull.rebase true           # this repo
git config --global pull.rebase true # every repo (overrides pull.ff only)
```

This book leaves `pull.ff` only as the global default so a surprise pull fails closed. Type `--rebase` when you mean it, until the habit is there.

After a rebase pull

```
git status
git log --oneline --graph --decorate -12
```

Your branch is ahead of 'origin/main' by N commits means you still need `git push`. If you had already pushed the old hashes, you need a force-with-lease (chapter 11).

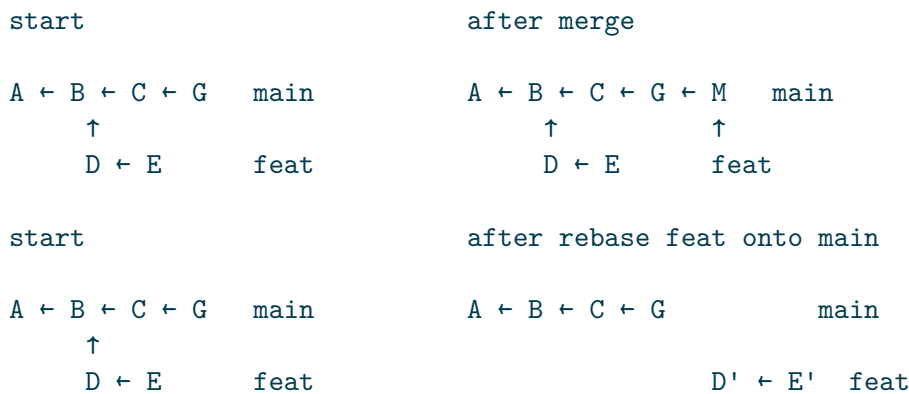
Rebase vs Merge

The Debate: Rebase vs Merge

After this page you can pick a default for `desk-api` and know what you are giving up.

Same trees, different graphs

Both operations can produce a working tree that contains both sides of the work. Merge records a join. Rebase copies a chain.



	Merge	Rebase
Hashes of your old commits	Unchanged	New
Extra commit	Often a merge commit	None if you fast-forward after
Conflicts	Once, at the join	Once per copied commit
Safe on published <code>main</code>	Yes	Usually no
Safe on your unpushed feature	Yes	Yes, and often cleaner

A default that scales

- **Feature branch you own:** rebase onto `main`, then fast-forward `main` (or open a pull request that the forge squash-merges).
- **`main` / shared long-lived branch:** merge or fast-forward only. Do not rebase commits other people have.
- **Hotfix already on `main`:** revert (chapter 10), do not rebase `main`.

Forges add a third option: **squash merge** (many commits become one on `main`). That is a server-side rewrite. Chapter 15 and chapter 31. Locally, rebase is how you squash *before* you push.

rerere, tests, Linux

Run the test command you actually have after every `--continue`. On Linux that is whatever `desk-api` uses (`python -m pytest`, a shell script, `make test`). Rebase does not run tests for you.

You can mix

Rebase the feature. Merge into `main` with `--no-ff` if the team wants a join node for `git log --first-parent`. That is Git Flow’s “feature into develop” shape. Trunk-based skips the join and fast-forwards.

There is no global winner. There is a default per branch class. Write it down in the repo (chapter 15). The hands-on page next squashes a messy feature on Linux.

Hands-On: Clean Up a Messy History

Hands-On: Clean Up a Messy History

Make three noisy commits on a feature branch, then squash them into one conventional commit with interactive rebase.

before	after fixup + amend
A ← wip ← more ← typo feat/health-header	A ← docs: mention a future health endpoint feat/health-header

Setup

```
cd ~/desk/desk-api
git switch main
git switch -c feat/health-header
```

Three messy commits

```
echo "# health" >> README.md
git add README.md
git commit -m "wip"

echo "Health endpoint later." >> README.md
git add README.md
git commit -m "more"

echo "See src/ping.py for now." >> README.md
git add README.md
git commit -m "typo fix later"

git log --oneline -5
```

Squash with GNU sed (Linux)

Keep the first of the three; fixup the rest:


```
GIT_SEQUENCE_EDITOR="sed -i '2,\$s/^pick /fixup /'" git rebase -i HEAD~3
git log --oneline -5
```

One commit remains, message still wip. Retitle:

```
git commit --amend -m "docs: mention a future health endpoint"
git log --oneline -3
```

Optional: rebase onto main

If main moved, replay this one commit:

```
git rebase main
git log --oneline --graph --decorate -8
```

Cleanup

```
git switch main
git merge --ff-only feat/health-header
git branch -d feat/health-header
```

If --ff-only refuses, main moved; rebase the feature first or merge.

Done when

- `git log` on main shows one docs: commit for this work, not three wips
- `git status` is clean
- The feature branch is deleted

Next chapter: copy a single commit onto another branch (cherry-pick) and undo a published commit (revert).

rebase -onto

rebase -onto

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Cut a commit chain off one base and replay on another.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

autosquash and fixup

autosquash and fixup

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git commit --fixup, git rebase -i --autosquash.`

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

rebase -exec

rebase -exec

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Run tests after every replayed commit.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

update-refs and keep-base

update-refs and keep-base

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--update-refs`, `--keep-base` for stacked branches.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Cherry-Pick and Revert

Theory

Theory: Cherry-Pick and Revert are Patches with New Hashes

Both operations take a **diff** (commit vs its parent, or the inverse) and apply it to **HEAD**, writing a new commit. They do not move the original node.

Invariants

1. Cherry-pick(C) apply `diff(parent(C), C)` onto **HEAD**.
2. Revert(C) apply `diff(C, parent(C))` onto **HEAD** (the inverse patch).
3. The original commit remains reachable if any name still points at it.
4. Revert of a **merge** needs `-m` to choose which parent is the mainline; Git then considers those changes “already merged” for future merges.
5. `-x` records cherry picked from in the message; it does not link objects.

<code>cherry-pick D onto main</code>	<code>revert C on main</code>
<code>feat A ← B ← C ← D ← E</code>	<code>A ← B ← C ← R</code>
<code>main A ← B ← C</code>	<code>R undoes C; C stays</code>
<code>after A ← B ← C ← D'</code>	

Why not reset on production

Reset moves a name backward. Clones that already have the old tip cannot fast-forward. Revert lets them fast-forward to a commit that *cancels* the bad one. That is the distributed-model consequence, not etiquette.

What can break

- Cherry-picking a merge without `-m`.
- Reverting a merge then trying to merge the same feature again (Git thinks it is in).
- Secrets: revert does not purge blobs from history.

How to read this chapter

Cherry-pick, hotfixes, revert, emergencies; then ranges/`-x`, empty picks.

Cherry-Pick

Cherry-Pick: Moving Commits Surgically

After this page you can copy one commit onto another branch without merging the whole feature.

Copy the patch, new hash

```
cd ~/desk/desk-api
git switch main
git cherry-pick <hash>
```

Git takes the **diff** of that commit against its parent, applies it onto **HEAD**, and writes a new commit. The new object has a new hash. Author is preserved; you are the committer.

feat	A ← B ← C ← D ← E		pick D only
main	A ← B ← C		
after	A ← B ← C ← D'	main	(D' patch of D, new hash)
feat	A ← B ← C ← D ← E		unchanged

```
git log -1 --format=fuller
```

Several commits

```
git cherry-pick abc123 def456
git cherry-pick main~3..feat/ping-json
```

A range A..B is commits reachable from B excluding A. Oldest is applied first.

Conflicts

Same markers as merge. Then:

```
git add src/ping.py
git cherry-pick --continue
git cherry-pick --abort
git cherry-pick --skip
```

Same loop as rebase, one commit at a time.

Empty pick

If `main` already contains the change, Git says the pick is empty. `--skip` or `--allow-empty`.

What cherry-pick is for

- A **hotfix** that lives on a release branch and also needs to be on `main`
- A commit that landed on the wrong branch
- Backporting a fix to an older **release/*** line

What it is a poor fit for: dragging a whole feature. Merge or rebase the branch instead so you get the whole chain and a honest graph.

Next page: hotfixes and backports as a desk scenario.

Hotfixes and Backports

Deploying Hotfixes Without Merging Features

After this page you can land a one-commit fix on `main` and on a release branch without dragging unreleased work.

The situation

`feat/big-refactor` is half done. Production (`main` or `release/1.x`) needs a one-line fix that you already committed on the feature branch.

Merging the feature would ship unfinished work. Cherry-pick the fix commit only.

```
feat/big-refactor  A ← B ← fix ← D ← E      (D,E not ready)
main              A ← B

cherry-pick fix    A ← B ← fix'              main
                  A ← B ← fix ← D ← E      feat (untouched)
```

```
cd ~/desk/desk-api
git log feat/big-refactor --oneline -5
# copy the hash of the fix

git switch main
git cherry-pick <hash>
```

Release branch

```
git switch -c release/1.2 main  # or an existing release line
git cherry-pick <hash>
```

Now both lines have the fix, as two commits with different hashes and the same author.

Backport the other direction

The fix landed on `main` first. An older release still needs it:

```
git switch release/1.1
git cherry-pick <hash-on-main>
```

If the older tree cannot apply the patch (the surrounding code moved), resolve like a rebase conflict or rewrite the fix by hand and commit normally. Cherry-pick is a tool, not a guarantee.

Label it in the message

```
git cherry-pick -x <hash>
```

-x appends (cherry picked from commit ...) to the message. Useful on release branches so `git log` shows the origin hash.

After the pick

Run whatever `desk-api` uses to check itself on Linux (`python -c 'from src.ping import ping; print(ping())'` is enough for the toy module). Then merge or push according to chapter 11. Do not merge `feat/big-refactor` until that feature is done.

Revert and Unmerge

git revert and Unmerging Changes

After this page you can undo a **published** commit by adding a new commit, including a merge.

Revert writes history forward

```
cd ~/desk/desk-api
git switch main
git revert HEAD --no-edit
git log --oneline -3
```

The bad commit is still in the graph. A new commit applies the inverse patch. Shared clones can fast-forward. This is the safe undo on `main`.

<code>reset --hard HEAD~1</code>	(local only)	A ← B	main
		C still in reflog	
<code>revert HEAD</code>	(shared)	A ← B ← C ← R	main
		R undoes C; C remains	

`--no-edit` keeps Git's default message (Revert "..."). Edit with `git revert HEAD` if you need a fix: subject.

Several commits

```
git revert --no-edit HEAD~2..HEAD
```

Reverts newest first (the range is exclusive on the left). Conflicts: `git add` and `git revert --continue`, or `git revert --abort`.

Revert a merge commit

A merge has two parents. Git needs to know which parent is “the mainline” to diff against:

```
git revert -m 1 <merge-hash>
```

`-m 1` means “keep first parent’s tree; undo what the merged branch brought in.” `-m 2` is the other parent (almost never what you want on `main`).

After reverting a merge, Git considers that feature’s commits **already merged**. Cherry-picking them back later can no-op. To re-land the feature, revert the revert, or rebase the feature into new hashes.

Revert vs reset vs restore

Goal	Command	History
Throw away uncommitted edits	<code>git restore</code>	Unchanged
Replace last unpublished commit	<code>git commit --amend / reset</code>	Rewritten
Undo a published commit	<code>git revert</code>	New commit

`git reset --hard` on shared `main` is a force-push incident. Revert instead.

Un-merge without `-m`

If the merge was never pushed and is `HEAD`:

```
git reset --hard HEAD~1
```

That moves `main` to first parent and drops the merge commit from the branch name. Only on a lab clone or an unpublished branch. Chapter 7 is reset in full. The merge commit stays in the reflog (chapter 17) until it expires.

Production Emergency Scenarios

Real Production Emergency Scenarios

After this page you have a short playbook for “production is wrong and Git is how we talk about it.”

Assume Linux, ~/desk/desk-api, and a main that is already on a remote. You are in a terminal, not a GUI.

```
published on origin/main?

no      reset / amend / rebase    (rewrite)

yes     revert or cherry-pick fix (forward)

secret in history    rotate now; filter-repo later
```

1. Bad commit is HEAD on main, already pulled by others

```
git switch main
git pull --ff-only
git revert HEAD --no-edit
# run tests
git push
```

2. Bad commit is in the middle of main

```
git log --oneline -20
git revert --no-edit <hash>
# tests
git push
```

If several commits are one logical change, revert the range newest-first, or revert the merge with `-m 1`.

3. Fix exists on a feature branch, production cannot wait

```
git log feat/whatever --oneline
git switch main
git cherry-pick -x <fix-hash>
# tests
git push
```

Leave the feature branch as-is. The pick is a new hash on `main`.

4. You committed a secret

`git revert` **does not** remove the secret from history; it only removes it from HEAD. Anyone who fetched still has the blob. Rotate the secret at the provider. History cleanup is chapter 20 (`git filter-repo`). For the next five minutes: revert or a follow-up commit that deletes the file, rotate, tell the team.

5. Merge to main was wrong, merge commit is HEAD, not pushed

```
git reset --hard HEAD~1
```

If it **was** pushed, revert with `-m 1` instead.

6. You are not sure what main is

```
git status
git log --oneline --graph --decorate -15
git reflog -10
```

Status, graph, reflog. Then pick revert (shared) or reset (local only).

Write the team a one-line message: hash reverted, why, who is rotating secrets if any. Git is the record; the pager is the people.

Ranges and $-x$

Ranges and -x

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- cherry-pick A..B, -x recorded origin.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Empty Picks and –skip

Empty Picks and `–skip`

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When the patch is already in the tree.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Remotes

Theory

Theory: A Remote is Another Repository Plus a Nickname

`origin` is not special to Git. It is a default name for a URL. Fetch copies objects and updates **remote-tracking refs**. Push asks the other repository to move a name it owns.

Invariants

1. `git fetch` does not move your `main`. It updates `refs/remotes/origin/main`.
2. `git pull` = fetch + merge or rebase of `@{u}` into `HEAD`.
3. Non-fast-forward push is rejected because it would drop commits the remote still names.
4. `--force-with-lease` succeeds only if the remote ref still matches what you last fetched.
5. Several remotes means several photographs of the same DAG, not several Git identities.

<code>laptop</code>	<code>origin (bare or forge)</code>
<code>refs/heads/main</code>	<code>refs/heads/main</code>
<code>refs/remotes/origin/main</code>	(fetch updates this)
<code>push</code>	(moves theirs if fast-forward)

Why PRs are not Git objects

A pull request is a forge row: “please integrate this ref into that ref,” plus comments and checks. The Git underneath is still fetch + merge/rebase/squash on the server.

What can break

- `git pull` on a dirty tree.
- Force-push of `main`.
- Two remotes with the same objects but different `main` (divergent policy).

How to read this chapter

Fetch/pull/push, extra remotes, force-with-lease, PRs, two-clone lab; then upstream, refsspecs, bundle, prune, fork vs clone.

Fetch, Pull, and Push

Remote Mechanics: Fetch, Pull, Push

After this page you can talk to another Git repository on Linux without confusing fetch with pull.

A remote is a named URL

```
cd ~/desk/desk-api
git remote add origin git@host:ada/desk-api.git
git remote -v
```

`origin` is a convention. The URL is SSH (chapter 2) or `https://`. For labs on one machine, a **bare** repo on disk is a valid remote:

~/desk/desk-api		origin
working tree + .git		(bare) objects + refs
git fetch	origin/main	refs/heads/main
git push	main	refs/heads/main
refs/heads/main		refs/remotes/origin/main

```
git init --bare ~/desk/desk-api.git
git remote add origin ~/desk/desk-api.git
```

`file://` and a path both work on Linux. Bare means no working tree — the shape a forge stores.

Fetch

```
git fetch origin
git fetch --all
git log --oneline --graph --decorate --all
```

Fetch downloads objects and updates **remote-tracking** names (`origin/main`). It does not move your `main`. Safe to run any time.

Push

```
git push -u origin main
```

`-u` sets **upstream** so later `git push` and `git pull` know the default. After that:

```
git push
git status
# Your branch is up to date with 'origin/main'
```

Push refuses if the remote has commits you do not. Fetch, integrate, push again.

Pull

```
git pull
```

Fetch plus integrate `@{u}` into HEAD. With `pull.ff only` (chapter 2), it fast-forwards or fails. `git pull --rebase` copies your local commits onto the updated upstream (chapter 9).

Prefer `git fetch` then look at `git log --oneline main..origin/main` before you integrate.

Tracking

```
git branch -vv
git rev-parse --abbrev-ref --symbolic-full-name '@{u}'
```

`@{u}` is the upstream of the current branch. `git push -u` is how it gets set.

Delete a remote branch

```
git push origin --delete feat/ping-json
```

Local delete is still `git branch -d`. The two are independent.

Multiple Remotes

Understanding Multiple Remotes

After this page you can push the same `desk-api` history to two places.

Extra remotes are normal

```
cd ~/desk/desk-api
git remote add origin git@forgejo.home:ada/desk-api.git
git remote add github git@github.com:ada/desk-api.git
git remote -v
```

Fetch both:

```
git fetch --all
git branch -r
```

`origin/main` and `github/main` are separate photographs.

```
git push origin
Forgejo

laptop desk-api

GitHub
git push github

refs/remotes/origin/main
refs/remotes/github/main
```

Push to both

```
git push origin main
git push github main
```

A convenience:

```
git remote add all git@forgejo.home:ada/desk-api.git
git remote set-url --add --push all git@forgejo.home:ada/desk-api.git
git remote set-url --add --push all git@github.com:ada/desk-api.git
git push all main
```

One `git push` all updates both push URLs. Fetch still uses the first URL of that remote unless you add more fetch URLs. Many people just type two pushes.

Independence preview

Chapter 22 is lock-in. The Git-level move is this page: the objects live in every clone. Two remotes means a forge outage does not strand `desk-api`. Issues and CI still live on each forge; Git history does not.

Rename and remove

```
git remote rename origin forgejo
git remote remove github
```

Removing a remote deletes remote-tracking refs for it. Local branches stay.

origin on a USB disk

```
git init --bare /media/ada/stick/desk-api.git
git remote add stick /media/ada/stick/desk-api.git
git push -u stick main
```

Linux mounts the stick; Git sees a path. Same protocol family as `~/desk/desk-api.git`.

Force Push

Force Push Dangers

After this page you can update a remote after a rebase without clobbering someone else's commits.

Why push refuses

You rebased `feat/ping-json`. Local hashes are new. The remote still has the old chain. A plain `git push` is a non-fast-forward and Git rejects it.

```
origin/feat    A ← B ← C ← D ← E
local feat     A ← B ← C ← G ← D' ← E'      (rebased onto new main)

plain push     rejected (not a fast-forward)
--force-with-lease
  if origin still at E → origin becomes E'
  if teammate added F → lease fails, fetch first
```

--force-with-lease

```
git push --force-with-lease origin feat/ping-json
```

Git checks that the remote ref still points at what you last fetched. If a teammate pushed in the meantime, the lease fails and you fetch again. That is the default force this book uses.

--force

```
git push --force origin feat/ping-json
```

Overwrites the remote ref even if it moved. Use on a branch only you own, and only when you meant to throw away the remote's extra commits. Never a habit on `main`.

main

Do not force-push `main` on a shared remote. Revert instead (chapter 10). Forges often deny force on the default branch (chapter 31).

Recover if you force-pushed by mistake

The old commits may still be on your laptop (`git reflog`) or another clone. Push them back:

```
git reflog
git push --force-with-lease origin <old-hash>:main # only if you own this recovery
```

That is an emergency. Coordinate on chat first.

Lease plus fetch

Always:

```
git fetch origin
git log --oneline --graph --decorate HEAD origin/feat/ping-json
git push --force-with-lease
```

Look at the graph. Then push.

Pull Requests and Code Review

Pull Requests, Code Review, and Handling Conflicts in Teams

After this page you can describe a pull request as Git operations plus a forge UI.

Git has no pull request object

A **pull request** (GitHub, Gitea, Forgejo, Bitbucket) or **merge request** (GitLab) is:

1. A branch you pushed (**feat/ping-json**)
2. A request that another branch (**main**) integrate it
3. Discussion, checks, and a button that runs **merge**, **squash merge**, or **rebase merge** on the server

Gerrit uses Change-Ids (chapter 27). SourceHut uses mailing lists (chapter 28). The Git underneath is still `fetch + merge` or `rebase`.

```
origin/main    A ← B ← C
origin/feat    A ← B ← C ← D ← E      ← you pushed this

forge PR       "merge feat into main"
               = merge, squash, or rebase on the server
```

Local shape

```
git switch -c feat/ping-json
# commits
git push -u origin feat/ping-json
```

On the forge: open a PR from **feat/ping-json** into **main**. Reviewers fetch that branch:

```
git fetch origin feat/ping-json
git log --oneline main..origin/feat/ping-json
git switch -c review-ping origin/feat/ping-json
```

Conflicts in a PR

The server's merge is the same three-way merge as chapter 8. Resolve **on Linux** in your clone:

```
git fetch origin
git switch feat/ping-json
git rebase origin/main          # or: git merge origin/main
# fix, add, rebase --continue
git push --force-with-lease
```

The PR updates because the branch name moved.

Review comments

Comments live on the forge. They are not Git objects. Closing a PR without merging leaves the branch on the remote until you delete it. Merging may delete the branch if the forge is set to.

Hands-on next

The next page uses two directories on one Linux machine as “you” and “origin,” so you can push and fetch without an account.

Hands-On: Two Clones

Hands-On: Fork, Clone, Branch, Commit, Push

Simulate a remote with a bare repo on Linux. No forge account required.

Origin (bare)

~/desk/desk-api (clone A, you) push/fetch	~/desk/desk-api.git (bare origin)	~/desk/desk-api-b (clone B, teammate) fetch/merge
---	--------------------------------------	---

```
mkdir -p ~/desk
git init --bare ~/desk/desk-api.git
```

Your clone

If ~/desk/desk-api already has history:

```
cd ~/desk/desk-api
git remote remove origin 2>/dev/null || true
git remote add origin ~/desk/desk-api.git
git push -u origin main
```

If you are starting empty:

```
git clone ~/desk/desk-api.git ~/desk/desk-api
cd ~/desk/desk-api
# if clone was empty, make the chapter 3 tiny project first, then:
git push -u origin main
```

A second clone (teammate or your own review checkout)

```
git clone ~/desk/desk-api.git ~/desk/desk-api-b
cd ~/desk/desk-api-b
git log --oneline -3
```

Feature on A, fetch on B

```
cd ~/desk/desk-api
git switch -c feat/readme-note
echo "Lab remote: ~/desk/desk-api.git" >> README.md
git add README.md
git commit -m "docs: note the local origin path"
git push -u origin feat/readme-note
```

```
cd ~/desk/desk-api-b
git fetch origin
git log --oneline main..origin/feat/readme-note
git merge origin/feat/readme-note
```

On a forge, that merge on B is the reviewer's local check; the **PR merge** happens on the server into main. Here, integrate on A:

```
cd ~/desk/desk-api
git switch main
git merge --ff-only feat/readme-note
git push origin main
git branch -d feat/readme-note
git push origin --delete feat/readme-note
```

```
cd ~/desk/desk-api-b
git fetch origin
git switch main
git merge --ff-only origin/main
```

Done when

- ~/desk/desk-api.git has main with the docs commit
- Both clones' main match `git rev-parse HEAD`
- Feature branch is gone locally and on origin

You just did clone, branch, commit, push, fetch, and integrate. A pull-request button is that last integrate, with a UI. Next: stash, tags, and cleanup on a single clone.

Tracking and Upstream

Tracking and Upstream

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `-u, @{u}, branch.autoSetupMerge.`

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Refspecs

Refspecs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Fetch and push refsspecs; + force; tags.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git bundle

git bundle

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Sneakernet / air-gap: `git bundle create` and `git clone file.bundle`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Prune and Rename

Prune and Rename

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `fetch.prune`, `git remote rename`, stale remote-tracking branches.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Forks vs Clones

Forks vs Clones

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Forge fork vs `git clone`; `origin` and `upstream` naming.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Stash, Tags, and Cleanup

Theory

Theory: Stash is Commits; a Tag is a Name Meant to Stay Put

`git stash` writes commit objects (index state + working-tree state) and points `refs/stash` at them. A **lightweight tag** is a ref. An **annotated tag** is an object (message, tagger, date) that then points at a commit.

Invariants

1. Stash does not change HEAD's commit; it makes the working tree match HEAD (plus options for untracked).
2. `stash@{n}` is reflog syntax on `refs/stash`.
3. Tags are not pushed by default (`git push origin v1.0.0`).
4. `git clean` deletes untracked files Git never hashed; there is no Git undo.
5. `git describe` finds the nearest reachable annotated tag and counts commits after it.

<code>stash push</code>	<code>annotated tag</code>
HEAD C (unchanged)	refs/tags/v0.1.0 → tag object → commit C
WD clean	
refs/stash → W (WIP commit)	
parent C	

Why stash vs a branch

A branch is a name you can push and PR. A stash is local, easy to drop, easy to lose in `stash clear`. If the WIP will survive the afternoon, `switch -c wip/...` and commit.

What can break

- `stash pop` with conflicts (entry may still drop).
- Moving a published tag (`-f`) while others have already fetched it.
- `git clean -f -x` deleting ignored `.env`.

How to read this chapter

Stash, tags, cleanup, lab; then `stash branch`, `describe`, `archive`.

Stash

Stash Mechanics, Including Patch Stashes

After this page you can park uncommitted work, switch branches, and bring the work back.

A stash is a commit Git keeps on the side

```
cd ~/desk/desk-api
echo "wip" >> README.md
git stash push -m "wip readme"
git status
git stash list
```

Working tree is clean. `stash@{0}` holds the diff. Switch branches, then:

before stash	after stash push
working dirty README	working clean
index clean	index clean
HEAD C	HEAD C
	stash@{0} a pair of commits (index + working tree)

```
git stash pop
# or
git stash apply
```

`pop` applies and drops. `apply` applies and keeps the stash entry.

Include untracked

```
git stash push -u -m "wip plus new files"
```

Without `-u`, untracked files stay in the way and can block a switch.

Patch stash (-p)

```
git stash push -p -m "only the ping hunk"
```

Git offers hunks, same as `git add -p`. Stash the ones that are in the way; leave the rest in the working tree.

Named and multiple

```
git stash push -m "experiment A"  
git stash push -m "experiment B"  
git stash list  
git stash apply stash@{1}  
git stash drop stash@{1}  
git stash show -p stash@{0}
```

Numbers shift when you drop. The message is how you find them.

Conflicts on apply

Markers appear. Add and continue as a normal working-tree conflict. The stash entry remains if you used `apply`; `pop` may still drop — prefer `apply` when you are unsure.

Stash vs commit

A commit on a throwaway branch is easier to push and to recover. Stash is for “I must switch in the next ten seconds.” If the WIP will last, `git switch -c wip/...` and commit.

Tags

Tags: Lightweight vs Annotated

After this page you can mark a commit as a release and push that name.

Two kinds

```
cd ~/desk/desk-api
git tag v0.1.0 # lightweight: a ref file
git tag -a v0.1.0 -m "desk-api 0.1.0" # annotated: a tag object
```

Lightweight is a name pointing at a commit (like a branch that you do not move).

Annotated is a real object: message, tagger, date, optional signature (chapter 20). Releases should be annotated.

lightweight		annotated	
refs/tags/v0.1.0	commit C	refs/tags/v0.1.0	tag object
			commit C
			tag object has message, tagger, date

If v0.1.0 already exists, Git refuses. `git tag -f` overwrites locally; moving a published tag surprises clones. Prefer v0.1.1.

List and show

```
git tag
git tag -l 'v0.1.*'
git show v0.1.0
```

Push

Tags are not pushed by default.

```
git push origin v0.1.0
git push origin --tags
```

--tags pushes all local tags. Prefer naming one tag for a release.

Checkout

```
git switch --detach v0.1.0
```

Detached HEAD (chapter 1). Build or inspect, then `git switch main`. To develop from a tag: `git switch -c hotfix/0.1.1 v0.1.0`.

Delete

```
git tag -d v0.1.0
git push origin --delete v0.1.0
```

Forges may still show a release page; that is forge metadata, not the Git tag.

Cleanup Commands

Cleanup Commands

After this page you can delete merged branches, drop stashes, and remove untracked files **on purpose**.

Merged branches

```
cd ~/desk/desk-api
git switch main
git branch --merged
git branch -d feat/readme-note
```

`--merged` lists names whose tips are reachable from `HEAD`. `-d` refuses otherwise; `-D` deletes anyway.

Remote leftovers:

```
git fetch --prune
git branch -r
```

`--prune` deletes remote-tracking names that the remote no longer has.

Stashes

```
git stash list
git stash drop stash@{0}
git stash clear          # every stash
```

`clear` is permanent for the stash reflog's practical purposes. Apply first if you need the diff.

Untracked files: `git clean`

```
git clean -n          # dry run
git clean -n -d       # include untracked directories
git clean -f -d       # delete
```

`-n` is mandatory the first time you run it on a tree you care about. `-f` is required (Git refuses to delete without it). `-x` also removes ignored files (build artifacts listed in `.gitignore`). That can delete `.env` if it is ignored — check `-n -x` first.

On Linux you can recover a deleted untracked file only from backups or the editor; Git never had a blob.

Garbage collection

```
git gc --aggressive --prune=now
```

Packs objects, drops unreachable ones older than the prune date. Rarely needed by hand; Git runs `gc` in the background. After a huge filter-repo (chapter 20) you will run it on purpose.

git fsck (peek)

```
git fsck --full
```

Reports dangling objects (often old amended commits) and real corruption. Dangling is normal. Missing blobs is chapter 21.

Hands-On: Stash and Switch

Hands-On: Stash, Work on Another Branch, Apply

Park a README edit, fix a typo on `main`, come back, apply the stash.

```
cd ~/desk/desk-api
git switch main
git status
```

Park work

```
echo "TODO: metrics" >> README.md
git stash push -m "metrics todo"
git status
git stash list
```

Other work

```
git switch -c fix/readme-typo
echo "typo-lab $(date -Iseconds)" >> README.md
git add README.md
git commit -m "docs: lab timestamp"
git switch main
git merge --ff-only fix/readme-typo
git branch -d fix/readme-typo
```

Restore the stash

```
git stash list
git stash apply stash@{0}
cat README.md
git add README.md
git commit -m "docs: TODO for metrics"
git stash drop stash@{0}
```



```
git stash list
git status
```

If apply conflicted, resolve markers, `git add`, commit, then drop.

Tag this point

```
git tag -a v0.1.0-lab -m "after stash lab"
git show v0.1.0-lab --quiet
git tag -d v0.1.0-lab
```

Done when

- TODO: metrics is committed on `main`
- `git stash list` is empty (or only older stashes you had before)
- status is clean

Next: hooks that run on this Linux clone before a commit exists.

stash branch

stash branch

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git stash branch` to avoid apply conflicts.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git describe

git describe

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Nearest tag plus commits; version strings from Git.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git archive

git archive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tarballs from a commit without the `.git` dir.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hooks

Theory

Theory: Hooks are Local Policy; the Server is Real Policy

A hook is an executable Git runs by **filename**. Exit non-zero aborts the Git operation. Client hooks are not cloned (by design: a clone must not execute a stranger's scripts). Therefore they are **courtesy**. Server **pre-receive** / **update** and CI are **enforcement**.

Invariants

1. `.git/hooks/pre-commit` runs on this clone only, unless `core.hooksPath` points at a shared dir.
2. `--no-verify` skips client `pre-commit` and `commit-msg`.
3. The web editor on a forge never runs your laptop hooks.
4. `core.hooksPath` and `init.templateDir` are how a Linux user shares hooks without committing `.git`.

```
git commit
  → pre-commit → commit-msg → object exists

git push
  → pre-push (client) → network → pre-receive (server) → CI
```

Why both

Client hooks fail in milliseconds (whitespace, WIP subject). CI fails in minutes (tests, secret scan) and cannot be skipped with `--no-verify`. Put the same *checks* in both; put *policy* on the server.

What can break

- Forgetting `chmod +x`.
- Hooks that print to stdout and corrupt `commit-msg` files.
- Assuming teammates have your hooks after `git clone`.

How to read this chapter

Types, writing, templates vs CI, lab; then `prepare-commit-msg`, `post-checkout/merge`, `--no-verify`.

Hook Types

Hook Types: Pre-Commit, Commit-Msg, Pre-Push

After this page you can name the client hooks this book uses and where they live on Linux.

A hook is a program Git runs

Executable files in `.git/hooks/`. Git looks at the filename. If `pre-commit` exists and is executable (`chmod +x`), Git runs it before it writes a commit object. Exit status 0 continues. Non-zero aborts.

```
git commit

pre-commit      exit  0    abort, no object
0

commit-msg     exit  0    abort, no object
0

commit object written, main slides forward

git push

pre-push       exit  0    abort, remote unchanged
0

objects uploaded
```

Sample files are `*.sample` and are **not** run. Copy and `chmod`:

```
cd ~/desk/desk-api
ls .git/hooks
install -m 0755 .git/hooks/pre-commit.sample .git/hooks/pre-commit
# then edit; or write your own
```

`install -m 0755` is the Linux way to copy and set the executable bit in one step.

Client hooks you will write

Hook	When	Typical job
pre-commit	After you type <code>git commit</code> , before the object exists	Lint, block secrets, <code>git diff --check</code>
commit-msg	After you save the message	Conventional Commits pattern
pre-push	After you type <code>git push</code> , before objects upload	Tests, block WIP branches

There are more (`prepare-commit-msg`, `post-commit`, `post-checkout`). Start with these three.

Server hooks (later)

A forge runs `pre-receive` / `update` / `post-receive` on the server when you push. Chapter 23 names them; chapter 31 uses them for policy. Your laptop hooks do not run on the server. A teammate can skip `pre-commit` with `git commit --no-verify`. Server hooks are the ones that actually enforce.

Environment

Hooks run with a POSIX shell on Linux unless the shebang says otherwise. Always start with:

```
#!/usr/bin/env bash
set -euo pipefail
```

Git sets `GIT_DIR` and runs with the working tree as `cwd` for client hooks (with nuances for worktrees, chapter 14). Print to `stderr` so you do not pollute commit output:

```
echo "pre-commit: running" >&2
```

Writing a Hook

Writing Your First Hook

After this page you have a `pre-commit` that refuses a commit containing a tab-only whitespace error, and a `commit-msg` that checks a conventional subject.

`pre-commit`

```
cat > ~/desk/desk-api/.git/hooks/pre-commit <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
git diff --cached --check
EOF
chmod +x ~/desk/desk-api/.git/hooks/pre-commit
```

`git diff --cached --check` flags leftover conflict markers and bad whitespace in the **index**. Exit status non-zero aborts the commit.

Try:

```
cd ~/desk/desk-api
printf 'bad \n' >> README.md    # trailing space
git add README.md
git commit -m "docs: trigger hook"
```

Fix the line, `git add`, commit again.

`commit-msg`

Git passes the path to a temporary file that holds the message:

```
cat > ~/desk/desk-api/.git/hooks/commit-msg <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
file="$1"
subject="$(head -n1 "$file")"
if [[ "$subject" == WIP* ]]; then
    echo "commit-msg: subject cannot start with WIP" >&2
    exit 1
fi
```



```
EOF
chmod +x ~/desk/desk-api/.git/hooks/commit-msg
```

A stricter conventional check can wait until the team agrees on types. Start by blocking WIP.

pre-push

Git passes the remote name and URL, and on stdin: local ref, local hash, remote ref, remote hash.

```
cat > ~/desk/desk-api/.git/hooks/pre-push <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
while read -r local_ref local_sha remote_ref remote_sha; do
    if [[ "$remote_ref" == refs/heads/main && "$local_sha" != "$remote_sha" ]]; then
        echo "pre-push: running tests before updating main" >&2
        # python -m pytest -q || exit 1
    fi
done
EOF
chmod +x ~/desk/desk-api/.git/hooks/pre-push
```

Uncomment the test command when `desk-api` has tests.

--no-verify

```
git commit --no-verify -m "docs: skip hooks"
```

Emergency only. Treat it as breaking the seatbelt.

Hook Templates and CI

Hook Templates: Sharing Hooks Across Repos

After this page you can point new repos at a shared hooks directory, and you know why CI still has to run on the server.

`.git/hooks` is local

Cloning does **not** copy hooks (Git ignores them in the payload on purpose: a clone would otherwise execute someone else's scripts). Each clone needs its own install step.

`core.hooksPath`

```
mkdir -p ~/desk/git-hooks
# put pre-commit, commit-msg, ... there, chmod +x
git config --global core.hooksPath ~/desk/git-hooks
```

Every repo on this Linux user account now runs those hooks. A repo can override with `git config core.hooksPath .githooks` (a directory **inside** the project that you *do* commit).

Project-local, committed:

```
mkdir -p ~/desk/desk-api/.githooks
# copy scripts, chmod +x, commit them
git config core.hooksPath .githooks
```

New clones still must run the `git config` once, or a Makefile / just recipe / bin/install-hooks script. Document that in README.md.

`init.templateDir`

```
mkdir -p ~/desk/git-template/hooks
# scripts in that hooks dir
git config --global init.templateDir ~/desk/git-template
```

`git init` and `git clone` copy the template's `hooks/` into the new `.git/hooks`. Good for your laptop. Teammates need the same config.

CI is the hook you cannot skip

A pre-commit hook is a courtesy. `git commit --no-verify` and a forge's web editor skip it. **CI on the forge** (chapter 32, GitHub Actions in Part VI) runs on the server after push. Put the same checks there: `git diff --check`, tests, secret scanning.

laptop	forge
<code>.git/hooks/pre-commit</code>	<code>pre-receive / CI job</code>
skippable	not skippable
<code>git commit</code>	<code>git push accepted</code>
<code>git push</code>	

Client hooks catch mistakes in milliseconds on Linux. CI is the policy. Use both.

Frameworks

pre-commit.com is a Python manager that installs a tiny `.git/hooks/pre-commit` and runs a YAML list of linters. Optional. Raw bash hooks in this chapter teach the mechanism; a framework is fine in production if the team already uses it.

Hands-On: A Pre-Commit Hook

Hands-On: Create a Pre-Commit Hook

Install a pre-commit that runs `git diff --cached --check` on desk-api.

```
cd ~/desk/desk-api

cat > .git/hooks/pre-commit <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
echo "pre-commit: git diff --cached --check" >&2
git diff --cached --check
EOF
chmod +x .git/hooks/pre-commit
test -x .git/hooks/pre-commit
```

Fail on purpose

```
printf 'trailing space \n' >> README.md
git add README.md
git commit -m "docs: should fail the hook"
```

Expect a non-zero exit and no new commit. `git log -1 --oneline` is unchanged.

Pass

```
git restore --staged README.md
git restore README.md
echo "hooks lab" >> README.md
git add README.md
git commit -m "docs: hooks lab"
git status
```

Clean status, new commit. Remove the hook when you are done with the lab, or leave it — `--check` is a good default:

```
rm -f .git/hooks/pre-commit
```

Leave it in place if you want the check on this clone.

Next: submodules (a repo inside a repo) and worktrees (a second working directory, same `.git`).

prepare-commit-msg

prepare-commit-msg

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Fill in a template before the editor opens.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

post-checkout and post-merge

post-checkout and post-merge

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Hooks after switch and merge (deps, submodules).

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

–no-verify

–no-verify

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When skipping hooks is justified; why CI still must run.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Submodules and Worktrees

Theory

Theory: Submodules Pin a Hash; Worktrees Share an Object Store

A **submodule** is a second repository whose current commit is recorded in the parent as a gitlink (mode 160000). A **worktree** is a second working directory for the *same* repository: one `.git`, several checkouts, **one branch per worktree**.

Invariants

1. The superproject stores a URL (`.gitmodules`) and a **gitlink hash**, not the child's files as blobs.
2. `git clone` of the parent does not fetch submodule objects unless `--recurse-submodules / submodule update --init`.
3. Two worktrees cannot check out the same branch (indexes would fight).
4. Worktree `.git` is a **file** pointing at the main repo's `worktrees/` metadata.

submodule	worktrees
desk-api tree	~/desk-api HEAD → main
src/desk-lib → gitlink 9fce	~/desk-api-hf HEAD → hotfix
desk-lib repo A ← 9fce	\ /
	shared objects/refs

Why not always submodule

If two projects always change together, you are paying pin-and-update tax for a monorepo's job. If they have separate release cycles, the pin is the feature.

What can break

- Committing in a submodule and forgetting to push it (parent points at a hash nobody has).
- Detached HEAD in the submodule by default after `update`.
- `rm -rf` a worktree directory without `git worktree remove`.

How to read this chapter

Mechanics, worktrees, when to use which, lab; then foreach/sync, deinit, locks.

Submodules

Submodule Mechanics: Add, Update, Clone

After this page you can attach another Git repository at a path and record **which commit** it should be.

A submodule is a nested repository plus a gitlink

`desk-api` might need a shared `desk-lib` history without merging the trees. A **submodule** stores a commit hash in the parent tree (a **gitlink**, mode 160000) and a URL in `.gitmodules`.

```
desk-api (superproject)
  README.md
  src/ping.py
  src/desk-lib/      gitlink 160000 → commit 9fceb02 of desk-lib
                     (nested .git)
```

```
desk-lib (its own history)
A ← B ← 9fceb02    ← the hash the parent pins
```

```
cd ~/desk/desk-api
git submodule add ~/desk/desk-lib src/desk-lib
git status
git commit -m "chore: add desk-lib submodule"
```

`.gitmodules` is a tracked file:

```
[submodule "src/desk-lib"]
  path = src/desk-lib
  url = /home/ada/desk/desk-lib
```

On Linux use an SSH URL for anything you will clone elsewhere (`git@host:ada/desk-lib.git`). A path only works on this machine.

Clone a superproject

```
git clone --recurse-submodules <url> desk-api
# or after a plain clone:
git submodule update --init --recursive
```

Without `--recurse-submodules` / `update --init`, the path exists as an empty directory.

Update the pin

```
cd src/desk-lib
git fetch
git switch main
git pull --ff-only
cd ../../
git add src/desk-lib
git commit -m "chore: bump desk-lib"
```

The parent records a **new gitlink hash**. Pulling the parent does not move the submodule until `git submodule update`.

Pitfalls

- Committing in the submodule and forgetting to push it: the parent points at a hash nobody else can fetch.
- Editing files in `src/desk-lib` while detached: `git switch -c` in the submodule first.
- Nested submodules: `--recursive` on clone and update.

If the team is small and `desk-lib` changes with `desk-api` every day, a **monorepo** (chapter 15) or a copy may hurt less. Submodules shine when the nested project has its own release cycle.

Worktrees


```
git worktree remove ~/desk/desk-api-pr
```

`remove` deletes the extra directory after the working tree is clean. `git worktree prune` cleans metadata if you `rm -rf` the directory by hand.

Detached review

```
git worktree add --detach ~/desk/desk-api-tag v0.1.0
```

Inspect a tag without leaving your feature branch in the main directory.

Hooks and GIT_DIR

Worktrees have `.git` as a **file** pointing at the main repo's `worktrees/...` metadata. Hooks still run from `core.hooksPath` or the main `.git/hooks`. Scripts that assume `.git/` is a directory should call `git rev-parse --git-dir`.

When to Use Which

When to Use Which

After this page you can pick submodule, worktree, or neither for a desk problem.

Worktree

Use a worktree when **one repository** needs two checkouts: a hotfix while a feature is dirty, a PR review, a tag build. Same remotes, same hooks, cheap on Linux disk.

Submodule

Use a submodule when **two repositories** must stay separate histories, with the parent pinning a commit of the child: a shared library, a firmware blob, a docs site with its own release train.

Copy the files

If `desk-lib` is three Python files and only `desk-api` uses them, put them in `desk-api`. Submodule overhead is real.

Monorepo

If `desk-web`, `desk-api`, and `desk-infra` always change together, one repository with three directories may be simpler (chapter 15). Worktrees still apply (two branches of the monorepo). Submodules do not.

Extra clone

`git clone` to a second path is fine for a throwaway experiment you might delete with `rm -rf`. You duplicate objects until `git gc`. Worktree is cleaner for a second **branch**.

Extra remote

Two forges is remotes (chapter 11), not submodules. Push the same history twice.

Rule of thumb

Same project, two directories → worktree. Two projects, one pin → submodule. Two projects, always in lockstep → one repo.

one repo, two checkouts	worktree
two repos, pin a commit	submodule
two repos, always together	one monorepo
two forges, same history	extra remote

Hands-On: Submodule and Worktree

Hands-On: A Repo with a Submodule and a Worktree

Two small Git repos on Linux, then a worktree of the parent.

Child library

```
mkdir -p ~/desk/desk-lib
cd ~/desk/desk-lib
git init
echo "DESK_LIB=1" > version.txt
git add version.txt
git commit -m "chore: desk-lib version pin"
```

Parent with submodule

```
cd ~/desk/desk-api
git submodule add ~/desk/desk-lib src/desk-lib
git status
git commit -m "chore: add desk-lib submodule"
ls src/desk-lib/version.txt
```

Worktree of the parent

```
git worktree add ~/desk/desk-api-wt -b feat/wt-lab
cd ~/desk/desk-api-wt
git status
echo "worktree lab" >> README.md
git add README.md
git commit -m "docs: worktree lab"
cd ~/desk/desk-api
git log --oneline feat/wt-lab -1
```

main in ~/desk/desk-api does not have that commit until you merge.

Cleanup

```
cd ~/desk/desk-api
git worktree remove ~/desk/desk-api-wt
git switch main
git merge --ff-only feat/wt-lab
git branch -d feat/wt-lab
```

Leave the submodule in place for later chapters, or remove it:

```
git rm src/desk-lib
git commit -m "chore: remove desk-lib submodule"
rm -rf ~/desk/desk-lib
```

Removing the submodule does not delete `~/desk/desk-lib` unless you `rm -rf` it.

Done when

- `git worktree list` shows only the main tree
- You know whether `src/desk-lib` is still a gitlink (`git ls-files -s src/desk-lib` shows mode 160000) or is gone

Next: Git Flow, GitHub Flow, trunk-based — how teams name these branches on purpose.

foreach and sync

foreach and sync

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git submodule foreach`, `git submodule sync`.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

deinit and absorbgitdirs

deinit and absorbgitdirs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- submodule deinit, absorbgitdirs.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Worktree Locks and prune

Worktree Locks and prune

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git worktree lock, prune`, why two checkouts cannot share a branch.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Branching Workflows

Theory

Theory: A Workflow is a Policy on Names, Not a Git Feature

Git Flow, GitHub Flow, and trunk-based development are **agreements** about which refs are long-lived, how long a feature name lives, and how `main` stays deployable. Git will merge whatever you ask. The workflow is which asks you allow.

Invariants

1. Long-lived branches **diverge**; every extra one is a merge or rebase tax.
2. Short-lived `feat/*` off `main` (GitHub Flow topology) is independent of GitHub the product.
3. Trunk-based requires **runtime flags** or very small commits so `main` can take unfinished code.
4. Monorepo vs polyrepo is a **repository boundary** decision; worktrees still apply; submodules usually do not in a monorepo.

Git Flow	<code>main + develop + release/*</code>	high ceremony
GitHub Flow	<code>main + short feat/*</code>	this book's desk default
Trunk-based	<code>main + flags</code>	high CI, tiny commits

Why write it down

A strategy that exists only in chat is not enforceable. Chapter 31 (governance) is how a forge *requires* reviews and checks. This chapter is the graph you are asking Git to keep.

What can break

- Git Flow on a two-person `desk-api` (ceremony without versions in the wild).
- Trunk-based without tests or flags.
- A monorepo whose CI always builds everything.

How to read this chapter

Git Flow, GitHub Flow/trunk, flags, mono/poly, choosing; then forking, stacked changes, release tags.

Git Flow

Git Flow: The Classic Model

After this page you can name Git Flow's long-lived branches and when the extra machinery is worth it.

The picture

main	production; tagged releases
develop	next release; features merge here
feat/...	branched from develop, merged back (often --no-ff)
release/x.y	branched from develop, hotfixes, then merge to main *and* develop
hotfix/...	branched from main, merge to main *and* develop

main is always what is in production. develop is the integration line. Features never commit straight to main.

```
graph TD
    main --> A
    A --> B
    B --> C
    C --> D
    D --> E
    E --> F
    F --> X
    X --> Y
    A --- H1
    H1 --- v1.2
```

The diagram illustrates the Git commit history. It shows a sequence of commits: A, B, C, D, E, F, X, and Y. The 'main' branch points to commit A, which is also labeled 'H1' and 'v1.2'. A backslash indicates a merge from commit B back to commit A. The 'develop' branch points to commit B. Commits C, D, E, and F are part of a feature branch. A backslash indicates a merge from commit F back to commit B. The 'feat/invoices' branch points to commit X, which is also labeled 'X' and 'Y'.

Commands (shape, not a plugin)

```
cd ~/desk/desk-api
git switch main
git switch -c develop
git switch -c feat/invoices develop
# ... commits ...
git switch develop
git merge --no-ff feat/invoices
git branch -d feat/invoices
```

--no-ff keeps a join node so `git log --first-parent develop` lists features.

What it costs

Two long-lived branches that **diverge**. Every hotfix is merged twice. Release branches exist for weeks. For `desk-api` with one or two people, this is a lot of graph.

What it buys

A support team that must ship 1.2.3 while `develop` already contains 2.0 work. Multiple versions in the wild. A release manager.

Use Git Flow when you **actually** have that. The original `git-flow` CLI is optional; the branch names are the model.

Chapter 31 is how a forge **enforces** this (protected `main`, required PRs). This page is the topology.

GitHub Flow and Trunk-Based

GitHub Flow and Trunk-Based Development

After this page you can contrast a short-lived PR branch with committing to `main` behind flags.

GitHub Flow (as a Git topology)

```
main          always deployable
feat/...      branched from main, opened as a PR, merged, deleted
```

There is no long-lived `develop`. Production is `main` (or a tag on it). CI on the PR is the gate.

GitHub Flow

```
main      A ← B ← C ← M      always deployable

feat/ping  D ← E      PR, then delete
```

```
git switch main
git pull --ff-only
git switch -c feat/ping-json
# commits, push, PR, merge on the forge
git switch main
git pull --ff-only
git branch -d feat/ping-json
```

The name “GitHub Flow” is a topology. You can do it on Forgejo or GitLab. The forge UI is a PR/MR; Git is short-lived branches off `main`.

Trunk-based

Engineers commit to `main` (the **trunk**) at least daily, often several times a day. Features that are not ready stay **dark** behind a flag (next page). Release branches, if any, are cut from `main` and live days, not months.

trunk-based

```
main      A ← B ← C ← D ← E ← F      tiny commits, flags hide unfinished work
```

```
release/1.2    short-lived, if you even cut one

git switch main
# small commit
git pull --rebase
git push
```

Feature branches, when used, last hours. Rebase onto `main` before merge. `--ff-only` after a rebase is the usual integrate.

Compare

	Git Flow	GitHub Flow	Trunk-based
Long-lived lines	<code>main + develop</code>	<code>main</code>	<code>main</code>
Feature life-time	Days–weeks	Hours–days	Hours or a flag
Release	<code>release/*</code> branch	Tag / deploy <code>main</code>	Tag / short <code>release/*</code>
Fits	Multi-version products	Web apps, desk tools	High-CI teams

`desk-api` in this book defaults to **GitHub Flow topology**: short `feat/` branches, `main` deployable. Trunk-based is the same graph with even shorter branches.

Enforcement (required reviews, required checks) is chapter 31. The next page is flags, which make trunk-based possible.

Feature Flags

Feature Flags and Deployment

After this page you can ship code to `main` that is not yet on for users.

A flag is a runtime switch

The commit on `main` contains the new code. Production **does not call it** until a config value is true. That is how trunk-based teams merge daily without releasing half-built ping JSON.

```
main      A ← B ← C ← flag-commit ← D      all on main

production
  DESK_PING_JSON=0   → old string path
  DESK_PING_JSON=1   → JSON path

Git shipped the code at flag-commit; the env var is the release.
```

Toy `desk-api` (Python, Linux):

```
# src/ping.py
import os

def ping():
    if os.environ.get("DESK_PING_JSON") == "1":
        return {"status": "ok"}
    return "ok"

export DESK_PING_JSON=1
python -c 'from src.ping import ping; print(ping())'
```

The flag lives in the environment, a file under `/etc`, or a later config service. Git holds the code; the flag holds the **release**.

What Git still does

- Small commits on `main` (or short `feat/` branches)
- Revert the commit if the code is wrong (chapter 10)
- The flag can stay off while you fix

A flag is not a substitute for `git revert` when the code crashes before the flag is read.

Cleanup

Every flag is debt. When `DESK_PING_JSON` is on everywhere, delete the old branch in the `if` and the env var. A follow-up commit: `chore: remove ping JSON flag`.

GitOps preview

Later parts deploy from Git. A flag in the **application** is still an env var on the cluster. A flag in **Git** (a value in `desk-infra`) is GitOps: changing the file and merging to `main` is the release. Same idea, different file.

Monorepos vs Polyrepos

Monorepos vs Polyrepos

After this page you can place `desk-web`, `desk-api`, and `desk-infra` in one Git repository or three, on purpose.

Polyrepo (this book’s default so far)

Three repositories, three histories, three remotes:

```
~/desk/desk-web
~/desk/desk-api
~/desk/desk-infra
```

Each clones, branches, and CI-runs on its own. A change that spans API and web is two pull requests. Versions of each other are documented (“web 1.2 needs api 1.4”) or ignored until something breaks.

Monorepo

One repository:

```
~/desk/desk/
  web/
  api/
  infra/
```

One `git clone`, one `main`, one PR can touch all three. CI must learn to run only what changed, or it becomes slow. Sparse checkout and partial clone (chapter 19) exist because monorepos grow.

Trade-offs

	Monorepo	Polyrepo
Cross-project change	One PR	Several PRs, several versions
Access control	One repo (paths, CODEOWNERS)	Separate remotes, easier per-team ACLs
CI	Must be smart or it is slow	Simple per repo

	Monorepo	Polyrepo
Submodules	Rarely needed	Common for shared libs
Git scale	Packfiles, sparse, LFS	Each repo stays small

Linux tooling

Path-based CI (`git diff --name-only origin/main...HEAD | grep '^api/'`) is how a monorepo stays fast. CODEOWNERS (chapter 31) assigns `api/` to one team. None of that requires GitHub; the files are in Git.

This book

Labs keep **three polyrepos** so platform chapters can push `desk-api` to GitLab and Forgejo independently. If your real desk is a monorepo, the Git commands are the same; only the directory layout changes. Worktrees still apply. Submodules usually do not.

Choosing a Strategy

Choosing a Strategy

After this page you can write a short default for `desk-api` and know where this book will enforce it later.

```
versions in the wild?   yes      Git Flow (main + develop + release/*)

                        no

several merges/day + flags?  yes      trunk-based

                        no

GitHub Flow             short feat/* off main      ← desk default
```

A default for this book's desk

- **Topology:** GitHub Flow — short `feat/` / `fix/` branches off `main`, delete after integrate
- **Integrate:** rebase the feature onto `main`, then `--ff-only` or a force squash
- **main:** always deployable; revert to undo (never rebase published `main`)
- **Repos:** polyrepo (`desk-web`, `desk-api`, `desk-infra`) until a chapter says otherwise
- **Host:** Linux; hooks and CI run there first

Write it in `README.md` or `CONTRIBUTING.md` in the repo. A strategy that exists only in someone's head is not a strategy.

When to change it

- Multiple supported versions in production → Git Flow release branches
- Several merges to `main` per engineer per day and strong tests → trunk-based plus flags
- One PR always touches all three desks → consider a monorepo
- A compliance officer needs a merge commit per change → `--no-ff` on `main`

What Git will not decide

Git will merge whatever you ask. Forges add buttons (squash, rebase, merge). Chapter 31 is branch protection, code owners, required checks. This chapter is the **agreement**. The next part (power

skills) is internals, reflog, bisect, and repair — the tools you want when the agreement meets a bad Tuesday.

The Forking Workflow

The Forking Workflow

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Personal fork, `upstream`, PRs from forks.

Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Stacked Changes

Stacked Changes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Dependent branches, rebase `-update-refs`, Gerrit preview.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Release Tags and Changelogs

Release Tags and Changelogs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Annotated tags, Conventional Commits → changelog.

i Note

Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Power Skills

Internals

Theory

Theory: Content Addressing is the Whole Trick

Git is a content-addressed filesystem plus names. Internals are not trivia; they are why hosting, LFS, and `fsck` look the way they do.

Invariants

1. Object name = hash of type + size + payload (SHA-1 historically, SHA-256 optional).
2. Blobs have no filename; trees bind names to hashes.
3. Packfiles are compression (deltas) of that object set, not the data model.
4. A server is constrained by quarantine, `receive.denyCurrentBranch`, and hook timing around these objects.

```
commit → tree → blob
      \→ parent commit
packfile = many objects + delta chains  (disk, not mental model)
```

What can break

- Editing files under `.git/objects` by hand.
- Assuming SHA-1 collision is a daily concern (it is a research/hosting concern).

How to read this chapter

Object store, hashes, packfiles/`git log`, server-facing internals, `cat-file` lab.

The Object Store

The Object Store

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Blobs, trees, commits, and tags in `.git/objects`.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Content Addressing

Content Addressing

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SHA-1, SHA-256, and content addressing.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Packfiles and git log

Packfiles and git log

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Packfiles, compression, and how `git log` works under the hood.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Internals That Affect Servers

Internals That Affect Servers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Internals that affect server design (preview of chapter 23, platform architecture).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: git cat-file

Hands-On: git cat-file

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Explore the object store with `git cat-file`.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Reflog

Theory

Theory: The Reflog is a Local Journal of Names

Every time a ref in this clone moves, Git can append a line to that ref's log. It is not replicated. It is not a backup of other clones.

Invariants

1. HEAD and each branch can have a reflog.
2. Default expire is on the order of 30 days (`gc.reflogExpire`).
3. `HEAD@{1}` is “where HEAD was last time,” not “parent commit.”
4. Recovering a deleted branch is `git switch -c name <hash-from-reflog>`.

```
main moved C → D → E
reflog  E HEAD@{0}
        D HEAD@{1}
        C HEAD@{2}      (local only)
```

What can break

- Expecting a teammate's reflog to contain your amended commit.
- `gc --prune=now` after expire.

How to read this chapter

How it works, recover a branch, the 30-day window, step-by-step scenarios.

How the Reflog Works

How the Reflog Works

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How the reference log works.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Recover Deleted Branches

Recover Deleted Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recover deleted branches in seconds.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The 30-Day Safety Window

The 30-Day Safety Window

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The 30-day safety window.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Recovery Scenarios

Recovery Scenarios

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Real recovery scenarios, step by step.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Bisect

Theory

Theory: Bisect is Binary Search on First-Parent History

If a test can label a commit good or bad, `git bisect` finds the first bad commit in $O(\log N)$ checkouts.

Invariants

1. You mark an old good and a current bad; Git checks out the midpoint.
2. `git bisect run ./test.sh` automates the oracle.
3. Skip unbuildable commits; they are not evidence.
4. The answer is a commit hash, not a file.

```
G G G G B B B B    (good/bad along a chain)
      ^ first bad
```

What can break

- A flaky test (non-monotonic good/bad).
- Bisecting a merge-heavy history without understanding first-parent.

How to read this chapter

Manual bisect, automate with tests, 2 AM story, lab.

Binary Search for Bugs

Binary Search for Bugs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Binary search to hunt bugs and regressions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Automate Bisect with Tests

Automate Bisect with Tests

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Automate with your test suite.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

2 AM Production Debugging

2 AM Production Debugging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- 2 AM production debugging example.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: git bisect

Hands-On: git bisect

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Find a regression with `git bisect`.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Performance and Scale

Theory

Theory: Scale is Reachability and Pack Shape

“Large” is commit count, tree width, blob size, and clone graph — not only disk bytes. Shallow, sparse, and partial clones cut different axes.

Invariants

1. Shallow (`--depth`) cuts **history** length.
2. Sparse checkout cuts **which paths** materialize in the working tree.
3. Partial clone cuts **which blobs** you download until needed (`--filter`).
4. Commit-graph and bitmaps speed `log` and `fetch` on the server and client.

<code>full clone</code>	<code>all commits, trees, blobs</code>
<code>shallow</code>	<code>last N commits</code>
<code>sparse</code>	<code>all objects, few WD paths</code>
<code>partial</code>	<code>delayed blobs</code>

What can break

- Shallow clone then needing `blame` across all history.
- LFS-sized files without LFS.

How to read this chapter

What large means, the three clone cuts, `git config` tuning.

Large Repositories

Large Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What “large” means and common bottlenecks.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Shallow, Sparse, and Partial Clones

Shallow, Sparse, and Partial Clones

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Shallow clones, sparse checkout, and partial clones.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Performance Tuning

Performance Tuning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Performance tuning with `git config`.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Git Security

Theory

Theory: Signatures Bind a Key to a Hash; History Still Contains Secrets

A signed commit proves a key holder produced that object. It does not encrypt the tree. A secret once hashed is in every clone that fetched it.

Invariants

1. SSH or GPG signing covers the commit/tag object.
2. Forges display verified if the key is uploaded.
3. `git filter-repo` rewrites hashes; everyone must reclone.
4. Pre-commit secret scanning is courtesy; rotation is mandatory.

```
commit hash    signed by key K
forge matches K to account
secret blob    still in pack until rewrite + rotate
```

What can break

- Signing with a key the forge does not know.
- Revert of a secret (blob remains).

How to read this chapter

Sign, secrets, audit trails, scanning, lab to remove a secret.

Signed Commits and Tags

Signed Commits and Tags

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Signed commits and tags: GPG and SSH.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Secrets in History

Secrets in History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets in history: removal and prevention.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Audit Trails

Audit Trails

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Audit trails at the Git layer.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Security Scanning

Security Scanning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Security scanning: pre-commit and CI integration.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: Remove a Secret

Hands-On: Remove a Secret

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Remove a committed secret.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Troubleshooting

Theory

Theory: fsck Walks the Graph; Repair is Restore Reachability

Corruption is missing or malformed objects. Dangling objects are normal (amended commits). Repair means getting a reachable graph again — often from another clone.

Invariants

1. `git fsck` reports missing, unreachable, and malformed objects.
2. Another clone of the same DAG is the best object source.
3. Reflog may still name “lost” commits locally.
4. Do not `rm -rf .git` until you have a copy of objects you need.

```
tip → commit → tree → blob    all hashes must exist
missing blob = unreadable file at that path in that commit
```

What can break

- Running random `rm` in `.git/objects`.
- Assuming `fsck` dangling = disaster.

How to read this chapter

`fsck`, error patterns, procedures, corruption lab.

fsck and Repair

fsck and Repair

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Corrupted repositories: `git fsck` and repair.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Common Error Patterns

Common Error Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Common error patterns: causes and fixes.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Recovery Procedures

Recovery Procedures

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recovery procedures.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: Corrupted Repo

Hands-On: Corrupted Repo

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Simulate a corrupted repo and recover.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hosting Platforms

GitHub Monoculture

Theory

Theory: Default is Gravity; Lock-in is Extra Objects Outside Git

GitHub became default by network effects, not by owning the Git protocol. Lock-in lives in identity, Actions YAML, Packages, issue graphs, and marketplace apps — data Git does not clone.

Invariants

1. A Git remote is replaceable; a social graph is not.
2. CI, packages, and issues are forge databases.
3. Multi-homing (two remotes) preserves Git objects only.
4. Regulation and data residency constrain *where* a forge may run.

```
Git objects      portable
PRs, Actions, Packages, stars  forge-specific
```

What can break

- Calling `git clone` a full backup of a GitHub org.

How to read this chapter

How GitHub won, lock-in, geopolitics, what “beyond” means, comparison rubric.

How GitHub Became Default

How GitHub Became Default

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How GitHub became the default.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Risks of Platform Lock-in

Risks of Platform Lock-in

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Risks of platform lock-in.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Regulatory and Geopolitical

Regulatory and Geopolitical

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The regulatory and geopolitical dimension.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

What Beyond GitHub Means

What Beyond GitHub Means

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What “beyond GitHub” actually means.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

A Framework for This Part

A Framework for This Part

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A framework for this part.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Platform Architecture

Theory

Theory: A Forge is Git Plus Metadata, Identity, and Jobs

`git init --bare` is hosting. A platform adds a database, authn/z, HTTP/SSH frontends, object storage, and workers. Operational complexity is the hidden product tax.

Invariants

1. Smart HTTP and SSH both speak the Git protocol; they differ in auth.
2. Repo bytes may live on local disk, NFS, or S3-style object stores.
3. Webhooks are the universal outbound glue.
4. Monolith vs microservices is a scaling and failure-domain choice.

```
clients --SSH/HTTP--> frontend ---> git storage
                                \---> metadata DB
                                \---> jobs / CI / hooks
```

What can break

- Treating GitLab Omnibus RAM as “just Git.”

How to read this chapter

Git vs hosting, components, protocols, storage, identity box, extensibility, styles, cost.

Git Alone vs Git Hosting

Git Alone vs Git Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git alone vs Git hosting.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Core Platform Components

Core Platform Components

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Core platform components.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The Protocol Stack

The Protocol Stack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SSH, HTTP, and Git smart protocols.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Storage Architectures

Storage Architectures

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Filesystem, distributed storage, and object stores.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Authn, Authz, and Identity

Authn, Authz, and Identity

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Authentication, authorization, and identity as a component.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Extensibility

Extensibility

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Webhooks, APIs, plugins.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Monolith vs Microservices

Monolith vs Microservices

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Architectural styles: monolith vs microservices.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Operational Complexity

Operational Complexity

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Operational complexity: the hidden cost.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitLab

Theory

Theory: GitLab Sells a Full SDLC on a (Mostly) Monolith

GitLab's theory is one application for Git, CI, registry, and security. Omnibus is that app plus dependencies as a package. Kubernetes is the same app shredded into services.

Invariants

1. FOSS vs Premium is feature gating on the same codebase.
2. Runners are the horizontal scale knob for CI, not the Rails app.
3. A small team can run CE; HA is a different product shape.

```
Omnibus: one node, many processes
```

```
Geo / K8s: DB, Gitaly, Sidekiq, registry, runners
```

What can break

- GitLab CE on a 2 GB VPS with CI enabled.

How to read this chapter

History, architecture, features, ops/license, lab, when it fits.

History and Philosophy

History and Philosophy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- History and philosophy.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitLab Architecture

GitLab Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Omnibus, monolith, Kubernetes.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitLab Feature Set

GitLab Feature Set

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CI/CD, registry, security, DevOps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Operations and Licensing

Operations and Licensing

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Operational requirements, scaling, FOSS vs Premium.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: GitLab

Hands-On: GitLab

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab CE compose or gitlab.com.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

When GitLab Is the Right Choice

When GitLab Is the Right Choice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When GitLab is the right choice.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gitea and Forgejo

Theory

Theory: A Single Binary is an Architectural Statement

Gitea/Forgejo keep Git hosting in one Go process, SQLite-capable. That is a complexity budget. The Forgejo fork is a governance and license theory, not a protocol fork.

Invariants

1. SQLite + local disk is enough until it is not (locking, size).
2. Actions-compatible CI is optional, not the product core. The runner is a second process.
3. Codeberg is Forgejo-as-a-service with community ops (chapter 29).
4. The Forgejo fork did not change the Git protocol. A remote is still SSH or HTTP.

```
one binary    git repos on disk
              SQLite/Postgres
              optional runners
```

What can break

- Expecting GitLab-class HA from a hobby compose file.
- Treating a current Gitea database as a drop-in for current Forgejo. The hard fork ended that promise.

How to read this chapter

Origins, fork, comparison, scale, two labs, when it fits.

Origins and Philosophy

Origins and Philosophy

After this page you can say what problem a single-binary forge is built to solve, and which project solved it first.

Two forks, one shape

Gogs was a lightweight Git server in Go. Development sat behind one maintainer. In 2016 contributors forked it and called the result **Gitea**: the same idea, with a group that could review and release.

The idea is small on purpose.

```
one Go process
  Git repositories as bare repos on disk
  issues, pull requests, users → one database
  HTTP on port 3000, and SSH for the Git remote
```

Default database is **SQLite**, a file next to the repositories. Postgres and MySQL are configuration, not a requirement. There is no separate Rails app, no Gitaly, no search cluster. Chapter 23 named those boxes. This product leaves most of them empty until you opt in.

What you get on the first afternoon

A Gitea or Forgejo instance gives a small team:

- repositories, with SSH and HTTP remotes (chapter 11)
- issues and pull requests
- a web UI to browse files and diffs
- organizations, teams, and repository permissions
- webhooks, when you want an outside CI to notice a push (chapter 32)

Packages, Actions, and a container registry exist. They are extra processes and extra disks. The core stays the binary plus the data directory.

What this shape refuses to be

GitLab's theory is one application for the whole lifecycle (chapter 24): Git, CI, registry, and security scanning in one product. Gitea's theory is the opposite budget. A desk, a lab, or a small company can run it on a machine that would struggle to boot GitLab Omnibus.

That budget is the point. The moment the database, the Git files, and CI runners all need to scale independently, you have left the afternoon install and entered chapter 36. The binary does not stop you. It also does not hide the work.

The next page is why Forgejo exists as a second project with this same shape.

The Forgejo Fork

The Forgejo Fork

After this page you can explain why Forgejo split from Gitea, and what that split changes for an operator.

Who holds the name

Gitea's code was community-run. In 2022 a for-profit company, Gitea Ltd, took the trademark and the project's direction. An open letter from contributors went unanswered. In October 2022 Codeberg and other operators published **Forgejo** as a fork.

Codeberg e.V., a non-profit in Berlin, holds the Forgejo domains. Forgejo has not registered a trademark. The public forge at codeberg.org runs Forgejo. Chapter 29 is that hosted instance. This chapter is the software you run yourself.

```
Gogs (single maintainer)
  Gitea, 2016           Gitea Ltd, MIT, company edition
    Forgejo, Oct 2022
      soft fork through 2023
      hard fork, early 2024      Codeberg e.V., GPL-3.0-or-later since v9
```

Soft fork, then hard fork

Through 2023 Forgejo was a **soft fork**: it tracked Gitea and stayed a drop-in. Changing the download URL was enough to upgrade a Gitea server onto Forgejo.

On 15 February 2024 the project became a **hard fork**. Stable Forgejo stopped promising to contain all of Gitea. The codebases diverge on purpose, so each side can change without rebasing onto the other. A Gitea database from the 1.22 era is the last generation where people treated the switch as a binary swap. Later pairs need the Forgejo upgrade guide for those two versions. Do not point a new Forgejo binary at a production Gitea data directory and hope.

What actually changed

The Git protocol did not change. Clients, remotes, and hooks from Parts I–III work against either UI.

What changed is governance and the license:

	Gitea	Forgejo
Steward	Gitea Ltd	Codeberg e.V.
License	MIT	MIT through v8; GPL-3.0-or-later from v9
Contributor terms	copyright assignment to the company	no copyright assignment
Paid edition	Enterprise	none
Developed on	GitHub	Forgejo

GPL-3.0-or-later means a modified Forgejo you ship to others comes with source under the same terms. MIT does not require that. Gitea can legally ship a closed Enterprise build on the same code. Forgejo's license is how the project refuses that path.

Day to day, issues, pull requests, and the settings screens still feel like the same product. Actions exist on both (Forgejo Actions and Gitea Actions), with a runner you install separately. Federation between Forgejo instances is a Forgejo project and is not something to design a company on until you have tried it on the version you run.

The next page puts Forgejo, Gitea, and GitLab in one table. The lab pins **Forgejo 15 LTS**, not whatever tag your registry calls latest.

Gitea vs Forgejo vs GitLab

Gitea vs Forgejo vs GitLab

After this page you can choose among the three from the constraint you actually have, not from a feature checklist.

Same afternoon, different contracts

Gitea and Forgejo are one architectural family. GitLab is a different product that also hosts Git. Versions below are the ones this book checked in September 2026.

	Forgejo 15 LTS	Gitea 1.27	GitLab CE
Steward	Codeberg e.V.	Gitea Ltd	GitLab Inc.
License of the server you run	GPL-3.0-or-later	MIT	MIT (CE). Paid tiers are a separate license
Process model	one Go binary	one Go binary	Omnibus: many processes (chapter 24)
Default database	SQLite file	SQLite file	Postgres
CI	Forgejo Actions, external runner	Gitea Actions, external runner	CI in the product
Paid tier	none	Enterprise	Premium and above
Public hosted twin	Codeberg	Gitea.com	GitLab.com
Support window	15.0 LTS through 15 July 2027. 16.0 is the short cycle (through 29 October 2026)	upstream releases	your own upgrade calendar

Forgejo 15.0.9 and 16.0.5 both shipped on 17 September 2026. The lab uses the **15** image tag so a reader is on the LTS line. Track 16 only if you intend to move every quarter.

What is not a difference

Push, fetch, branches, and review comments are Git plus a database row. A developer who learned chapters 8–11 does not relearn them here. The UI names a pull request; Gerrit (chapter 27) names a change. Under both, the bytes are still a branch on a remote.

Webhooks out of either lightweight forge are the same idea as GitLab's: HTTP to your CI when a ref moves. Chapter 32 is that integration. This chapter does not stand up a runner.

How to read the license row

Pick Forgejo when the operator must be able to show that the forge itself stays free software, with a non-profit holding the name and no Enterprise fork of the same tree.

Pick Gitea when the MIT terms are a requirement (you need to ship a closed product derived from the forge) or you are already paying for Gitea Enterprise and the company build is the point.

Pick GitLab when the requirement is CI, a container registry, and security scanning inside one installer, and you can afford the machine that installer expects. A 2 GB lab VM is a Forgejo conversation. It is a bad GitLab conversation.

The next page is how the lightweight shape is deployed, and where SQLite stops being the right database.

Deployment and Scaling

Deployment and Scaling

After this page you can sketch a Forgejo or Gitea layout and name the first limit you will hit.

One directory is the whole server

The container's data directory holds the bare repositories, the database, and the config. On the standard image that directory is `/data`. SQLite, when you leave the database unset, is a file in that tree (`gitea.db`, including on Forgejo: the filename stayed). Config is `app.ini` under the same tree.

<code>./data/</code>	(bind mount → <code>/data</code>)
<code>git/repositories/</code>	bare repos
<code>gitea/gitea.db</code>	SQLite, until you move it
<code>gitea/conf/app.ini</code>	generated on first start

Environment variables of the form `FORGEJO__section__KEY` write `app.ini` for you. Gitea uses the same pattern with the prefix `GITEA__`. A value already stored in `app.ini` is not cleared by unsetting the variable. Edit the file, or start from an empty data directory.

Two images ship:

Image	Data mount	SSH inside the container
<code>codeberg.org/forgejo/forgejo</code>	<code>/data</code>	port 22
<code>codeberg.org/forgejo/forgejo</code>	<code>/var/db/gitea</code>	port 2222

The lab uses the first. Rootless is the same product with a different mount and a different internal SSH port. Mirror `data.forgejo.org` when `codeberg.org` is not reachable from the build machine.

Where it stops being small

SQLite has one writer. A desk, a class, or a company of tens of people is the design center. A busy public forge is not. The move is mechanical:

1. Keep Git bytes on local disk. NFS for the repository directory is a corruption risk unless you already know that filesystem's locking story. Chapter 36 is the storage chapter.

2. Move the database to Postgres when SQLite lock time shows up in the logs. Set `FORGEJO__database__DB_TYPE=postgres` and a host. Do this before the file is large, and take a backup first (chapter 37).
3. Put LFS and package blobs on object storage when the data disk fills with artifacts, not when the Git packs do.
4. Scale CI by adding runners, not by cloning the Forgejo process. Actions are optional. The server is still one writer of its own database.

There is no supported active-active pair of Forgejo nodes over one SQLite file. High availability here means backups, a second disk, and a practiced restore. GitLab's multi-node shape is a different product.

Upgrades

Pin an image tag (`:15` or `:15.0`), not `latest`. Moving from major N to N+1 is a manual upgrade with a backup of `/data` in front of it. Patch releases inside 15.0 replace the container and keep the directory.

Gitea's image is `docker.io/gitea/gitea:1.27`. Same directory layout, same port story, `GITEA__` environment names. Do not run both compose projects against one data directory.

The hands-on page brings Forgejo up, pushes `desk-api`, then shows the Gitea file as the same lab with the image line changed.

Hands-On: Gitea and Forgejo

Hands-On: Gitea and Forgejo

Bring up Forgejo 15 on Linux, push `desk-api`, tear it down, then read the Gitea file as the same lab with one image line changed.

Examples were checked against the Forgejo 15 LTS docs and Gitea 1.27. You need Docker, or Podman with `podman compose`. The commands below say `docker compose`. On Podman, substitute `podman compose`. Do not point this lab at a Forgejo you already use.

Forgejo

Host port 3000 is HTTP. Host port 2222 is Git SSH, so the lab does not take the machine's own `sshd` on port 22. If either host port is already taken, stop. Another forge is probably there.

```
mkdir -p ~/desk/forgejo-lab
cd ~/desk/forgejo-lab
mkdir -p data
cat > compose.yml <<'EOF'
services:
  server:
    image: codeberg.org/forgejo/forgejo:15
    environment:
      USER_UID: "1000"
      USER_GID: "1000"
      FORGEJO__server__DOMAIN: localhost
      FORGEJO__server__ROOT_URL: http://localhost:3000/
      FORGEJO__server__SSH_PORT: "2222"
      FORGEJO__server__HTTP_PORT: "3000"
    restart: unless-stopped
    volumes:
      - ./data:/data
    ports:
      - "3000:3000"
      - "2222:22"
EOF
docker compose up -d
docker compose ps
```

`USER_UID` should be your numeric id (`id -u`) if 1000 is not you, or the data directory will be owned by the wrong account. Wait until the HTTP port answers:

```
curl -fsS -o /dev/null -w '%{http_code}\n' http://127.0.0.1:3000/
```

A 200 or a 303 is enough. Open `http://localhost:3000/` in a browser. The installation page asks for the administrator. Create that account. Leave the database on SQLite. Submit, then sign in.

Create a repository named `desk-api`. Empty is fine. The UI shows two clone URLs. The SSH one should contain `:2222`. If it says port 22, `SSH_PORT` did not land in `app.ini` and the compose environment block needs another look.

Push desk-api

`~/desk/desk-api` is the project from chapter 3. Add this forge as a second remote (chapter 11). Use the HTTP URL from the UI if you have not loaded an SSH key into the forge yet.

```
cd ~/desk/desk-api
git remote add forgejo http://127.0.0.1:3000/ADMIN/desk-api.git
git push -u forgejo main
```

Replace `ADMIN` with the username you just created. The password is the one from the installation page. After the push, the file list in the browser matches `git log --oneline` locally.

SSH, once the UI has your public key:

```
git remote set-url forgejo ssh://git@127.0.0.1:2222/ADMIN/desk-api.git
```

Tear it down

```
cd ~/desk/forgejo-lab
docker compose down
```

`./data` is still on disk. `docker compose down` does not delete that directory. Delete `~/desk/forgejo-lab` when you are finished with the lab. Leave `~/desk/desk-api` in place, and remove the remote if you do not want it:

```
cd ~/desk/desk-api
git remote remove forgejo
```

The Gitea file

Do not start this while the Forgejo container is still bound to 3000 and 2222. Same lab, other project:

```
mkdir -p ~/desk/gitea-lab/data
cd ~/desk/gitea-lab
cat > compose.yml <<'EOF'
services:
  server:
    image: docker.io/gitea/gitea:1.27
    environment:
      USER_UID: "1000"
      USER_GID: "1000"
      GITEA__server__DOMAIN: localhost
      GITEA__server__ROOT_URL: http://localhost:3000/
      GITEA__server__SSH_PORT: "2222"
      GITEA__server__HTTP_PORT: "3000"
    restart: unless-stopped
    volumes:
      - ./data:/data
    ports:
      - "3000:3000"
      - "2222:22"
EOF
```

`docker compose up -d`, the same installation page, a repository, a push, `docker compose down`. The screen is Gitea's. The Git remote is still just a URL.

Done when

- Forgejo's UI showed `desk-api` after `git push`
- the SSH clone URL contained port 2222
- both compose projects are down, and `~/desk/desk-api` still has its history
- you can point at the one line that differs between the two files (the image, and `FORGEJO__` versus `GITEA__`)

When Gitea or Forgejo Fits

When Gitea or Forgejo Fits

After this page you can decide whether this shape is the forge for the job in front of you.

Choose it when

The requirement is Git hosting for a desk, a course, or a company that can name every user. You want issues and pull requests, you can live with CI as a webhook or a separate runner, and you would rather administer one data directory than a platform of services.

Forgejo is the default in this book for that job:

- the steward is a non-profit, and the license from v9 on is GPL-3.0-or-later
- 15.0 is a long-term line (security support through 15 July 2027)
- Codeberg is the same software, operated by someone else, when you do not want a server (chapter 29)

Gitea is the right lightweight forge when MIT is the constraint, or when a support contract with Gitea Ltd is the constraint. The operator experience is the sibling of Forgejo, not a different architecture.

Run it yourself when the repositories must stay on a machine you control. Use Codeberg when the code can live on a public community forge and you only need an account.

Choose something else when

- You need merge trains, a built-in registry, dependency scanning, and GitLab CI on day one. That is chapter 24, and it is a larger machine.
- Review must be a stack of patchsets with a Change-Id. That is Gerrit (chapter 27), not a pull-request UI with a different coat of paint.
- The company standard is already GitHub or GitLab.com and the lock-in discussion in chapter 22 has been accepted. A second forge “for practice” is this lab. A second forge “for production beside the real one” is two metadata stores and one confused team.
- You expect two nodes to share one SQLite file and call it high availability. Stop at the previous page and read chapter 37 before you draw that diagram.

A decision you can write down

small team, self-hosted, free-software forge	→ Forgejo 15 LTS
small team, MIT or a Gitea contract	→ Gitea 1.27
hosted community, no server to patch	→ Codeberg
CI and registry inside the installer	→ GitLab

Whichever line you pick, the developer workflow does not restart. `desk-api` is still a repository, the remote is still a URL, and force-push rules from chapter 11 still apply on `main`. Part V is what changes when you are the operator of that URL for other people: identity, branch protection, and backups. The binary will not do those for you.

Bitbucket

Theory

Theory: Bitbucket is Git Hosting Inside an Issue Database

Atlassian's theory is that code hosting is a satellite of Jira. Cloud vs Data Center is who operates the satellite.

Invariants

1. Jira keys in branch names and commits are the integration surface.
2. Data Center is the self-host shape; Cloud is the SaaS shape.
3. Lock-in is the Atlassian user directory and issue graph.

```
Jira < links > Bitbucket repos
      same org, same identity
```

What can break

- Evaluating Bitbucket without counting Jira.

How to read this chapter

Cloud vs DC, architecture, Jira, enterprise features, comparison, lab, when it fits.

Cloud vs Data Center

Cloud vs Data Center

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bitbucket Cloud vs Bitbucket Data Center.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Bitbucket Architecture

Bitbucket Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Architecture and deployment models.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Jira and Atlassian

Jira and Atlassian

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Jira and the Atlassian ecosystem.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Enterprise Features

Enterprise Features

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Branch permissions, Code Insights, audit logs.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Bitbucket Compared

Bitbucket Compared

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Comparison with GitLab and GitHub.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: Bitbucket Cloud

Hands-On: Bitbucket Cloud

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bitbucket Cloud lab.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When Bitbucket Fits

When Bitbucket Fits

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When Bitbucket fits.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gerrit

Theory

Theory: The Change is the Object, Not the Branch

Gerrit stores a **change** with a Change-Id, patch sets, and an approval predicate. Branches are an output of submit, not the collaboration primitive.

Invariants

1. Each push of the same Change-Id is a new patch set (new Git commits).
2. Submit is a server-side integrate (merge/rebase/cherry-pick) after labels.
3. Android/Chromium scale is why this model exists.

```
refs/for/main → change 123 patch set 3  
labels Code-Review+2 Verified+1 → submit → main
```

What can break

- Treating Gerrit like GitHub PRs (one branch per feature forever).

How to read this chapter

Model, Change-Ids, CI, ops, practice, lab, when it fits.

What Gerrit Is

What Gerrit Is

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The review-centric model.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Change-Ids, Drafts, Approvals

Change-Ids, Drafts, Approvals

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Change-Ids, drafts, and approval chains.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gerrit and CI

Gerrit and CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Integration with Jenkins, CI, and other tools.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gerrit Operations

Gerrit Operations

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Performance and complexity.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gerrit in Practice

Gerrit in Practice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Android, Kubernetes, and other major projects.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: Gerrit

Hands-On: Gerrit

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gerrit compose — one change, one review.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

When Gerrit Is the Right Choice

When Gerrit Is the Right Choice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When Gerrit is the right choice.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

SourceHut

Theory

Theory: Tools in a UNIX Pipeline, Not a Platform Product

SourceHut's theory is that email, git, CI, and tickets can be separate programs with glue. The mailing list is a first-class review transport.

Invariants

1. `git send-email` / patches are the review objects.
2. Each `sr.ht` service is independently hostable.
3. Hook scripts replace a marketplace.

```
git.sr.ht --patch--> lists.sr.ht --job--> builds.sr.ht
```

What can break

- Expecting in-browser PR gadgets as the default.

How to read this chapter

Philosophy, suite, architecture, comparison, lab, when it fits.

SourceHut Philosophy

SourceHut Philosophy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tools, not platforms.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The sr.ht Suite

The sr.ht Suite

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git, issues, mailing lists, CI.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

SourceHut Architecture

SourceHut Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Python, mail-driven workflows, hook scripts.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

SourceHut Compared

SourceHut Compared

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Comparison with feature-rich platforms.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hands-On: SourceHut

Hands-On: SourceHut

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- sourcehut.org or a documented subset self-host.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

When SourceHut Makes Sense

When SourceHut Makes Sense

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When SourceHut makes sense.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Niche Platforms

Theory

Theory: Niche is a Different Default, Not a Toy

Codeberg, Gogs, GitBucket, Phorge, GHES, and git-without-a-UI occupy corners: community ops, ancestry, JVM, review heritage, on-prem GitHub, or no web forge at all.

Invariants

1. GHES is still GitHub lock-in, on your power bill.
2. Gogs is the ancestor story for Gitea.
3. A forge-less host (gitolite, Soft Serve) is still Git hosting.

```
UI-heavy  GitLab ... Gitea ... cgit/gitolite  UI-minimal
```

What can break

- Picking a niche forge for features it does not claim.

How to read this chapter

Codeberg, GitBucket, Gogs, Phorge, GHES, summary table.

Codeberg

Codeberg

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Codeberg and community-run instances.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitBucket

GitBucket

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitBucket and GitHub-compatible alternatives.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gogs

Gogs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gogs: the ancestor of Gitea.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Phorge and Arcanist

Phorge and Arcanist

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Phabricator / Phorge and Arcanist.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitHub Enterprise Server

GitHub Enterprise Server

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Self-hosted GitHub Enterprise Server.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Summary of Niche Platforms

Summary of Niche Platforms

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Summary of niche platforms.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Operating a Platform

Identity

Theory

Theory: Authn is Who; Authz is What a Ref May Do

A forge maps credentials to a principal, then to permissions on repos, orgs, and refs. SSH keys, OAuth, SAML, and LDAP are different *proofs* of the same principal problem.

Invariants

1. Password/SSH/OAuth/SAML all produce a session or git credential.
2. RBAC is groups $\times$ resources $\times$ verbs (clone, push, admin, force).
3. Fine-grained tokens are more blast-radius-aware than a single password.
4. MFA protects the *account*, not a stolen deploy key unless you require it.

```
proof (key/token/SAML) → principal → role → refs + APIs
```

What can break

- Deploy keys with write access in CI forever.

How to read this chapter

Methods, LDAP, RBAC, keys at scale, MFA.

Password, SSH, OAuth, SAML

Password, SSH, OAuth, SAML

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Password, SSH keys, OAuth, SAML.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

LDAP and Directory Services

LDAP and Directory Services

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- LDAP, Active Directory, and enterprise identity.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

RBAC and Fine-Grained Permissions

RBAC and Fine-Grained Permissions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- RBAC and fine-grained permissions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

SSH Keys at Scale

SSH Keys at Scale

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SSH key management at scale.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

MFA and Hardening

MFA and Hardening

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Multi-factor authentication and hardening.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Governance

Theory

Theory: Governance is a Server-Side Predicate on Refs

Branch protection, CODEOWNERS, and required checks are hooks and config that reject a push or a merge button. The Git objects may be fine; the *name movement* is denied.

Invariants

1. Developer workflow (ch 15) is advice; this chapter is enforcement.
2. CODEOWNERS is path → reviewer, evaluated on the proposed tree.
3. Monorepo governance is path-scoped; polyrepo is repo-scoped.

```
push/merge request
  → protected branch?
  → required signatures/checks/owners?
  → accept or reject
```

What can break

- Required reviews with no CODEOWNERS on a 200-person monorepo.

How to read this chapter

Enforcement, protection, mono/poly operator view, automation.

Enforcement Implications

Enforcement Implications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Branching models revisited: enforcement implications.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Protecting Branches

Protecting Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Protecting branches; code owners and review requirements.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Monorepo vs Polyrepo (Operator)

Monorepo vs Polyrepo (Operator)

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Monorepo vs polyrepo trade-offs (operator view).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Governance Automation

Governance Automation

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Conventional commits, structured history, and governance automation.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

CI/CD

Theory

Theory: CI is a Function from Git Events to Jobs

A webhook or native parser turns a ref movement into a graph of jobs. Runners are the computers. GitOps later inverts this: the cluster pulls desired state from Git.

Invariants

1. Webhooks are the universal outbound event.
2. Native CI (Actions, GitLab CI, Gitea Actions) stores YAML in Git.
3. External CI (Jenkins, Drone) stores jobs elsewhere, triggered by Git.
4. Runners expand horizontally; the control plane does not always.

```
push/PR → event → pipeline YAML → queue → runner → status back to forge
```

What can break

- Untrusted PRs on privileged runners (injection).

How to read this chapter

Webhooks, native, external, runners, GitOps preview.

Webhooks

Webhooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Webhooks: the universal glue.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Native CI

Native CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab CI, GitHub Actions, Gitea Actions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

External CI

External CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Jenkins, Drone, Cirrus, Buildkite.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Pipelines and Runners

Pipelines and Runners

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pipeline architecture, runners, executors, and build infrastructure.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitOps Preview

GitOps Preview

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git as the source of truth (full treatment in Part VII).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Packages

Theory

Theory: A Registry is an Object Store with Language Semantics

Container images and language packages are not Git blobs. They need retention, GC, and authn that match the forge's principals.

Invariants

1. A digest (sha256) is the real name of an image; a tag is a moving ref.
2. Retention is a cost and a supply-chain control.
3. CI should push by digest; humans talk in tags.

```
git tag v1.2.0          oci image registry.example/app@sha256:...  
both are names over content-addressed bytes
```

What can break

- Immutable tags that were mutated (`latest`).

How to read this chapter

Registries, language managers, retention, self-host architecture.

Container Registries

Container Registries

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab Registry, Harbor, and alternatives.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Language Package Managers

Language Package Managers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Language-specific package managers.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Retention and CI Integration

Retention and CI Integration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Artifact storage, retention policies, and CI integration.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Self-Hosted Registry Architecture

Self-Hosted Registry Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Self-hosted registry architecture.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Security

Theory

Theory: Supply Chain is Trust in What You Did Not Write

Git signatures, SBOMs, dependency scans, and secret detection are layers on *provenance*. A platform hardens the perimeter; it cannot see an unsigned `curl | bash` in a Dockerfile unless you look.

Invariants

1. Verify on the server what ch 20 taught on the client.
2. SBOM is a graph of what you shipped, not a scanner by itself.
3. A supply-chain incident is usually a trusted path (CI, dep, maintainer).

```
source (signed) → CI (OIDC) → registry (digest) → runtime
                SBOM + scan attach evidence to this path
```

What can break

- Signing commits but not pinning Actions by digest.

How to read this chapter

Server-side git security, scanning, SBOM, secrets, incidents, hardening.

Signed Commits on the Server

Signed Commits on the Server

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Signed commits, tags, verification on the platform.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Dependency Scanning

Dependency Scanning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Dependency scanning and vulnerability management.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

SBOM

SBOM

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Software Bill of Materials (SBOM).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Secrets Detection

Secrets Detection

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets detection and management.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Supply Chain Attacks

Supply Chain Attacks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Lessons from real incidents.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Platform-Level Hardening

Platform-Level Hardening

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Platform-level security hardening.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Federation

Theory

Theory: Git is Already Distributed; Forges are Not

Every clone is a replica of objects. Federation of *issues and PRs* needs extra protocols (ForgeFed, ActivityPub) or boring mirroring.

Invariants

1. Mirroring is practical federation of Git objects.
2. Cross-instance PRs are still experimental.
3. Radicle and similar try to drop the forge as a single host.

```
clone A ↔ clone B      (Git, solved)
forge A  ?  forge B      (issues/PRs, unsolved at scale)
```

What can break

- Assuming Codeberg and GitHub issues sync because remotes do.

How to read this chapter

Git as primitive, federated hosting, readiness.

Git as a Decentralization Primitive

Git as a Decentralization Primitive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git's native distribution as a decentralization primitive.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Federated Hosting

Federated Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Federated hosting and cross-instance collaboration.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Are We Ready?

Are We Ready?

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Are we ready for decentralized code hosting?

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Storage and Scale

Theory

Theory: Git Storage is Packs Plus a Database of Metadata

Repo growth hurts fetch time, gc, and inode counts. The forge DB (users, issues) scales on a different axis than packfiles.

Invariants

1. Local disk vs NFS vs Ceph vs S3 changes latency of `git fetch`.
2. Gitaly-style git servers shard repos, not tables.
3. CDNs help clone of *public* objects; auth still hits origin.

```
packfiles / alternates / object storage
metadata DB (issues, users) separate scale plan
```

What can break

- NFS for lots of tiny Git objects without cache.

How to read this chapter

Growth, storage options, fetch, DB, CDN/geo, benchmarks.

Repository Growth

Repository Growth

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When size becomes a problem.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Storage Options

Storage Options

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Local, NFS, Ceph, S3-compatible.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Clone and Fetch Performance

Clone and Fetch Performance

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Cloning and fetch performance.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Database Scaling

Database Scaling

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Database scaling for Git platforms.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

CDN and Geo-Distributed Hosting

CDN and Geo-Distributed Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CDN strategies and geo-distributed hosting.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Benchmarks

Benchmarks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Benchmarks and performance profiles.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

HA and DR

Theory

Theory: Backup is Another Clone Plus Metadata Dump

Git objects backup with `git clone --mirror`. Issues, PRs, packages, and CI logs need the forge's dump. HA is keeping both classes online.

Invariants

1. A mirror clone is not a point-in-time of the issue tracker.
2. Geo-redundant Git needs object replication *and* DB replication.
3. DR tests are restores, not only snapshots.

```
objects --mirror--> cold clone
postgres dump --> object store
both required for a forge restore
```

What can break

- Backing up `/data/git` and forgetting the DB.

How to read this chapter

Backup, PITR, geo, DR plan, repo health alerts.

Backup Strategies

Backup Strategies

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bare clones, dumps, and replication.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Point-in-Time Recovery

Point-in-Time Recovery

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Point-in-time recovery and versioning.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Geo-Redundant Deployments

Geo-Redundant Deployments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Geo-redundant and multi-region deployments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Disaster Recovery Planning

Disaster Recovery Planning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Disaster recovery planning.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Monitoring Repository Health

Monitoring Repository Health

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Monitoring and alerting for repository health.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Observability

Theory

Theory: If You Cannot See Ref Movement, You Cannot Audit

Operational metrics (latency, error rate, disk) keep the forge up. Audit logs keep it honest. They are different streams.

Invariants

1. Golden signals for a git frontend: fetch/push latency, 5xx, queue depth.
2. Audit: who moved which ref, who changed access.
3. SOC 2 / ISO / GDPR ask for *retention and access* of those logs.

```
metrics (SRE)      logs (debug)      audit (compliance)
all three, different retention
```

What can break

- DEBUG application logs as your only audit trail.

How to read this chapter

Metrics, logs, audit, tracing, frameworks, detection.

Operational Metrics

Operational Metrics

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What to measure and why.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Log Aggregation

Log Aggregation

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Log aggregation and analysis.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Audit Logging

Audit Logging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Audit logging for compliance.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Tracing User Actions

Tracing User Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tracing user actions across the platform.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Compliance Frameworks

Compliance Frameworks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SOC 2, ISO 27001, GDPR.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Security Monitoring

Security Monitoring

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Security monitoring and incident response.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Enterprise and Air-Gap

Theory

Theory: Air-Gap is a Closed Universe of Bytes

Offline forges must import Git, packages, container images, and CI images as **files**. TCO includes humans who carry those files.

Invariants

1. `git bundle` and registry exports are the transport.
2. Data residency is a legal constraint on *where disks live*.
3. SLAs are people and spare capacity, not a YAML flag.

```
connected world --bundle/image--> sneakernet --> air-gapped forge
```

What can break

- CI that `docker` pulls from the internet on an air-gapped runner.

How to read this chapter

Air-gap, hybrid, residency, support/SLA, TCO.

Air-Gapped Deployments

Air-Gapped Deployments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Air-gapped and offline deployments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Multi-Cloud and Hybrid

Multi-Cloud and Hybrid

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Multi-cloud and hybrid architectures.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Data Residency

Data Residency

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Compliance and data residency.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Support and SLAs

Support and SLAs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Enterprise support and SLAs.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Capacity and TCO

Capacity and TCO

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Capacity planning and TCO analysis.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Migration and Future

Theory

Theory: Migration is Mapping Names, Then Objects, Then Metadata

Git history is the easy part (`push --mirror`). Users, PRs, CI, and packages are lossy maps. A decision framework is which losses you accept.

Invariants

1. Object hashes can stay; URLs and issue numbers usually cannot.
2. Interoperability is shared Git + webhooks, not identical UIs.
3. AI features will attach to whatever forge has the social graph.

Git objects	lossless	new forge
PRs/issues	best effort	importer
Actions YAML	rewrite	new CI

What can break

- Promising zero-loss PR comments without reading the importer docs.

How to read this chapter

Strategies, preserving history, interoperability, trends, decision, future.

Migration Strategies

Migration Strategies

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tools and processes.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Preserving History

Preserving History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Preserving history, permissions, and metadata.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Interoperability

Interoperability

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Interoperability between platforms.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Emerging Trends

Emerging Trends

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AI-assisted development, collaboration, ecosystem evolution.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

A Decision Framework

A Decision Framework

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Making your choice: a decision framework.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The Future of Code Hosting

The Future of Code Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The future of code hosting.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitHub Actions

Fundamentals

Theory

Theory: A Workflow is a Graph of Jobs Triggered by Git Events

Actions YAML in Git is both source and config. An event (push, PR, schedule) instantiates jobs, each a sequence of steps on a runner.

Invariants

1. Jobs in a workflow may run in parallel unless **needs** says otherwise.
2. The GITHUB_TOKEN is a short-lived credential scoped to the repo/job.
3. Contexts (**github**, **secrets**, **matrix**) are the runtime environment.

```
event → workflow → jobs → steps → runner VM/container
```

What can break

- Unpinned third-party actions on **main**.

How to read this chapter

Why Actions, anatomy, runtime, authoring/debug, build/validate.

Automate Tasks with Actions

Automate Tasks with Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Automate tasks in GitHub with Actions.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Workflow Anatomy

Workflow Anatomy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Events, jobs, steps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Runtime and Contexts

Runtime and Contexts

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The workflow runtime and environment contexts.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Authoring and Debugging

Authoring and Debugging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Authoring, debugging, and validating workflows.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Building and Validating Code

Building and Validating Code

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Building and validating your code.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Custom Actions and Releases

Theory

Theory: Composite, JS, and Container Actions are Functions

A custom action is a reusable function with inputs/outputs. Releases are Git tags plus bits in GitHub Packages or attached files.

Invariants

1. Pin by commit SHA, not by moving tag, for third-party actions.
2. OIDC lets jobs prove identity to clouds without long-lived keys.
3. Matrix is cartesian expansion of a job.

```
uses: org/action@<sha>      pinned
uses: org/action@v3          moving tag  (weaker)
```

What can break

- `pull_request_target` with untrusted checkout.

How to read this chapter

Building actions, reuse/matrix/cache, releasing, security notes.

Building Custom Actions

Building Custom Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Building custom GitHub Actions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Reuse, Matrix, Cache, Secrets

Reuse, Matrix, Cache, Secrets

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Reuse, matrix, cache, artifacts, secrets, and environments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Releasing Software

Releasing Software

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Releasing software with Actions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Actions Security Notes

Actions Security Notes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pinning, GITHUB_TOKEN, OIDC (full supply-chain is chapter 34).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitOps

What Is GitOps

Theory

Theory: Desired State Lives in Git; the Cluster Converges

GitOps's four pillars: Git as source of truth, declarative desired state, pull-based agents, continuous reconciliation. Push-based CI (`kubectl apply` from a pipeline) is CIOps, not GitOps.

Invariants

1. If it is not in Git, it is drift.
2. The agent, not the human, writes to the cluster API.
3. Preview environments are extra desired states (extra Git refs or folders).

```
Git desired --pull--> reconciler --apply--> cluster actual
                        compare + converge
```

What can break

- Manual `kubectl` on a GitOps cluster (the next reconcile fights you).

How to read this chapter

Pain, pillars, benefits, push vs pull, env repos, trunk vs PRs, vs other *Ops, when.

DevOps Pain Points

DevOps Pain Points

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Manual kubectl, script proliferation, security gaps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The Four Pillars of GitOps

The Four Pillars of GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The four pillars of GitOps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Deploy Faster, Recover Faster

Deploy Faster, Recover Faster

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Deploy faster, recover faster, secure deployments, self-documenting systems.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Push vs Pull

Push vs Pull

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Push-based vs pull-based deployments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Environment Repositories

Environment Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Environment repositories; multiple apps and environments; preview environments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Trunk vs Pull Requests

Trunk vs Pull Requests

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Trunk-based development vs pull requests in a GitOps world.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

GitOps vs Other Ops

GitOps vs Other Ops

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitOps vs DevOps, CIOps, NoOps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

When to Use GitOps

When to Use GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When to use GitOps vs traditional CI/CD; common misconceptions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Flux Quickstart

Theory

Theory: Flux is a Set of Controllers that Watch Git

Flux Operator installs controllers that reconcile GitRepository, Kustomization, HelmRelease objects. The cluster is the actor.

Invariants

1. App repo vs env repo is a naming convention for desired state.
2. A GitHub Action may *update a tag in Git*; Flux *applies* it.
3. This book's deep GitOps path continues with Argo CD (different UX, same theory).

```
Actions: build image, commit tag
Flux:    see tag, helm upgrade / apply
```

What can break

- Storing cluster secrets as plain YAML in a public repo.

How to read this chapter

Repos, CD with Actions, Flux Operator, why Argo is next.

Application and Environment Repos

Application and Environment Repos

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Application and environment repositories.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

CD Pipeline with Actions

CD Pipeline with Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Continuous delivery pipeline with GitHub Actions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Flux Operator on Kubernetes

Flux Operator on Kubernetes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Install and pull-based deployment.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The Future of GitOps

The Future of GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The future of GitOps (and why Part VIII is Argo CD).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Argo CD

Concepts and Architecture

Theory

Theory: Reconciliation is Compare Desired to Actual, Forever

Argo CD's Application is a pointer at Git (desired) plus a cluster (actual). The controller loop diffs and optionally syncs. Immutable infrastructure: replace, do not SSH-mutate.

Invariants

1. Five-step loop: detect, compare, present, sync (if policy), record.
2. Repo server renders manifests (Helm/Kustomize/plain) in a sandbox.
3. Sync status and health are different questions.

```
Git → repo-server → desired manifests  
cluster API → actual  
controller: diff + sync policy
```

What can break

- Editing live resources that auto-sync will revert.

How to read this chapter

History, reconciliation, loop, components, apps/projects, path to cluster, desired vs actual.

Where Argo CD Came From

Where Argo CD Came From

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Where Argo CD came from, and why that gap existed.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Reconciliation and Immutable Infrastructure

Reconciliation and Immutable Infrastructure

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What Argo CD actually does.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The Reconciliation Loop

The Reconciliation Loop

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The five-step reconciliation loop; declare outcomes, not steps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Argo CD Components

Argo CD Components

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- API server, application controller, repo server, Redis, Dex, ApplicationSet controller, Web UI, CLI.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Applications and Projects

Applications and Projects

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Applications and projects; how Git becomes manifests.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

From Git to the Cluster

From Git to the Cluster

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How a change travels from Git to the cluster.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Desired vs Actual State

Desired vs Actual State

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Desired vs actual state, sync status, and sync history.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Install and Lab

Theory

Theory: The Control Plane is Just Another Workload

Argo CD runs in the cluster (or a management cluster). Install flavor (full, core, Helm, Kustomize) is which components you accept.

Invariants

1. A management cluster can drive many workload clusters.
2. The CLI and UI are API clients; RBAC applies to both.
3. kind/K3s on Linux is enough to learn the loop.

```
laptop kubectl → k3s → argo-cd namespace → in-cluster (or external) apps
```

What can break

- Exposing the Argo UI to the internet with admin:admin.

How to read this chapter

Lab shape, k3s, install methods, UI/CLI/clusters.

Lab Shape

Lab Shape

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Two servers, a cluster, and an app to deploy.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

K3s and kubectl

K3s and kubectl

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Installing K3s (or kind); kubeconfig; kubectl.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Install Methods

Install Methods

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Full, minimal (core), custom, Helm, Kustomize.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

UI, CLI, and Clusters

UI, CLI, and Clusters

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Web UI, CLI, token login, adding clusters.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Applications and Sync

Theory

Theory: Sync Policy is When the Loop May Write

Manual sync is a human-gated apply. Auto-sync is the loop applying whenever Git moves. Self-heal reapplies when the cluster drifts. Prune deletes objects gone from Git.

Invariants

1. Every field on `spec.syncPolicy` is a permission the controller has.
2. Server-side apply vs client-side merge is a conflict-management choice.
3. History limits bound how much old sync state you keep.

```
Git move → (auto-sync?) → apply → (prune?) → health
```

What can break

- Auto-prune on a shared namespace you do not fully own.

How to read this chapter

First app, policy, prune, retries/options, namespace/labels, recap.

Your First Application

Your First Application

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CLI vs declarative manifests.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Sync Policy

Sync Policy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Auto-sync, self-healing, jitter, cluster drift.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Pruning

Pruning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pruning, exceptions, propagation policy, and garbage collection.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Retries and Sync Options

Retries and Sync Options

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Retries, backoff, ignore differences, selective sync, replace vs merge, server-side apply.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Namespace, Labels, History

Namespace, Labels, History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Namespace automation, labels, history limits.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Recap Exercise

Recap Exercise

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recap exercise.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Clusters and Diff

Theory

Theory: Diff Algorithms Decide What “Out of Sync” Means

Argo compares live Kubernetes objects to rendered Git. Ignore differences, jq paths, and server-side apply all change that predicate.

Invariants

1. Webhook vs polling is how fast Git is noticed, not how sync works.
2. Health is controllers’ opinion (Deployment available), not Git.
3. External clusters need credentials in Argo’s DB/secret store.

```
desired YAML vs live JSON → sync status
live conditions           → health status
```

What can break

- Ignoring /spec/replicas while HPA owns replicas (or vice versa).

How to read this chapter

Registration, poll vs webhook, diffs, health.

Cluster Registration

Cluster Registration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- In-cluster vs external clusters; registration and discovery.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Polling vs Webhooks

Polling vs Webhooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git polling vs webhook-based synchronization.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Diff Algorithms

Diff Algorithms

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Desired vs actual state: diff algorithms and change detection.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Application Health

Application Health

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Application health and monitoring.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hooks and Waves

Theory

Theory: A Sync is a Phased Transaction with Optional Jobs

PreSync/Sync/PostSync hooks are Kubernetes Jobs/Pods Argo runs around apply. Waves order ordinary resources. A failed hook can fail the sync.

Invariants

1. `generateName` lets hooks rerun; a fixed `name` plus `BeforeHookCreation` is a trap.
2. Waves are integers on annotations, not a separate API.
3. Hook cleanup policy is part of the theory (delete on success/failure).

```
PreSync jobs → apply waves 0..N → PostSync jobs
```

What can break

- A hook that never completes (no backoff awareness).

How to read this chapter

Phases, names/waves, failures, anti-patterns.

Sync Phases

Sync Phases

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How a sync runs in phases; a first PostSync smoke test.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Names and Sync Waves

Names and Sync Waves

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `generateName` vs `name`; hook cleanup; ordering with sync waves.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

When a Hook Fails

When a Hook Fails

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What happens when a hook fails.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Hook Anti-Patterns

Hook Anti-Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Hook anti-patterns and the `BeforeHookCreation` trap.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Users and RBAC

Theory

Theory: Local Users are Passwords; RBAC is Casbin Strings

Creating a login and granting a role are two operations. **p** lines are permissions; **g** lines are group membership. Projects scope which repos and clusters an app may use.

Invariants

1. SSO (Dex) maps external identity into the same RBAC.
2. Project is a hard sandbox (source repos, destinations).
3. CLI tokens are users too.

```
account + p, g lines + project = what you may sync
```

What can break

- Admin in `policy.csv` and SSO both on, conflicting.

How to read this chapter

Creating a user, p/g, verifying grants.

Creating a User

Creating a User

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Creating a user is two unrelated operations.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

The p Line and the g Line

The p Line and the g Line

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Accounts; the p line and the g line; project scoping.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Verifying Grants

Verifying Grants

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Verifying grants.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Helm

Theory

Theory: Helm is a Function from Chart+Values to YAML

Argo renders Helm in the repo server. Helm hooks are not Argo hooks. Values can come from Git, multiple sources, or extras.

Invariants

1. The rendered YAML is what is synced, not the chart tarball as a black box (unless you choose that).
2. Versioning the chart *and* the values is two Git objects.
3. Helm hook annotations can surprise Argo if mixed carelessly.

```
chart + values.yaml --helm template--> desired YAML --sync--> cluster
```

What can break

- Helm hooks that create Jobs Argo then prunes.

How to read this chapter

Helm vs plain, charts/repos, deploy/values, hook comparison.

Helm vs Plain Resources

Helm vs Plain Resources

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Helm vs plain Kubernetes resources.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Charts and Repositories

Charts and Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Creating, packaging, and versioning charts; Helm repositories.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Deploying Charts and Values

Deploying Charts and Values

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Deploying and upgrading charts; overriding values; multiple sources.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Helm Hooks vs Argo CD Hooks

Helm Hooks vs Argo CD Hooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Helm hooks vs Argo CD hooks.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

ApplicationSets and Patterns

Theory

Theory: ApplicationSets are Generators of Applications

An ApplicationSet template plus a generator (list, git, matrix, PR, cluster) produces many Application objects. App-of-apps is the same idea with a parent Application.

Invariants

1. Generators must be deterministic enough to converge.
2. One Application per cluster/tenant/branch is a naming policy.
3. Build→registry→bump tag in Git is the promotion loop.

```
generator (list of clusters) × template → Application objects
parent app → child apps                (app of apps)
```

What can break

- A generator that creates and destroys apps on every reconcile (flap).

How to read this chapter

ApplicationSets, app-of-apps, multi-branch, build-to-update.

ApplicationSets

ApplicationSets

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The problem ApplicationSets solve; list generator and beyond.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

App of Apps

App of Apps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The app-of-apps pattern; repository patterns for GitOps.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Multi-Branch CI/CD

Multi-Branch CI/CD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- One application per Git branch.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Build to Application Update

Build to Application Update

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Build → registry → Helm package → application update.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Notifications, Secrets, and Blue-Green

Theory

Theory: Notifications Observe the Loop; Secrets Must Not Live as Plain Desired State

Slack fires on Application events. Secrets need an external system (Sealed Secrets, SOPS, ESO) so Git stays the *index* of secret *references*. Blue-green at the Argo CD layer is usually two apps or a Rollouts object.

Invariants

1. Notification config is itself desired state (or a cluster controller).
2. Sealed/SOPS ciphertext can live in Git; plaintext must not.
3. Blue-green without Rollouts is two Destinations or two services.

```
Application event → notification controller → Slack
Git ciphertext    → controller                → cluster Secret
```

What can break

- Base64 secrets in Git called 'encryption'.

How to read this chapter

Slack, secret strategies, blue-green.

Slack Notifications

Slack Notifications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Slack notifications for sync events.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Secret Management

Secret Management

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secret management strategies.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Blue-Green with Argo CD

Blue-Green with Argo CD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Blue-green deployments with Argo CD.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Argo vs Alternatives

Theory

Theory: Engines Differ in UX and Ecosystem, Not in Pull-vs-Push Physics

Flux, Argo CD, Jenkins X, Spinnaker, Tekton occupy different UI and CD-vs-CI mixes. The GitOps physics (reconcile desired from Git) is shared with Flux.

Invariants

1. Argo CD is application-centric UI + multi-cluster.
2. Flux is Kubernetes-API-centric, GitOps Toolkit CRDs.
3. Spinnaker is pipeline-centric, heavier.

CI (build)	CD (converge cluster)
Tekton/Jenkins	Argo CD / Flux / Spinnaker

What can break

- Running two reconcilers on the same objects.

How to read this chapter

Comparisons, choosing an engine.

Argo CD vs the Alternatives

Argo CD vs the Alternatives

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Argo CD vs Flux, Tekton, Jenkins X, Spinnaker.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Choosing a GitOps Engine

Choosing a GitOps Engine

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Choosing your GitOps engine.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Argo Rollouts

Introducing Rollouts

Theory

Theory: Progressive Delivery is Reducing Blast Radius Over Time

Kubernetes RollingUpdate is maxUnavailable/maxSurge, not traffic-aware. Canary and blue-green split *traffic* or *replicas* and add pause/analysis gates.

Invariants

1. A Rollout replaces a Deployment as the object that owns ReplicaSets.
2. Canary: some users (or pods) see the new version first.
3. Blue-green: two full stacks; cut traffic when ready.

RollingUpdate	gradual pods, mixed traffic
Canary	gated traffic/replicas
Blue-green	two stacks, one cutover

What can break

- Calling a 50% RollingUpdate a canary (no analysis, no traffic split).

How to read this chapter

Beyond RollingUpdate, canary/blue-green, risk.

Beyond RollingUpdate

Beyond RollingUpdate

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Where Kubernetes RollingUpdate stops and progressive delivery begins.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Canary and Blue-Green Fundamentals

Canary and Blue-Green Fundamentals

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary and blue-green fundamentals.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Gradual Releases and Risk

Gradual Releases and Risk

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gradual releases and deployment risk.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Resources and Lifecycle

Theory

Theory: A Rollout Owns Stable and Canary ReplicaSets

The controller creates ReplicaSets labeled stable/canary, shifts weights or replica counts through steps, and consults AnalysisTemplates.

Invariants

1. Experiments are extra ReplicaSets for isolated tests.
2. AnalysisRun is an instance of an AnalysisTemplate.
3. Abort returns traffic/replicas to stable.

```
Rollout → RS stable  (old)
        → RS canary  (new)
        → AnalysisRun(s)
```

What can break

- Editing the underlying Deployment that Rollout is supposed to own.

How to read this chapter

Rollout vs Deployment, stable/canary, analysis/experiments.

Rollout vs Deployment

Rollout vs Deployment

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Rollout vs Deployment; the controller; ReplicaSets.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Stable and Canary Versions

Stable and Canary Versions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stable and canary versions; the rollout lifecycle.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Analysis Templates and Experiments

AnalysisTemplates and Experiments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AnalysisTemplates, experiments, and resource relationships.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Rollouts Lab

Theory

Theory: The Lab is a Tiny Cluster that Still Reconciles from Git

kind/k3s + Argo CD + Rollouts on Linux is enough to see steps, pauses, and aborts. Git remains desired state.

Invariants

1. A Service must select the right pods for replica-based canaries.
2. Traffic-based canaries need a mesh (Istio) in later chapters.
3. Verify with `kubectl` and the Rollout status, not only Git.

```
Git → Argo CD → Rollout CR → ReplicaSets + Services
```

What can break

- Applying Rollout YAML by hand and via Argo to the same object.

How to read this chapter

Stack, manifests, verify.

Rollouts Lab Stack

Rollouts Lab Stack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Docker, kind, Argo CD, Argo Rollouts.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Manifests and Verify

Manifests and Verify

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Services, rollout manifests, GitOps deployment, verifying state.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

First Canary

Theory

Theory: A Step is a Promise About Traffic or Replicas, Then a Pause

Canary steps are a state machine: `setWeight/setCanaryScale`, `pause`, `analysis`. Promotion advances; abort rewinds to stable.

Invariants

1. Pause without analysis is a human gate.
2. Inspection (`kubectl argo rollouts get`) reads that state machine.
3. Replica-based weight approximates traffic if pods are homogeneous.

```
setWeight 20 → pause → setWeight 50 → analysis → setWeight 100
```

What can break

- No pause and no analysis (just a slow `RollingUpdate`).

How to read this chapter

Steps, `promote/abort/inspect`.

Canary Strategy and Steps

Canary Strategy and Steps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary strategy and steps; traffic progression and pauses.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Promote, Abort, Inspect

Promote, Abort, Inspect

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Promotion, abortion, and rollout inspection.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Istio Canaries

Theory

Theory: Traffic Split is a Mesh API, Not a Replica Count

Istio VirtualService weights send a percentage of requests to canary subsets. Replica count can stay equal while traffic is 1%.

Invariants

1. DestinationRule subsets must match pod labels.
2. Replica-based routing request-based routing under uneven load.
3. `setCanaryScale` exists because HPA + replica canaries fight (later chapter).

```
Gateway → VirtualService  90% stable subset  
                        10% canary subset
```

What can break

- Subsets that do not match Rollout's pod labels.

How to read this chapter

Istio objects, weights, replica vs traffic.

Istio Traffic Management

Istio Traffic Management

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gateway, VirtualService, DestinationRule.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Subsets and Weighted Traffic

Subsets and Weighted Traffic

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stable and canary subsets; weighted canary traffic.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Replica-Based vs Traffic-Based

Replica-Based vs Traffic-Based

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Replica-based vs traffic-based routing.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Analysis and Prometheus

Theory

Theory: Analysis is a Predicate on Metrics Over a Window

AnalysisTemplate defines queries and success/failure conditions. Inline analysis blocks a step; background analysis can abort later.

Invariants

1. Prometheus is an oracle, not the rollout controller.
2. Failure conditions should be monotonic enough to trust.
3. Automatic abort is the point; otherwise it is a dashboard.

```
step → AnalysisRun → Prom query → success | fail → next | abort
```

What can break

- Success condition that is true on no data.

How to read this chapter

Templates, conditions, inline vs background.

AnalysisTemplate and AnalysisRun

AnalysisTemplate and AnalysisRun

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AnalysisTemplate and AnalysisRun; Prometheus queries.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Success, Failure, Aborts

Success, Failure, Aborts

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Success and failure conditions; automatic aborts.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Inline vs Background Analysis

Inline vs Background Analysis

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Inline vs background analysis.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Experiments

Theory

Theory: An Experiment is a Bounded Extra Version

Experiments run a template for a duration, optionally with analysis, without becoming stable. They are for A/B or soak tests beside the main canary.

Invariants

1. Failed experiments should not promote.
2. They consume cluster capacity (extra pods).
3. They are not a substitute for canary steps.

```
stable + canary (rollout) + experiment pods (temporary)
```

What can break

- Leaving experiments running (cost, drift).

How to read this chapter

Side by side, analysis, vs AnalysisTemplates.

Running Versions Side by Side

Running Versions Side by Side

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Running versions side by side; experiment templates.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Analysis During Experiments

Analysis During Experiments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Analysis during experiments; lifecycle and failed experiments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Experiments vs Analysis Templates

Experiments vs AnalysisTemplates

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Experiments vs AnalysisTemplates.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

HPA and Canary

Theory

Theory: HPA Owns Replicas; Canary Steps Also Own Replicas

Two controllers cannot both own `spec.replicas` without a rule. `setCanaryScale` and traffic-based splits exist to stop HPA from shrinking the canary out of existence.

Invariants

1. Replica-based canary + HPA on the Rollout is the subtle bug.
2. Traffic-based canary lets HPA scale the whole set more safely.
3. Metrics-server must exist for HPA to do anything.

```
HPA --replicas--> Rollout <-- canary step replica math
```

What can break

- HPA on both stable and canary Deployments independently.

How to read this chapter

The problem, metrics, `setCanaryScale`.

HPA with Replica-Based Canaries

HPA with Replica-Based Canaries

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A subtle problem: HPA with replica-based canaries.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Metrics Server and Canary Scaling

Metrics Server and Canary Scaling

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Metrics server, CPU load, and canary scaling.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

setCanaryScale

setCanaryScale

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Istio routing and `setCanaryScale`.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Blue-Green

Theory

Theory: Blue-Green is Two Stacks and One Cutover

Active and preview Services point at old vs new ReplicaSets. Promotion swaps who is active. Analysis can run on preview before the swap.

Invariants

1. Capacity is $\sim 2\times$ during the preview window.
2. Rollback is swapping Services back, not a Git revert alone (Git should still record the desired Rollout spec).
3. StatefulSets are a later limitation chapter.

```
Service active → RS blue
Service preview → RS green    promote: swap active
```

What can break

- Preview service accidentally public without auth.

How to read this chapter

Services, analysis, promote/rollback.

Active and Preview Services

Active and Preview Services

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Active and preview services; the blue-green lifecycle.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Promotion Analysis

Promotion Analysis

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pre-promotion and post-promotion analysis.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Promote and Rollback

Promote and Rollback

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Manual and automatic promotion; rollback behavior.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Rollout Notifications

Theory

Theory: Notifications are Observers of Rollout Events

Triggers fire on events (step, abort, promote). Templates format Slack messages. Scope is namespace or cluster.

Invariants

1. Mis-scoped notifications spam or miss.
2. Secrets for Slack tokens are cluster secrets, not Git plaintext.
3. Troubleshooting is “did the event fire” then “did the webhook accept.”

```
Rollout event → trigger → template → Slack webhook
```

What can break

- One cluster-scoped subscription for every namespace.

How to read this chapter

Architecture, Slack config, troubleshooting.

Notification Architecture

Notification Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Notification architecture and scope (namespace vs cluster).

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Triggers, Templates, Slack

Triggers, Templates, Slack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Triggers, templates, subscriptions, and Slack configuration.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Troubleshooting Notifications

Troubleshooting Notifications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Troubleshooting notifications.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Migrating to Rollouts

Theory

Theory: workloadRef Lets a Rollout Adopt a Deployment

Migration is changing the controller that owns ReplicaSets without changing the pod spec. `workloadRef` points at an existing Deployment. StatefulSets often cannot follow this path.

Invariants

1. Git (Argo) must stop managing the old Deployment's replica field.
2. Aborts during migration need a rehearsed rollback.
3. Stateful identity (ordinals, PVCs) does not map to canary ReplicaSets cleanly.

```
Deployment --workloadRef--> Rollout takes over RS  
Git desired: Rollout, not Deployment replicas
```

What can break

- Both Deployment and Rollout controllers racing on the same RS.

How to read this chapter

`workloadRef`, conversion, canary migration, StatefulSet limits.

Migration and workloadRef

Migration and workloadRef

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Migration strategies and workloadRef.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Direct Conversion

Direct Conversion

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Direct conversion and Argo CD synchronization.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Canary Migration

Canary Migration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary migration, aborts, retries, scaling down the old Deployment.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

StatefulSet Limitations

StatefulSet Limitations

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- StatefulSet limitations.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Capstone

Theory

Theory: One Loop from Commit to Careful Traffic

Git commit → forge CI → image digest in Git → Argo CD sync → Rollout steps → analysis → stable. That is the book's end-to-end mental model.

Invariants

1. Every stage is a name over content-addressed bytes (Git hash, image digest).
2. Humans change Git; controllers change clusters.
3. Larger environments add more clusters, ApplicationSets, and stricter analysis — not a different physics.

```
commit → CI → digest in Git → Argo → Rollout → traffic → stable
```

What can break

- A snowflake kubectl at any stage (breaks the loop).

How to read this chapter

End-to-end model, larger environments.

A Working Mental Model

A Working Mental Model

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A working mental model, end to end.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Larger Environments

Larger Environments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Progressive delivery in larger environments.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Appendices

Understanding Questions and Answers

Understanding Questions and Answers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Four focused questions with detailed answers after each part.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Git Competency Checklists

Git Competency Checklists

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Level 1: 6 essential skills everyone needs.
- Level 2: 5 power skills (reflog, interactive rebase, cherry-pick, bisect, Conventional Commits).
- Level 3: 5 advanced specialist topics.

Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Real-World Scenario Playbook

Real-World Scenario Playbook

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Sanjay's deleted branch panic → recovered with reflog.
- Ajay's messy history → cleaned with interactive rebase.
- Maya's critical hotfix → deployed with cherry-pick.
- The team's mysterious production bug → fixed with bisect.
- Professional commit messages.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).

Lab Environment Reference

Lab Environment Reference

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Docker, kind/K3s, kubectl, Helm, Argo CD/Rollouts setup.
- Heredoc and shell conventions.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Command Quick Reference and Glossary

Command Quick Reference and Glossary

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Command quick reference and glossary.

i Note

Scaffolding for the frozen TOC. Linux is the primary host. See [content/_planning/SYLLABUS.md](#).